



To catch and reverse a quantum jump mid-flight

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(Thesis: arXiv:1902.10355)



Nintendo Corp.

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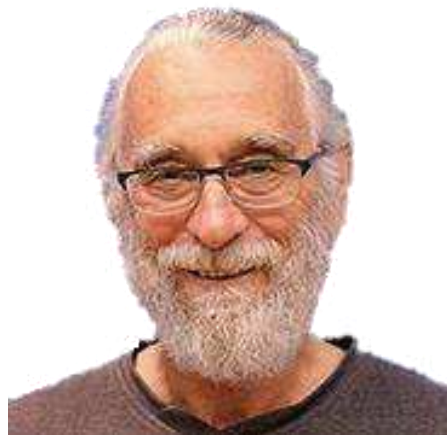
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M.H. Devoret

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Yale Institute for Nanoscience
and Quantum Engineering

Is it possible to get an advance warning signal
that a quantum jump is about to occur?



H.J. Carmichael

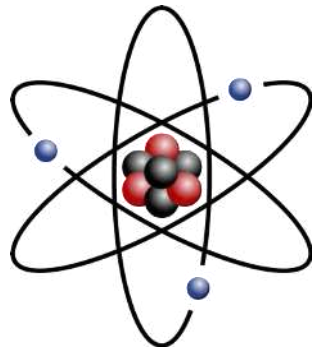


Outline

Jumps in a quantum system

original observation

quantum trajectory prediction

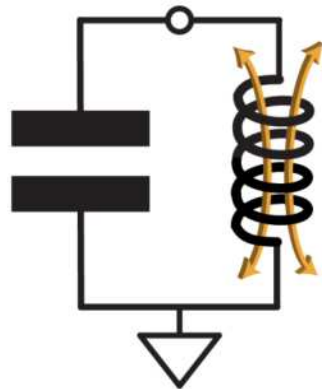


Art: Indoleces

Circuit quantum electrodynamics realization

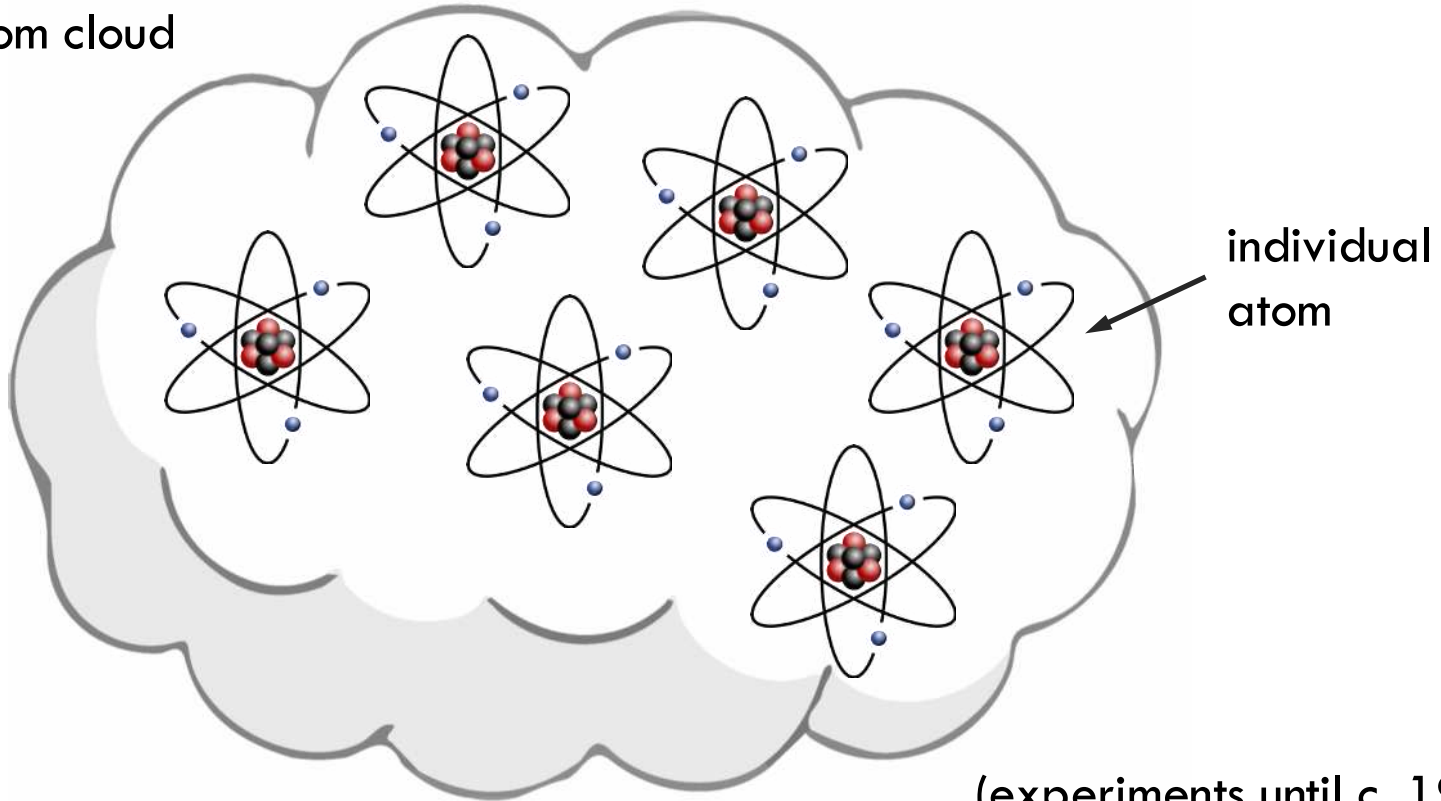
experimental apparatus

results on jump coherence and determinism



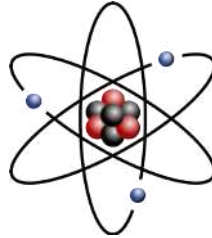
Ensemble of quantum systems

e.g., atom cloud

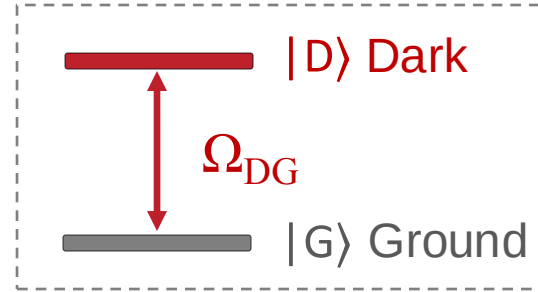
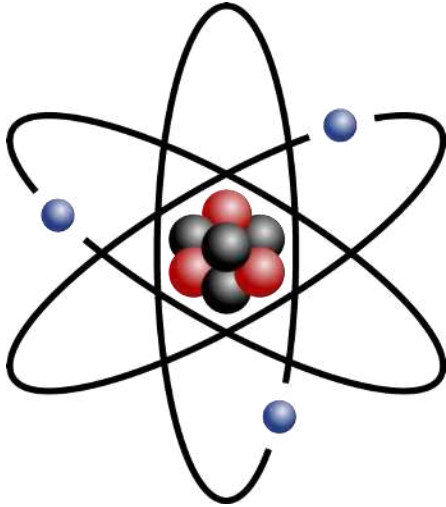


(experiments until c. 1980)

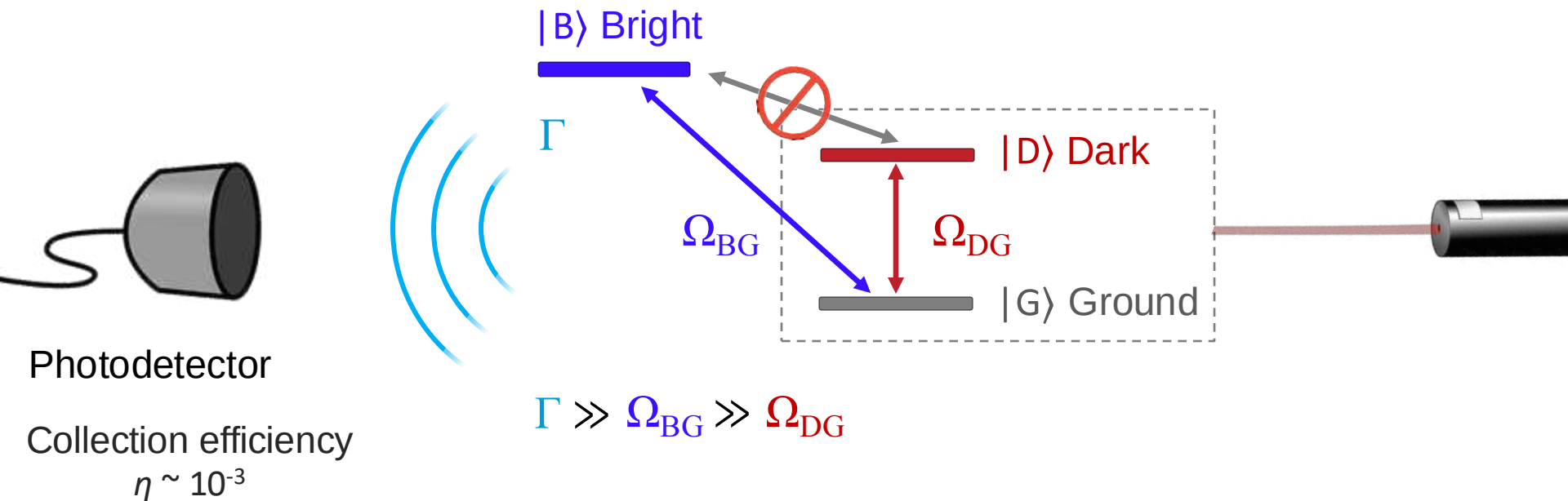
Individual quantum system



Quantum jumps of an *individual* atom

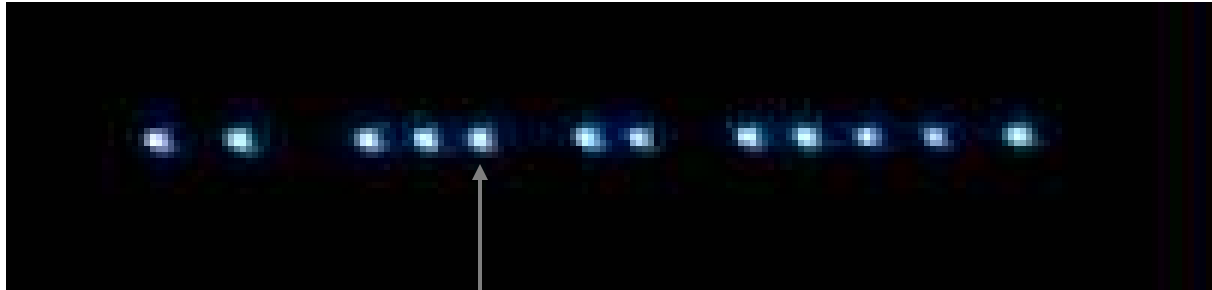
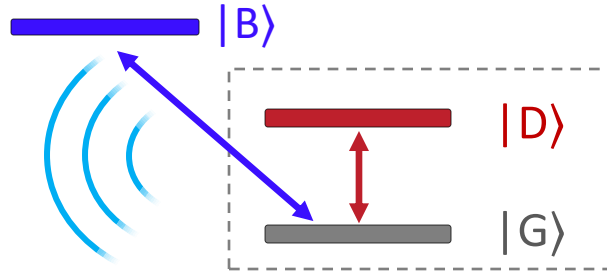


Observation of jumps in an *individual* atom



Exp: Berquist *et al.*, PRL (1986); Sauter *et al.*, PRL (1986); Nagourney *et al.*, PRL (1986)
Thy: Cook & Kimble, PRL (1985); Cohen-Tannoudji & Dalibard, EPL (1986), ...

Quantum jumps of *individual* ions



Single Ca^+ ion

(real-time)

Principle of quantum jumps observation

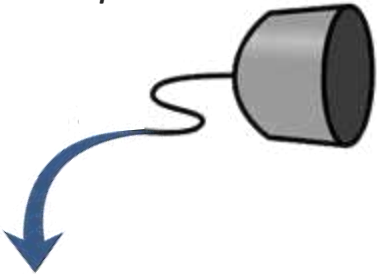
Photodetector
 $\eta \sim 10^{-3}$

$$\Gamma \gg \Omega_{BG} \gg \Omega_{DG}$$

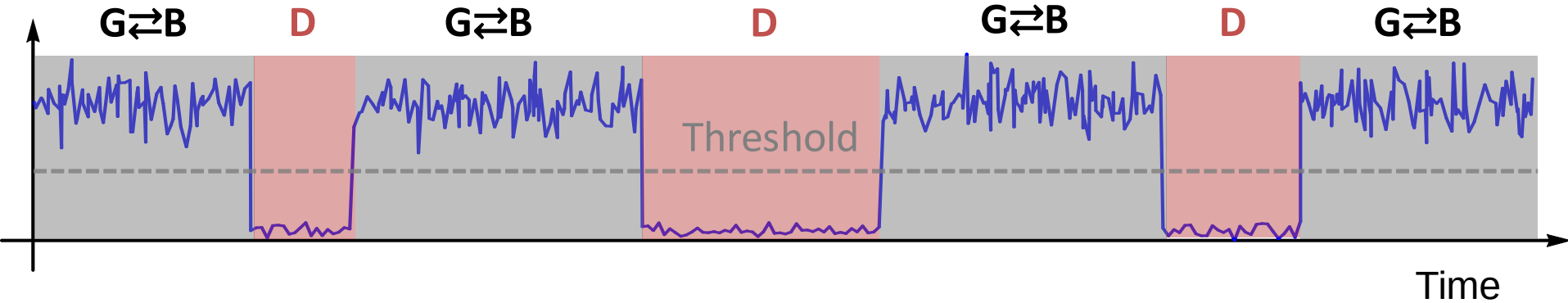
$|B\rangle$ Bright

$|D\rangle$ Dark

$|G\rangle$ Ground



Measurement record (fluorescence)



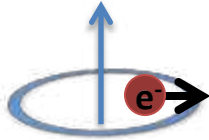
Observations of quantum jumps since 1986

Trapped massive particles

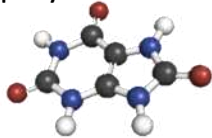
Single *ion* [1-3] (1986)



Single *electron* [4] (1992)
($1e^-$ cyclotron oscillator)

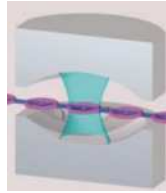


Single *molecule* [5] (1995)
(large polyatomic molecule)

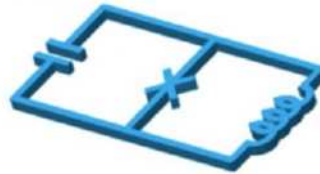


Light quanta

Microwave cavity
(*CQED*) [6] (2007)

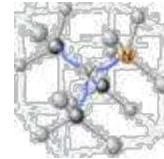


Superconducting qubit
& cavity (*cQED*) [9-11] (2011)

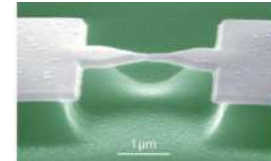


Solid state and mesoscopic

Nitrogen-vacancy center
in diamond [7,8] (2010)



Atomic-point contact
(Andreev states) [12] (2015)



And others!

[1] PRL 56, 2797 (1986)

[2] PRL 57, 1696 (1986)

[3] PRL 57, 1699 (1986)

[4] PRL 83, 1287 (1999)

[5] Nature 373, 132 (1995)

[6] Nature 446, 297 (2007)

[7] Science 329, 542 (2010)

[8] Nature 477, 574 (2011)

[9] PRL 106, 110502 (2011)

[10] Science 339, 178 (2013)

[11] Nature 511, 444 (2014)

[12] Science 349, 1199 (2015)

“If all this damned *quantum jumping* were really to stay,
I should be sorry I ever got involved with quantum theory.”

- E. Schrödinger



The case of the fully efficient observer

Hypothetical efficiency $\eta = 1$

$$BW_{\text{det}} \gg \frac{\Omega_{\text{BG}}^2}{\Gamma} \gg \Omega_{\text{DG}}$$

$$\Gamma \gg \Omega_{\text{BG}} \gg \Omega_{\text{DG}}$$

$|B\rangle$ Bright

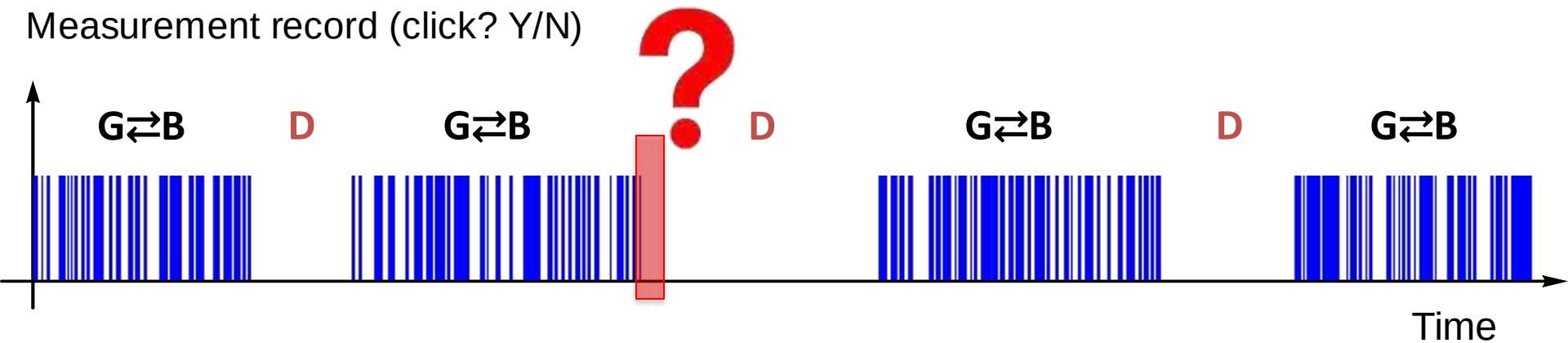


$|D\rangle$ Dark

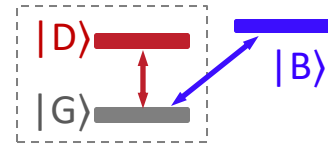
Ω_{DG}

$|G\rangle$ Ground

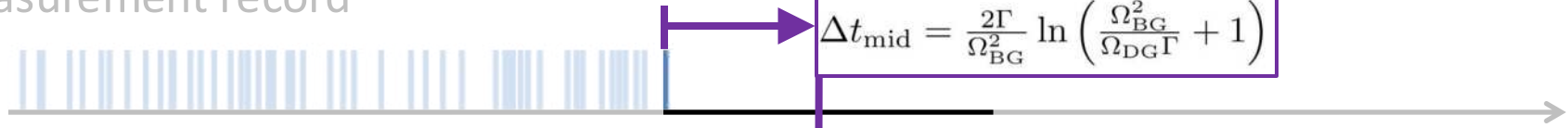
Measurement record (click? Y/N)



No-click trajectory

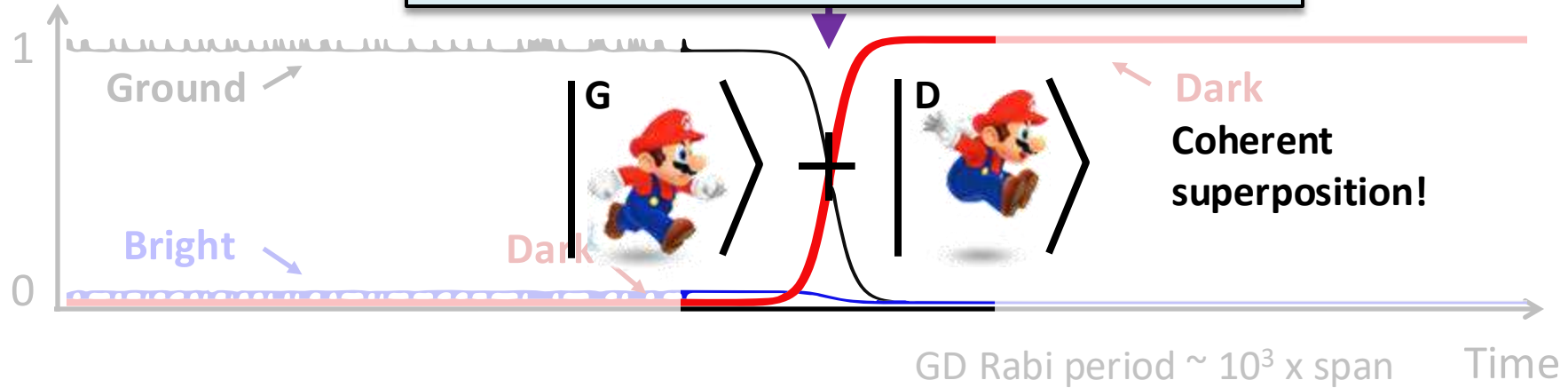


Measurement record



Inferred population

Requires not missing a **SINGLE** click!



Carmichael, SUSSP71 (2015); Ruskov et al., PRB (2007); Related: Porrati & Putterman, PRA (1987); Mabuchi & Zoller PRL (1996); Plenio & Knight, RMP (1998); Katz et al., Science (2006) & PRL (2008)

Principle of our cQED experiment

Related quantum circuit experiments on trajectories of 2-level atoms:

Hacohen-Gourgy *et al.*, PRL (2018)

Cottet *et al.*, PNAS (2017)

Hacohen-Gourgy *et al.*, Nature (2016)

Slichter *et al.*, NJP (2016)

Naghiloo *et al.*, Nat. Comm. (2016)

Campagne-Ibarcq *et al.*, PRL (2016)

Weber *et al.*, Nature (2014)

de Lange *et al.*, PRL (2014)

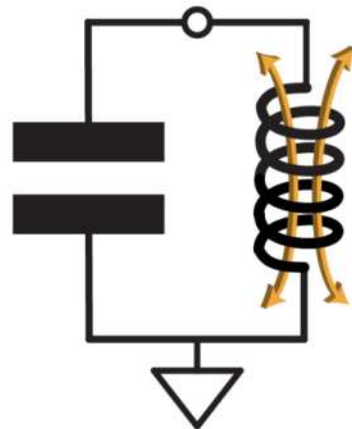
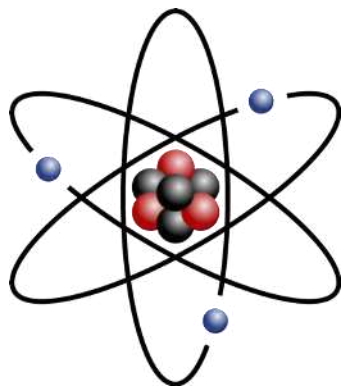
Murch *et al.*, Nature (2013)

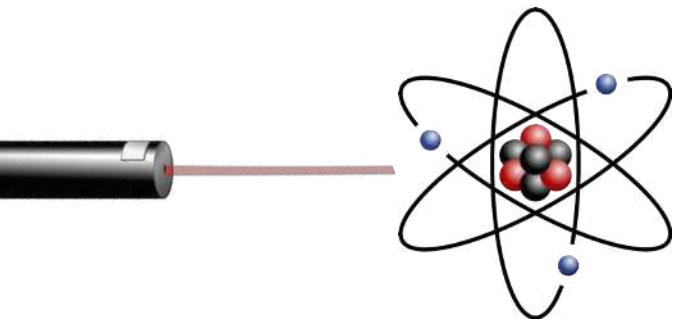
Campagne-Ibarcq *et al.*, PRX (2013)

Katz *et al.*, PRL (2008)

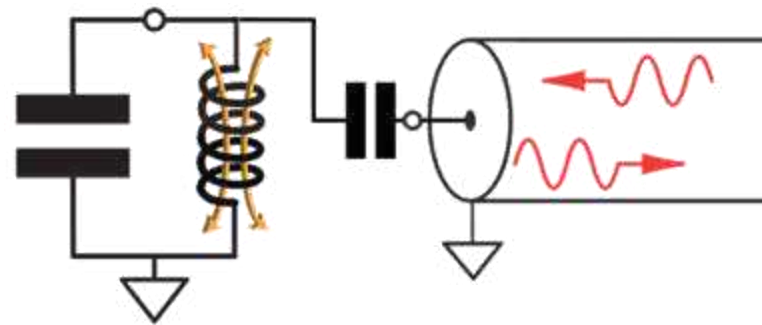
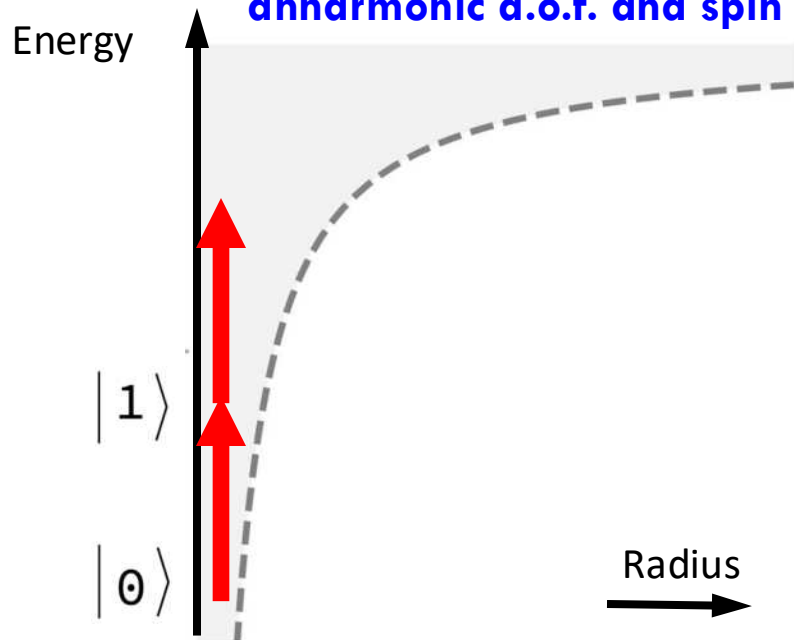
Katz *et al.*, Science (2006)

...

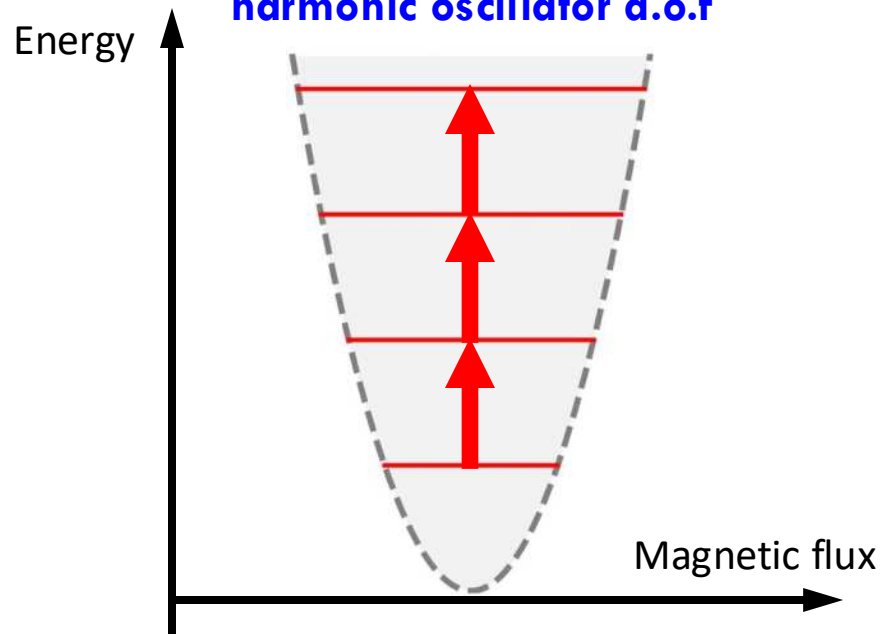


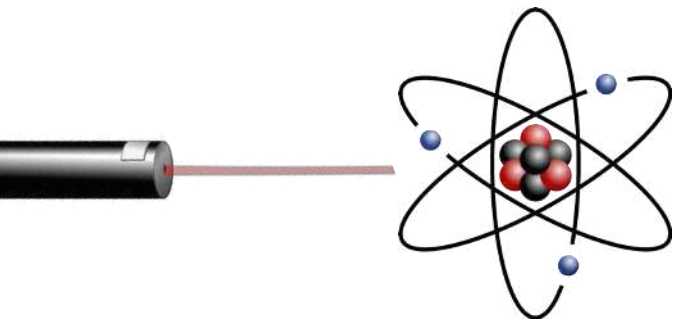


anharmonic d.o.f. and spin

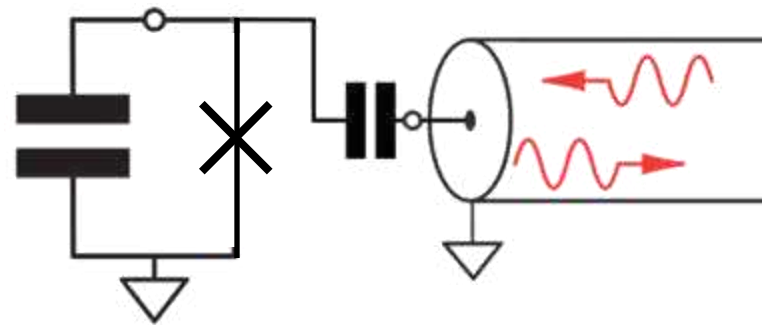
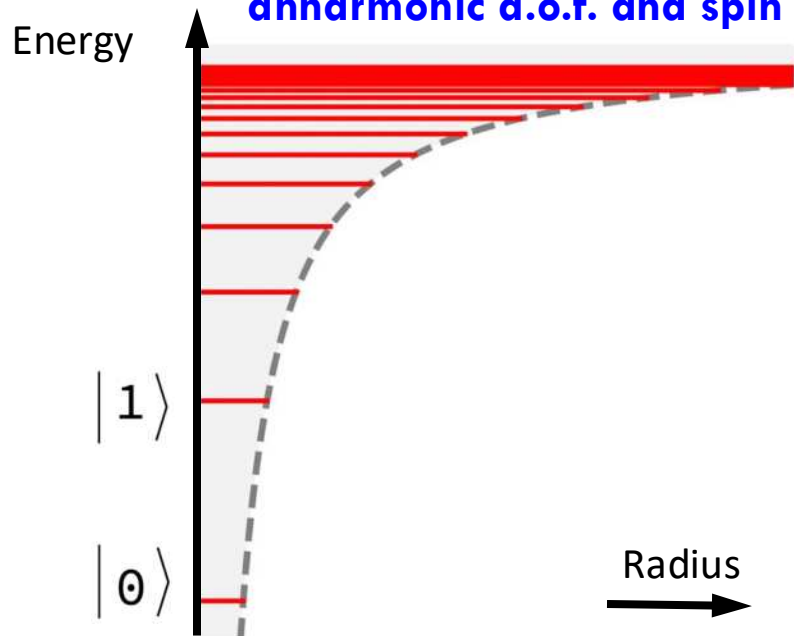


harmonic oscillator d.o.f

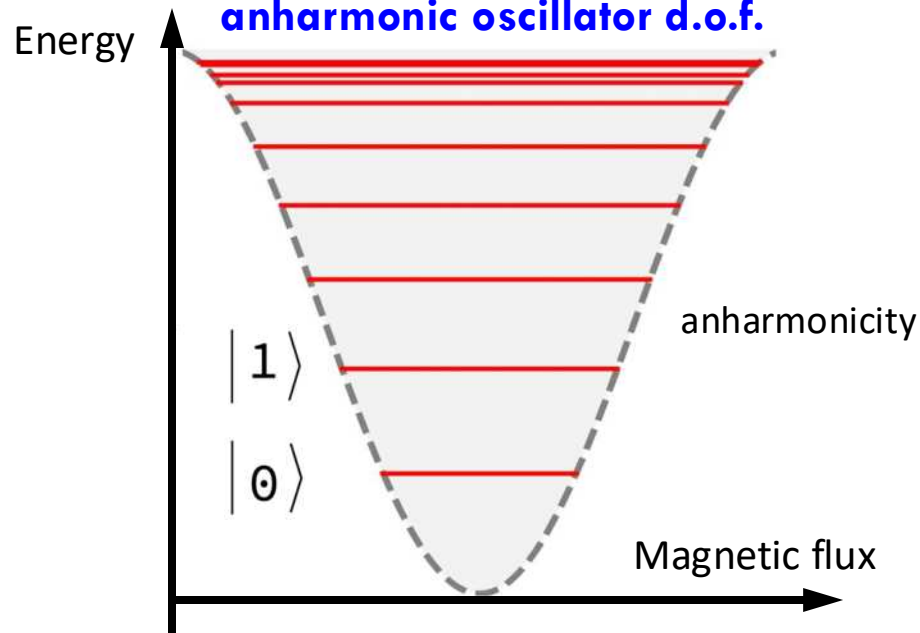




anharmonic d.o.f. and spin

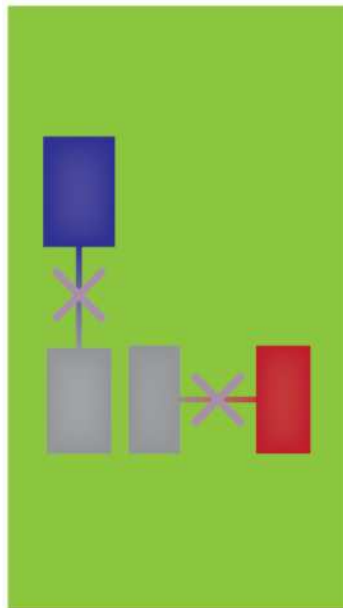


anharmonic oscillator d.o.f.

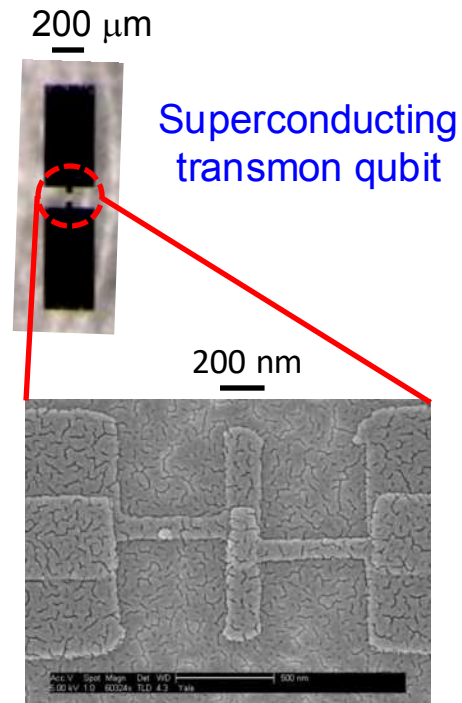


Quantum superconducting circuit architecture

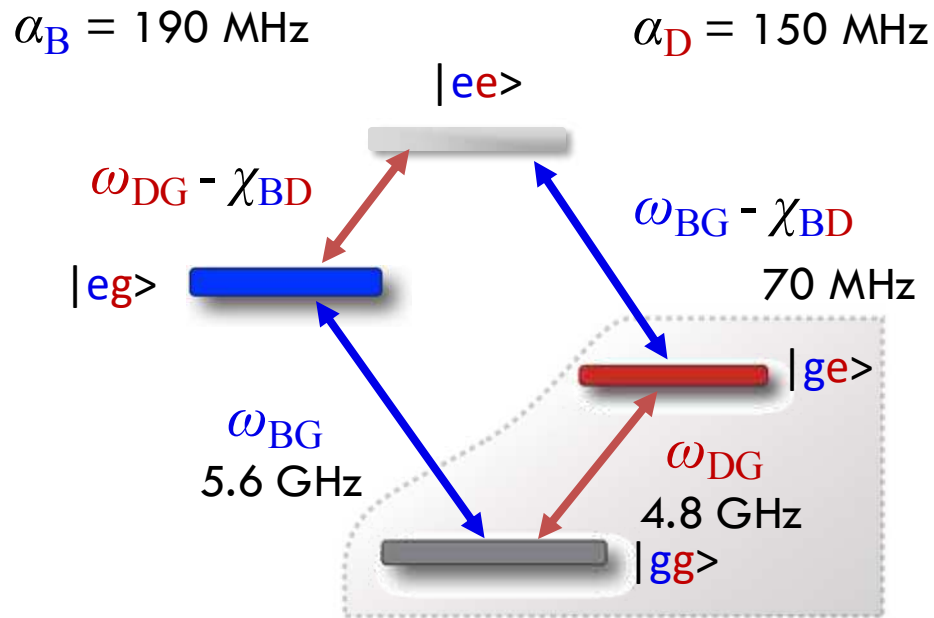
Schematic representation of chip



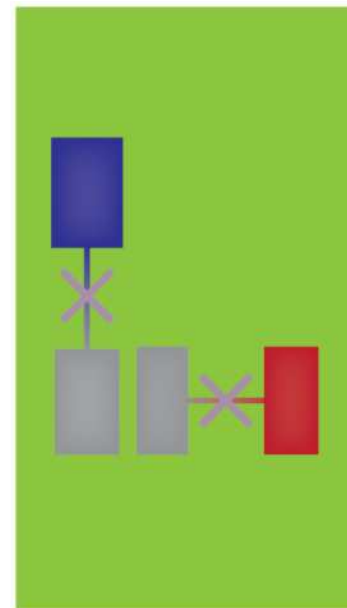
Substrate
(not to scale)



cQED implementation of V-system



Schematic representation of chip



Substrate
(not to scale)

Related work: Gambetta *et al.*, PRL (2011)

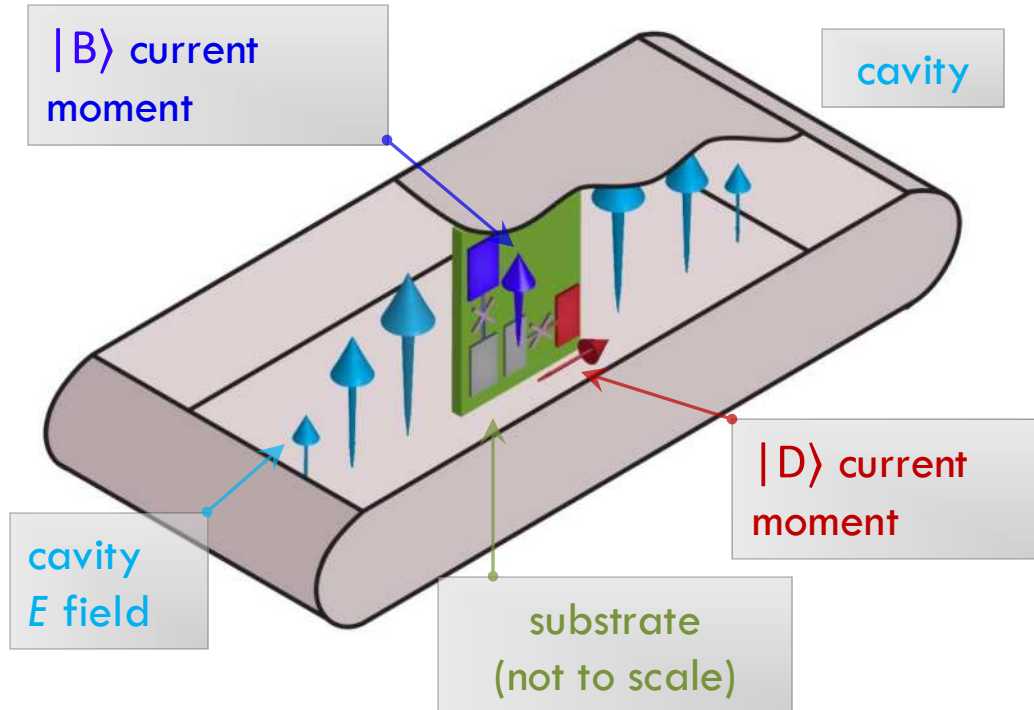
Srinivasan *et al.*, PRL (2011) Dumur *et al.*, PRB (2015)

Tanay Roy, ... Dr. Rajamani Vijayaraghavan

Zhang *et al.*, Nature JQI (2017) ...

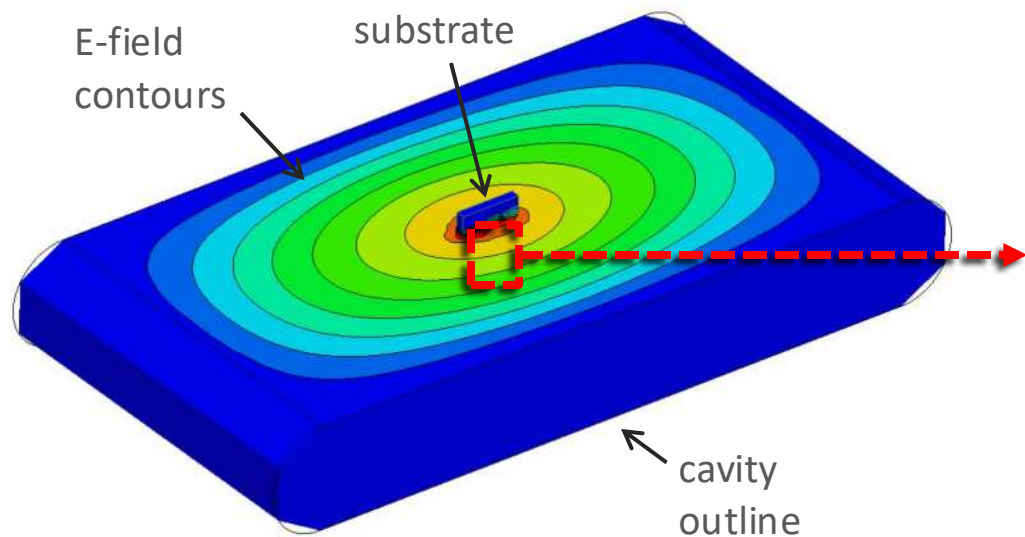
cQED implementation of V-system

Schematic representation of design



Energy participation ratio (EPR) approach*

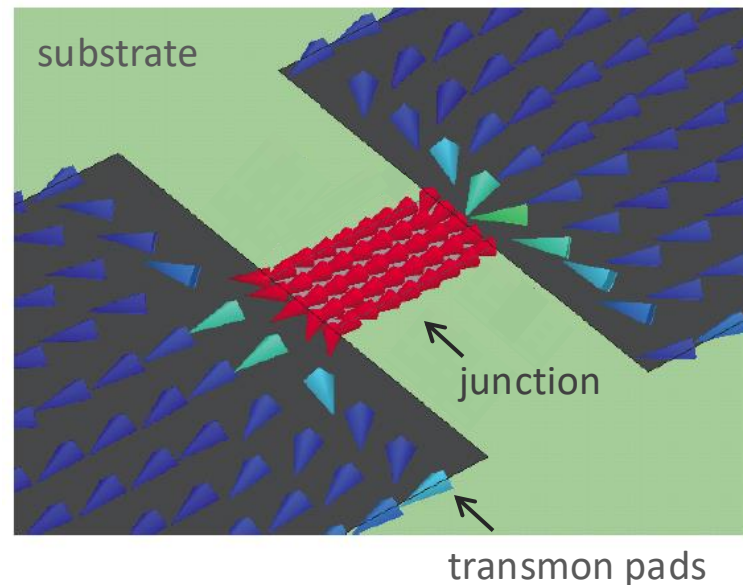
Cavity mode (9.0 GHz)



0  Max

E-field magnitude

Qubit modes (linearized)

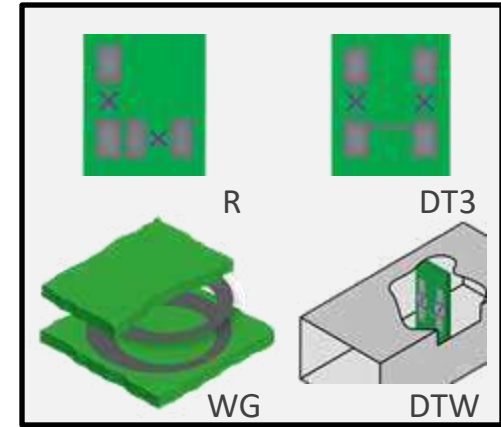
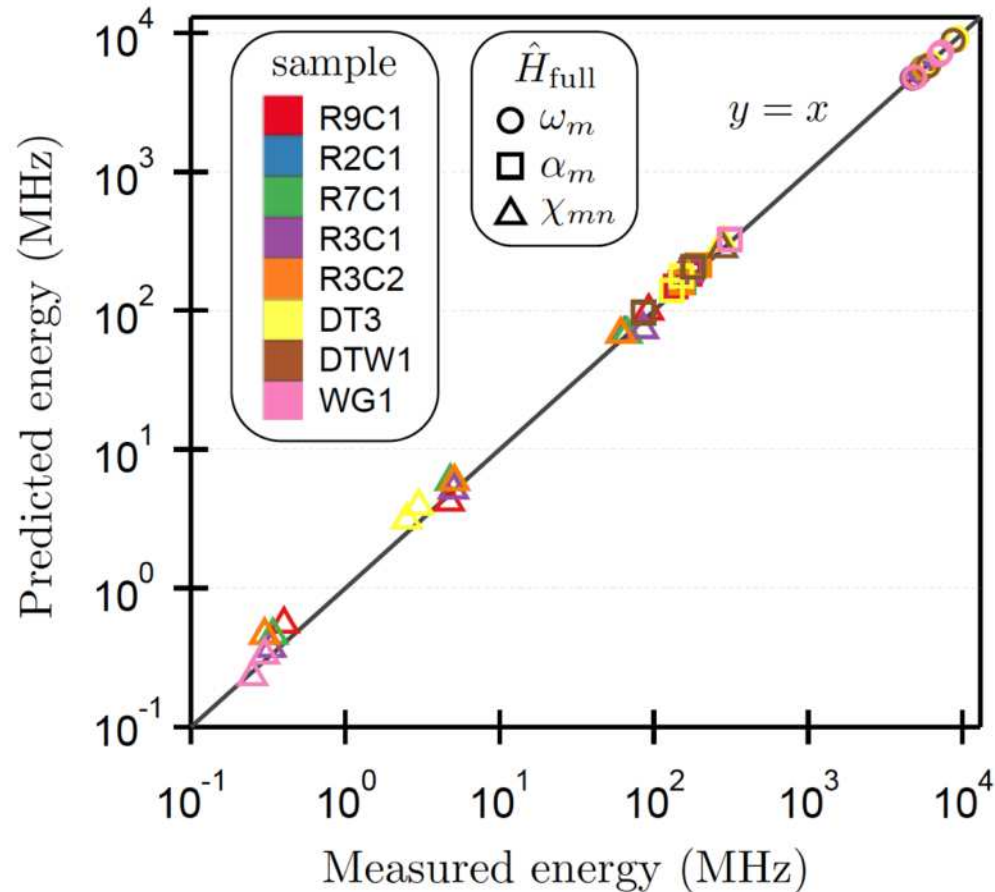


0  Max

Current-density magnitude

* Mineev arXiv:1902.10355 (2019) Related: Nigg *et al.* (2012); Bourassa *et al.* (2012); Solgun & DiVincenzo ...

EPR theory vs. experiment: agreement over 5 orders of magnitude



R: Minev *et al.* (2018)

WG: Minev *et al.* (2013, 2016) DT3,

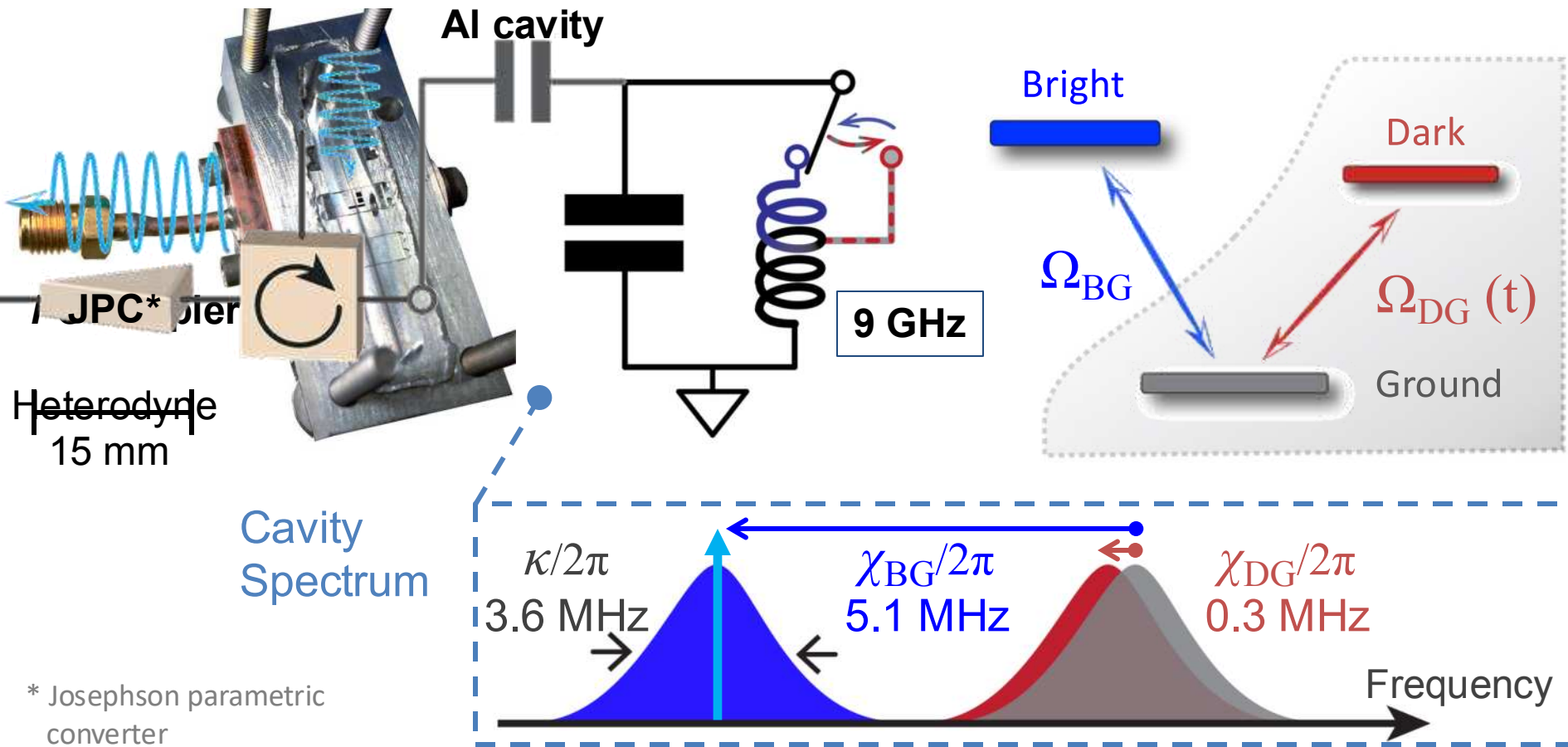
DTW:

Minev *et al.* (2019)

www.github.com/zlatko-minev/pyEPR

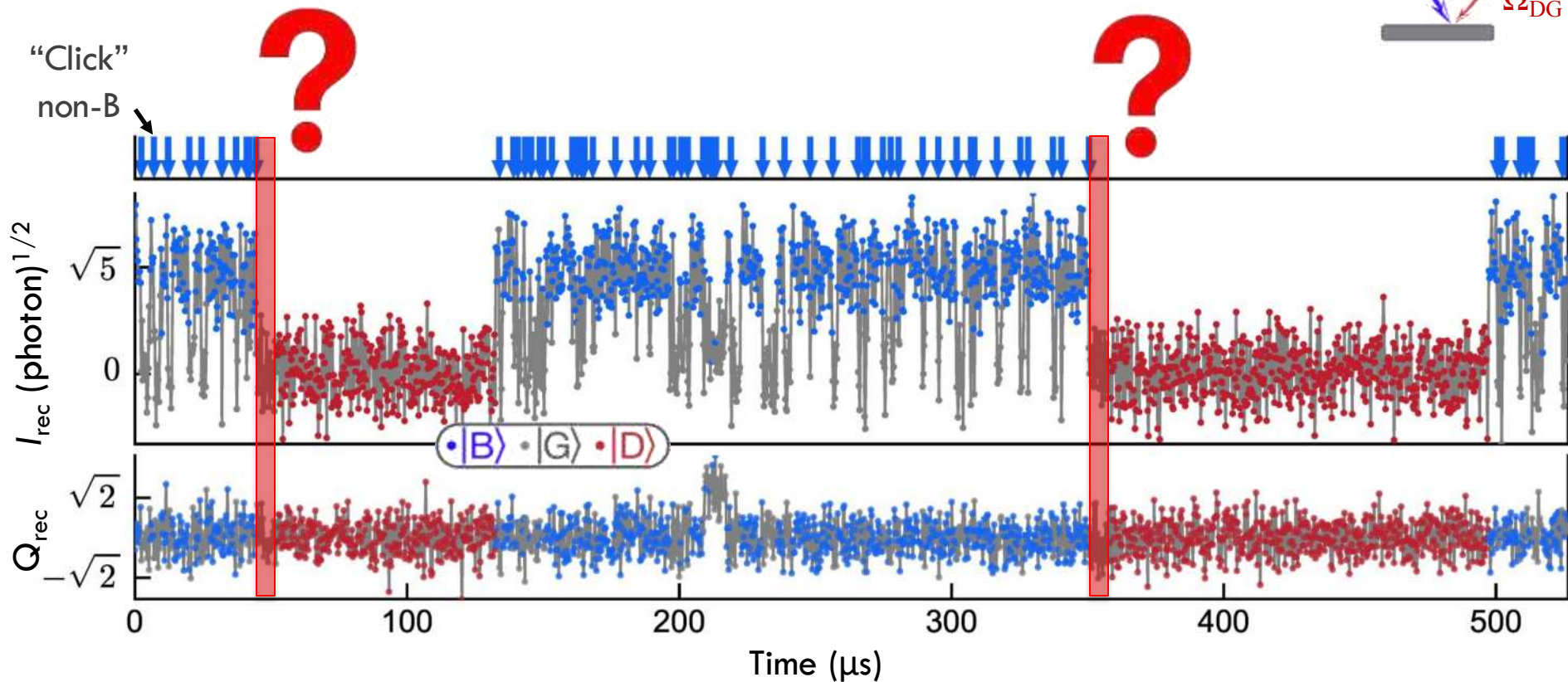
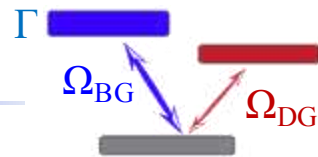
* Minev arXiv:1902.10355 (2019)

Measuring B / non-B with 90% efficiency



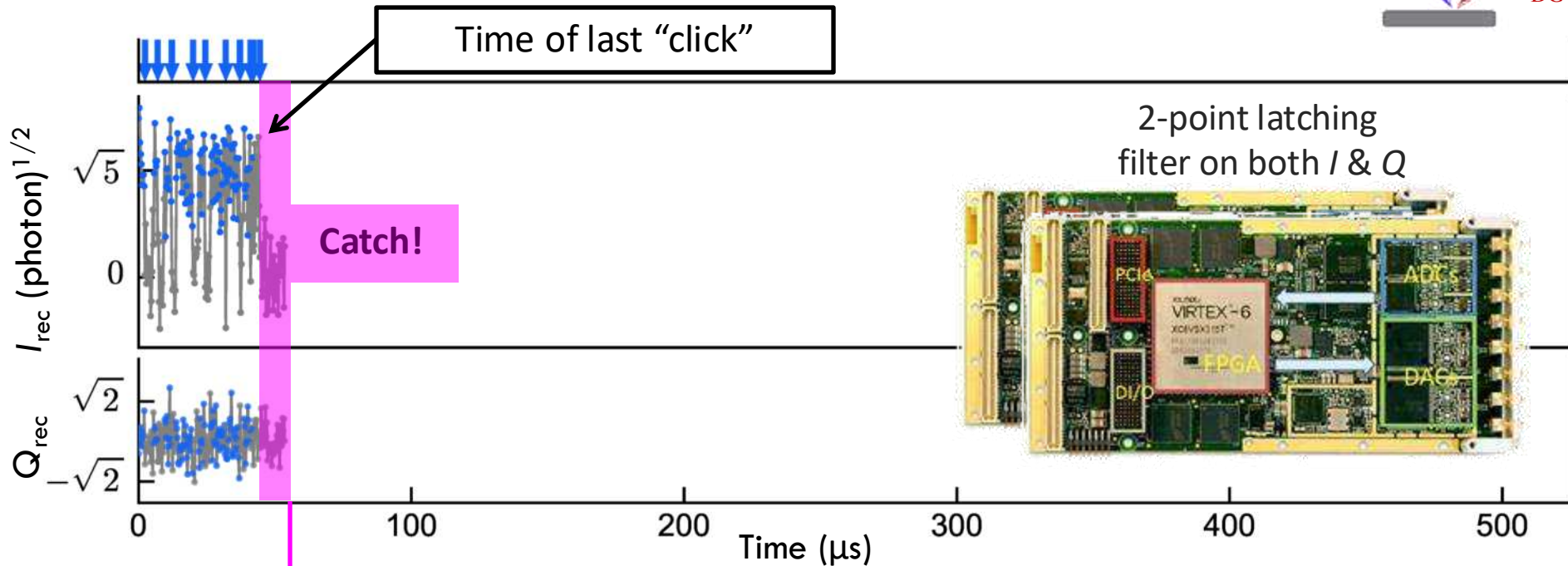
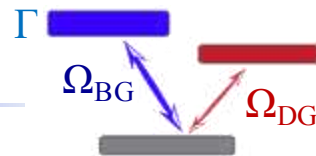
Unconditioned quantum jumps

Inferred state & click times



Real-time detection of a quantum jump

Catch protocol



Msmt. rate
 $\Gamma_m^{-1} \sim 8 \text{ ns}$

$\ll$

Integration time
 $2.6 \times 10^2 \text{ ns}$

$\ll$

GB jump timescale
 10^3 ns

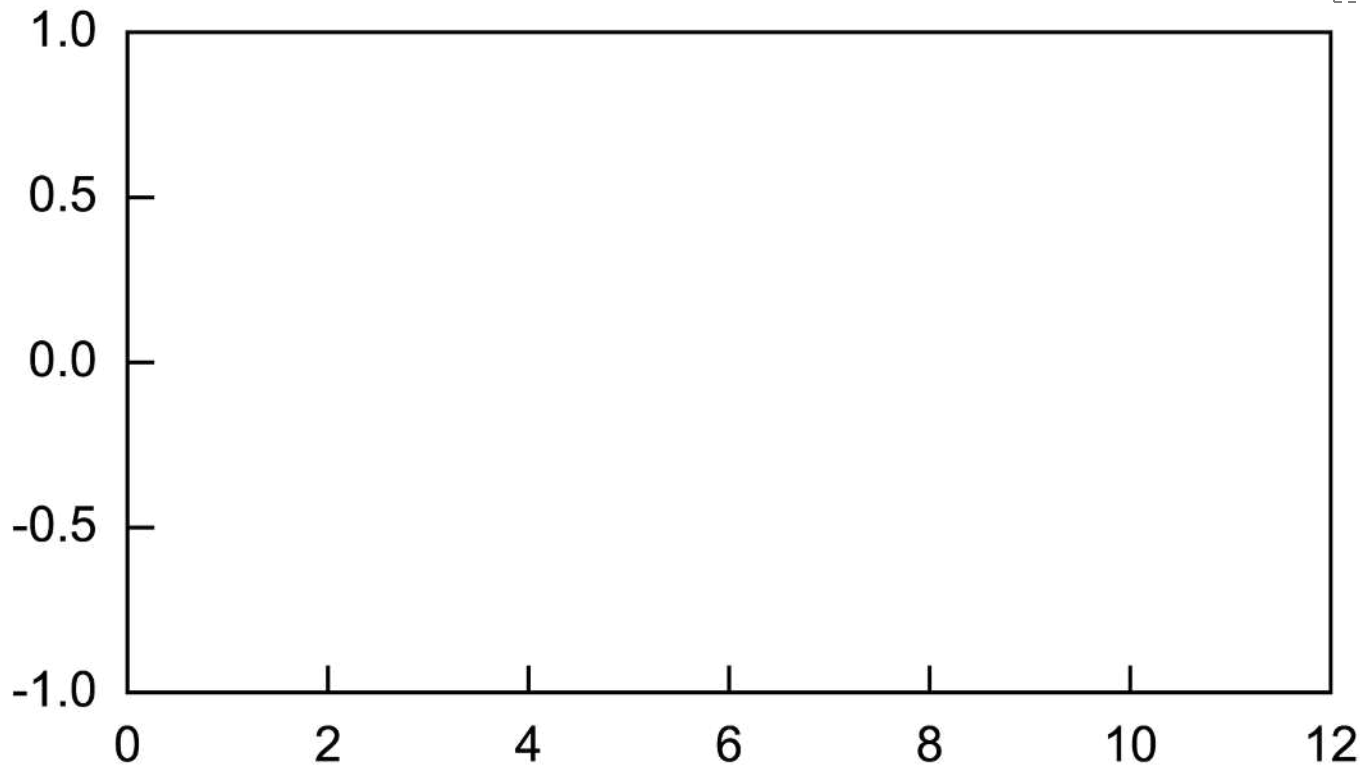
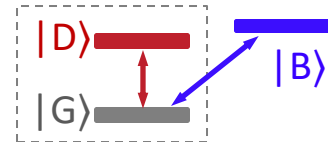
$\ll$

t_{mid}

$\ll$

GD jump timescale
 $\sim 10^5 \text{ ns}$

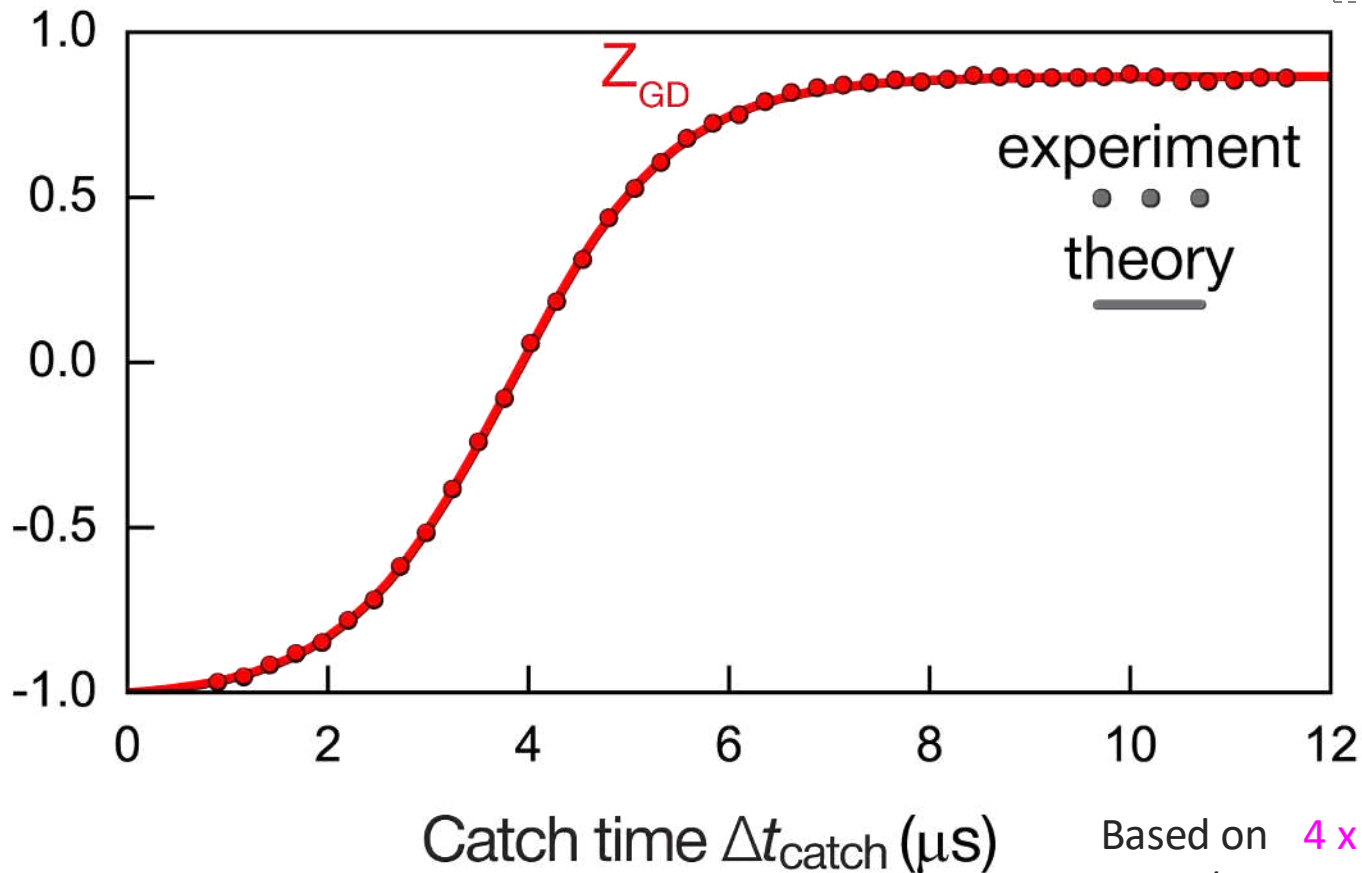
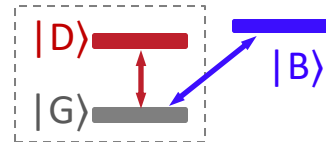
Catching the quantum jump mid-flight



Catch time Δt_{catch} (μs)

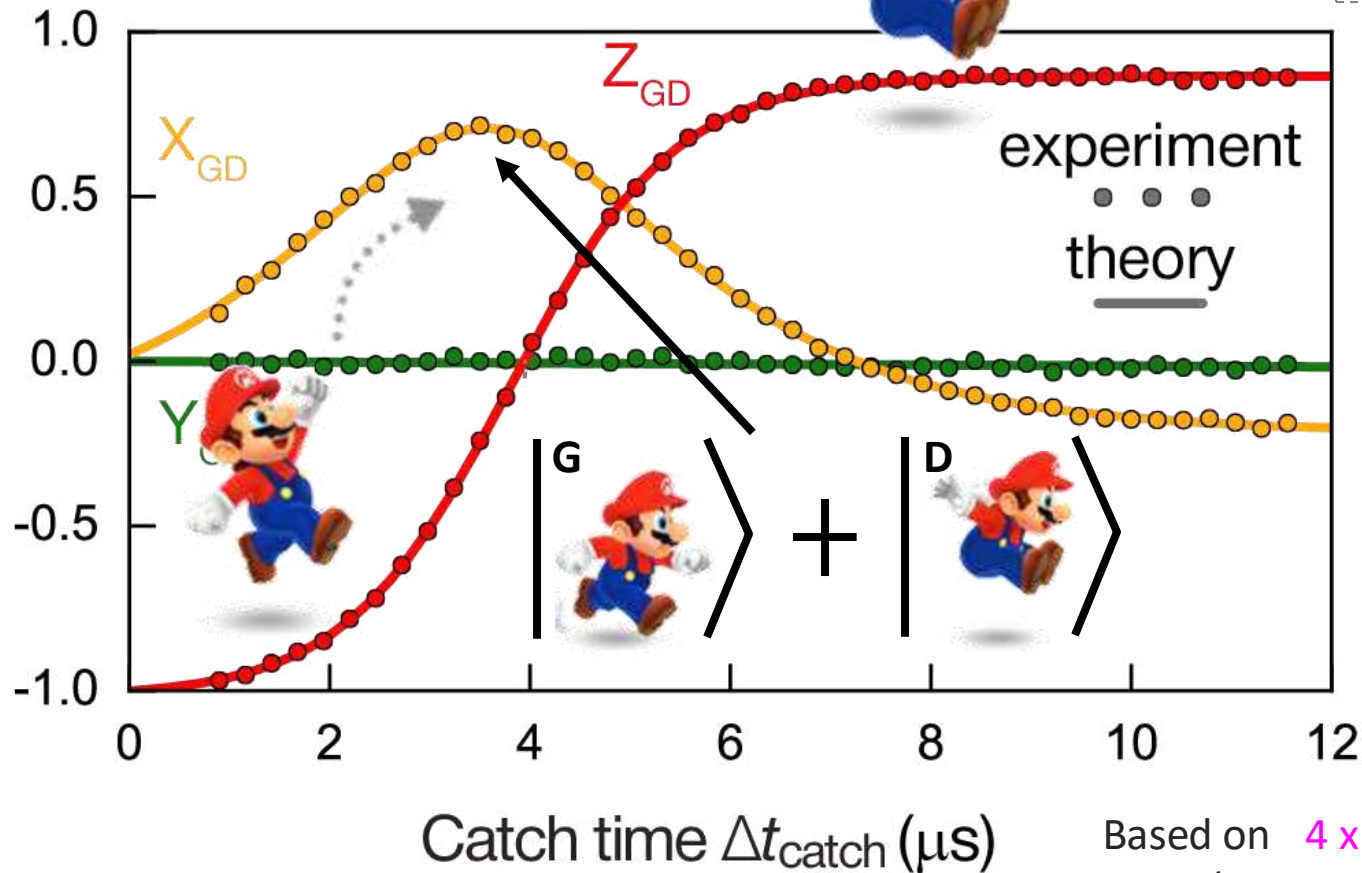
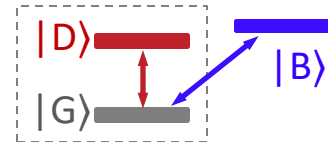
Based on 4×10^6 catch events
(10 min. tracking time)

Catching the quantum jump mid-flight



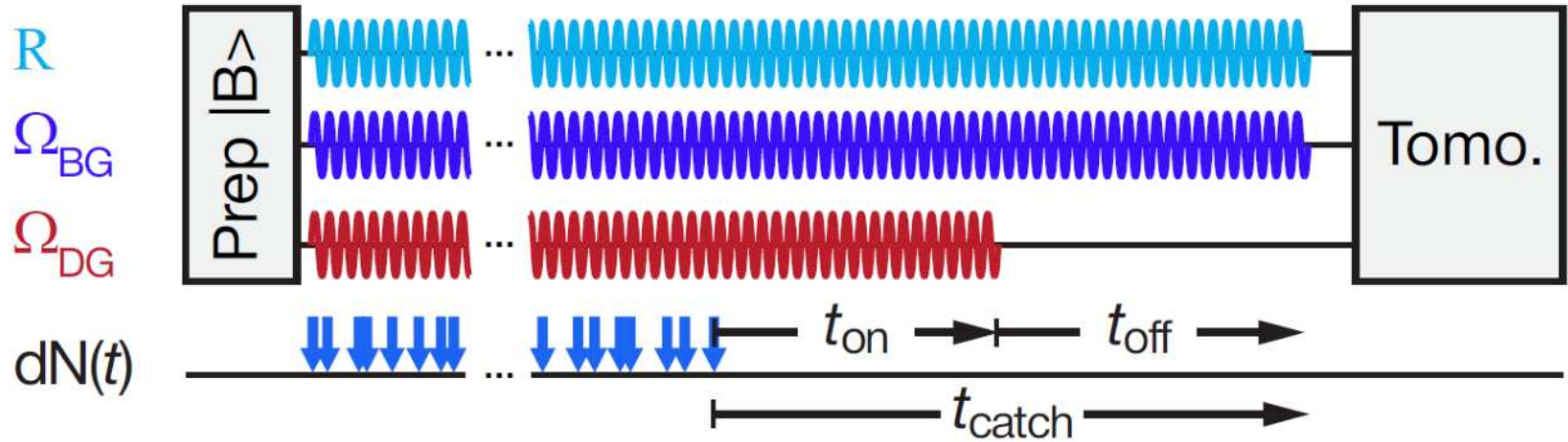
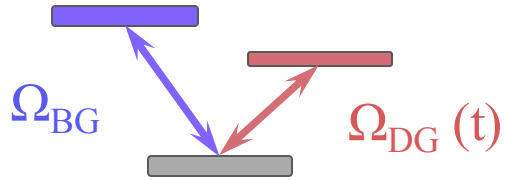
Based on 4×10^6 catch events
(10 min. tracking time)

Catching the quantum jump mid-flight

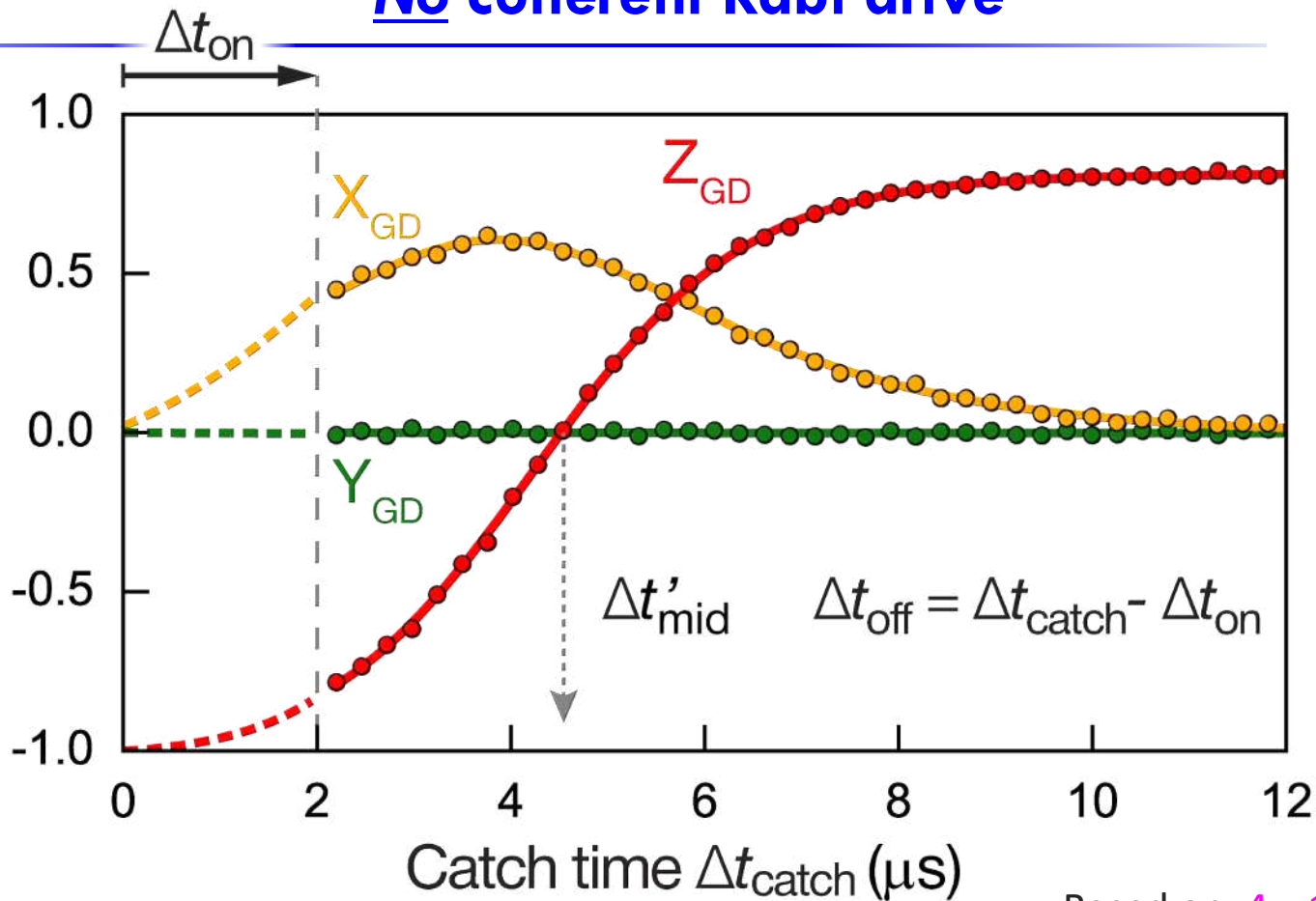
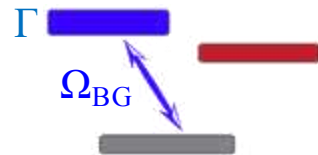


Based on 4×10^6 catch events
(10 min. tracking time)

Catch in *absence* of Dark drive

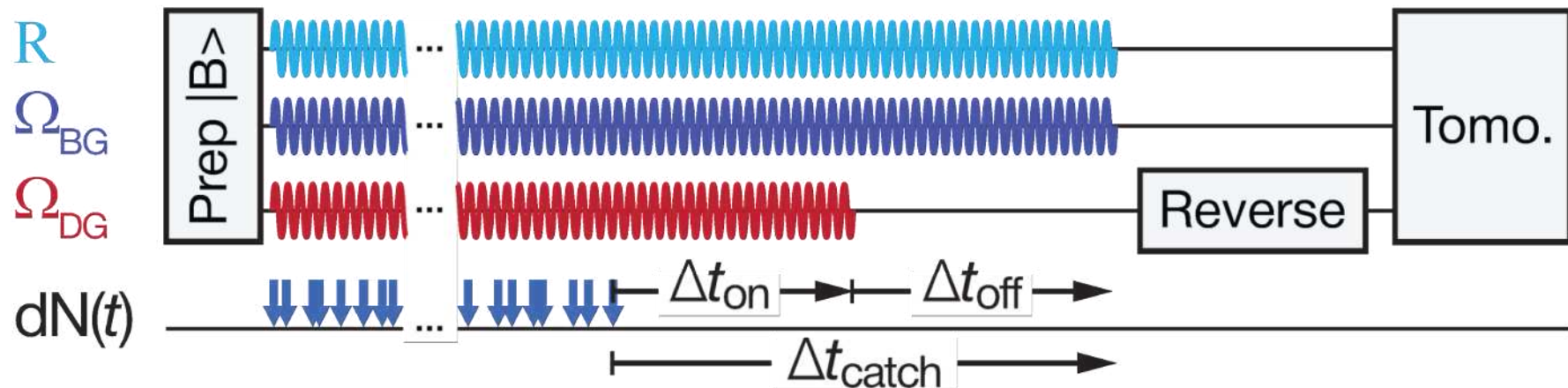


No coherent Rabi drive



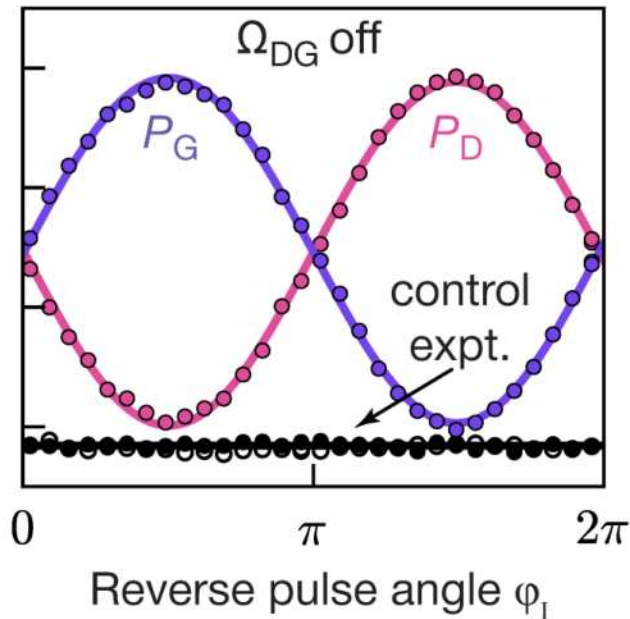
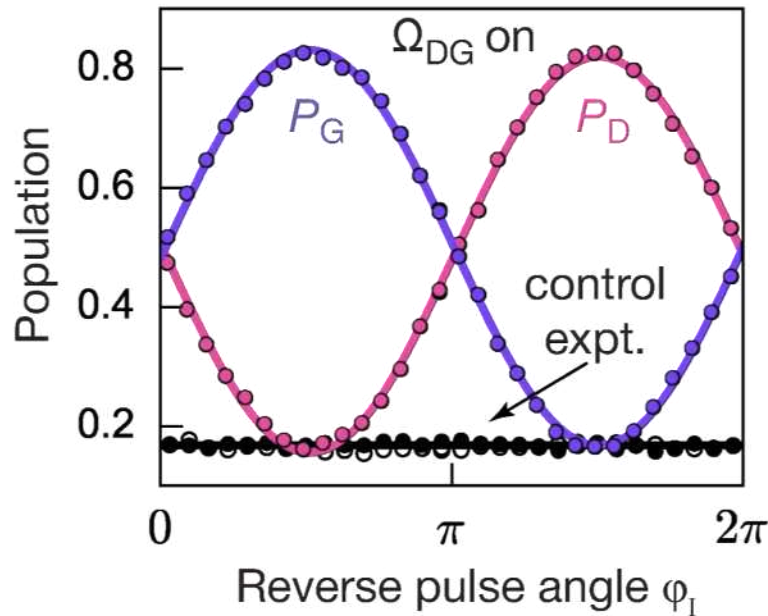
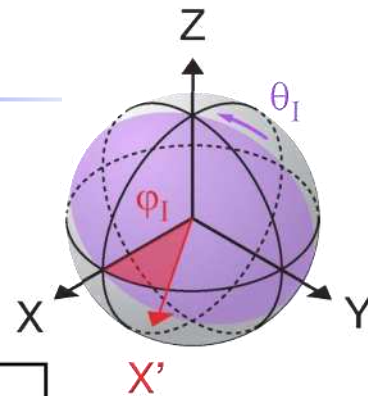
Based on 4×10^6 catch events

Reversing the quantum jump mid-flight

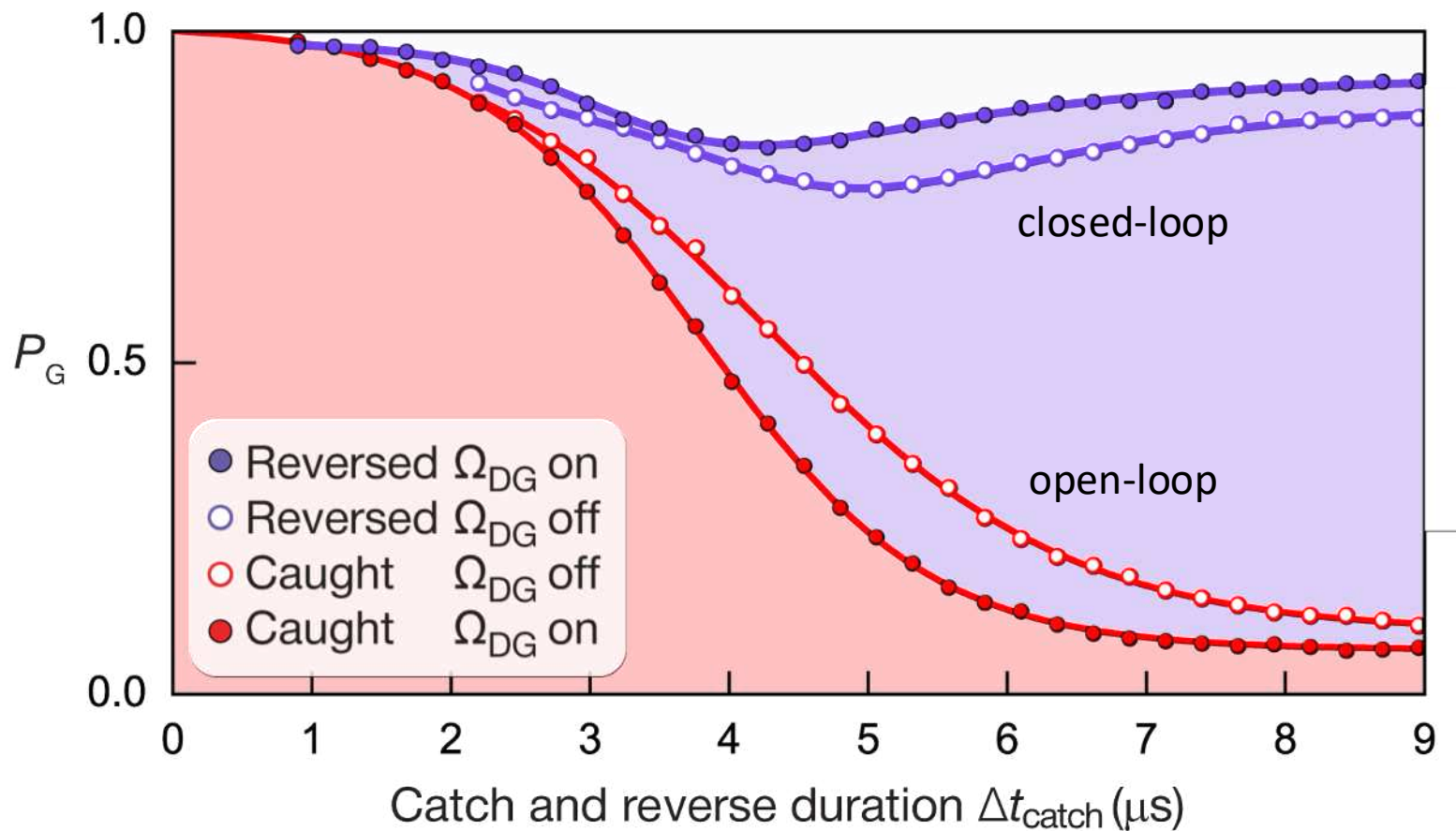


Experimental reversal of jump

No click until $\Delta t_{\text{mid}} =$ advance warning of individual jump

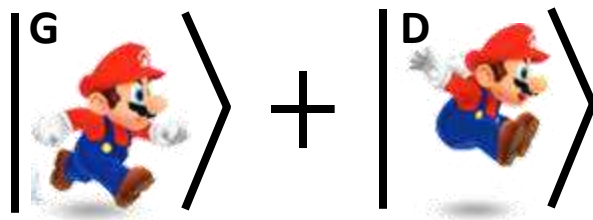


Catching and reversing the quantum jump mid-flight



Conclusions & Future directions

Observed jump “internal coherence” using fast electronics and feed-forward control



Caught individual jumps in mid-flight and reversed them

Quantum trajectory efficiently explains details of exp. results

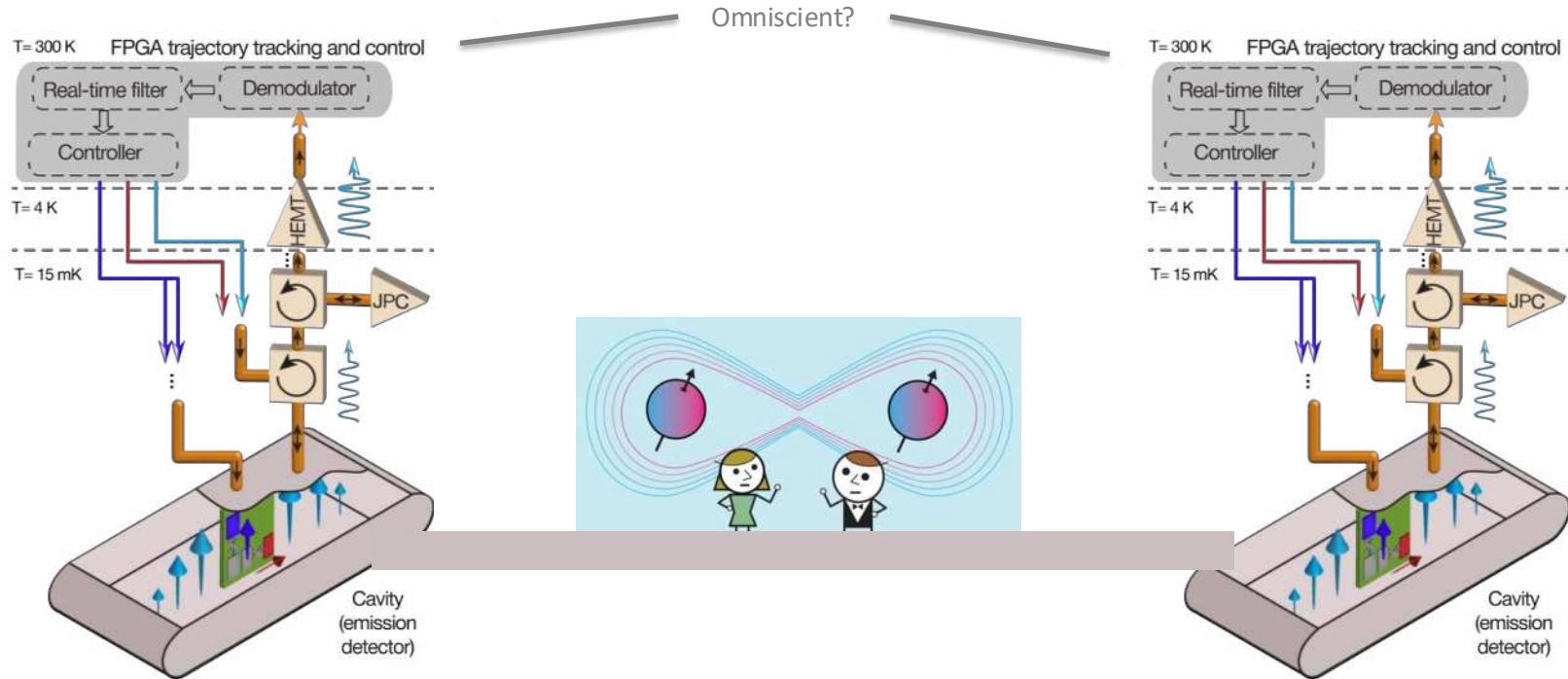
Nature 570, 200 (2019) Minev, Ph.D. Thesis (2018)

Continuous syndrome measurement for autonomous quantum error correction

Follow ups: [arXiv:1808.00726](https://arxiv.org/abs/1808.00726), [arXiv:1810.03225](https://arxiv.org/abs/1810.03225), ...

**Next experiment
idea for non-local /
Wigner / Bell**

Next experiment idea for non-local / Wigner / Bell





Open Labs

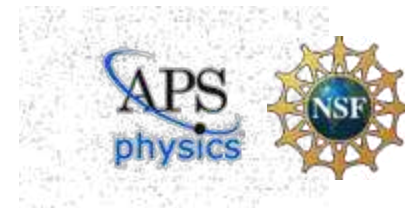
Est. 2012



Yale-Jefferson
Award

Open Labs

Science outreach and career pathways



Yale
University



University of
Pennsylvania



Princeton
University



Columbia
University

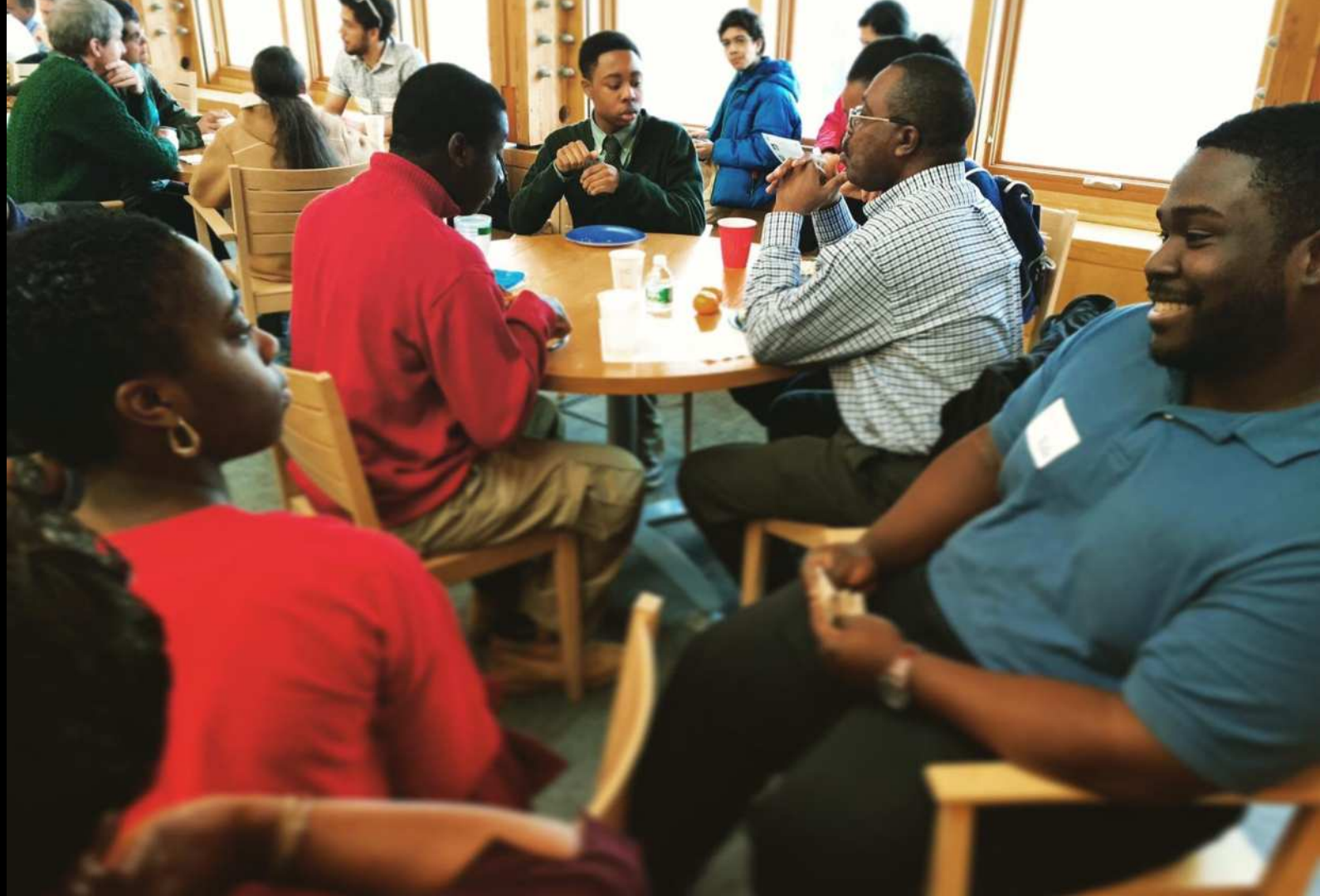


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Outreach and career pathways



Step into the shoes of a scientist at the cutting edge...

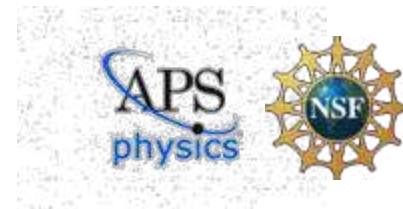
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Science outreach and career pathways



Yale
University



University of
Pennsylvania



Princeton
University



Columbia
University



Harvard
University

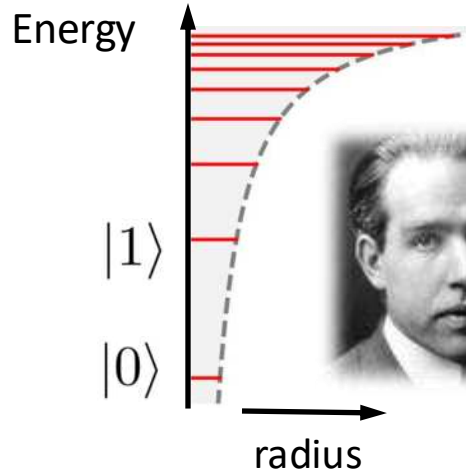
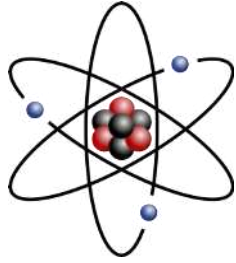


...

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EXTRA SLIDES

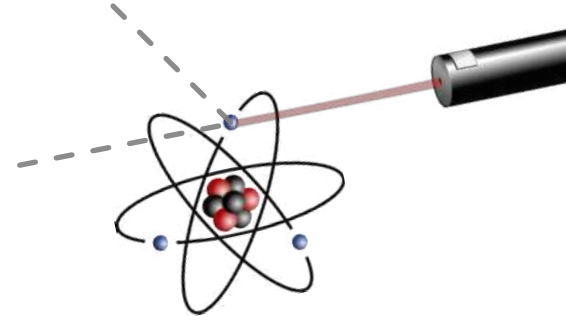
Quantum physics



discrete



random



$|0\rangle$ or $|1\rangle$??

Letter | Published: 03 June 2019

To catch and reverse a quantum jump mid-flight

Z. K. Minev , S. O. Mundhada, S. Shankar, P. Reinhold, R. Gutiérrez-Jáuregui, R. J. Schoelkopf, M. Mirrahimi, H. J. Carmichael & M. H. Devoret 

Nature **570**, 200–204 (2019) | [Download Citation](#) 



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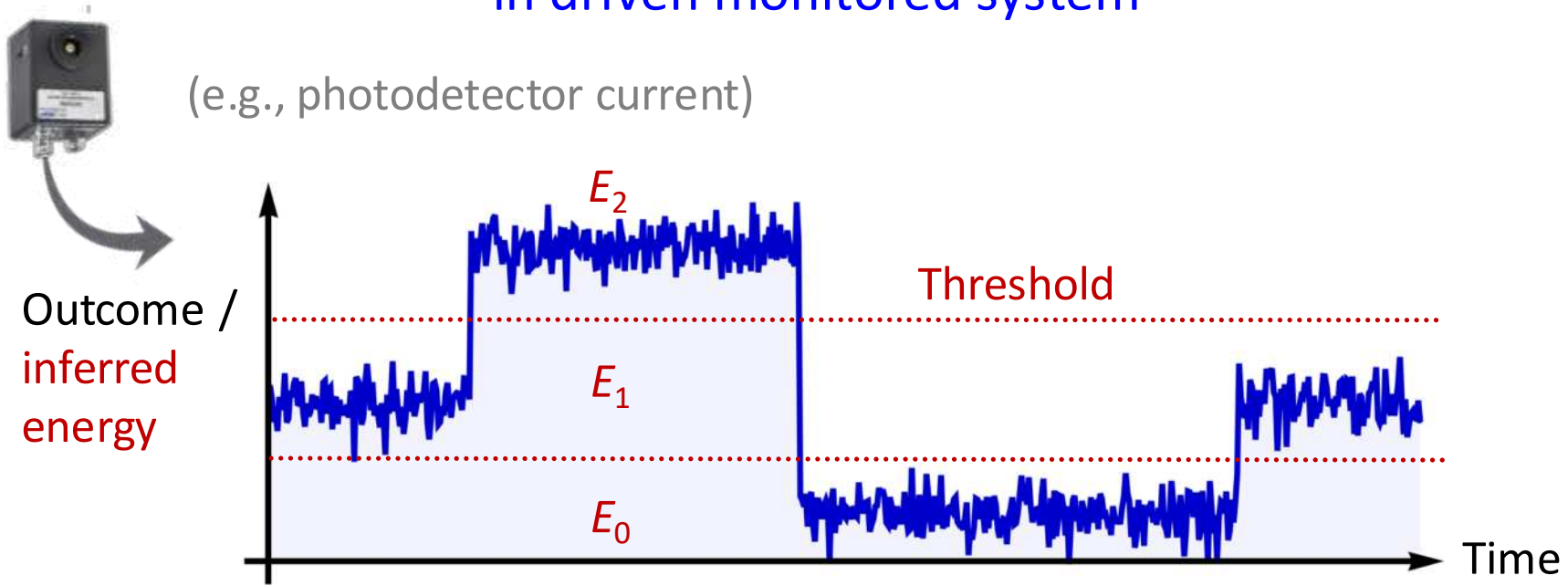
Популярная Механика

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Quanta magazine

Operational definition of quantum jumps in driven monitored system



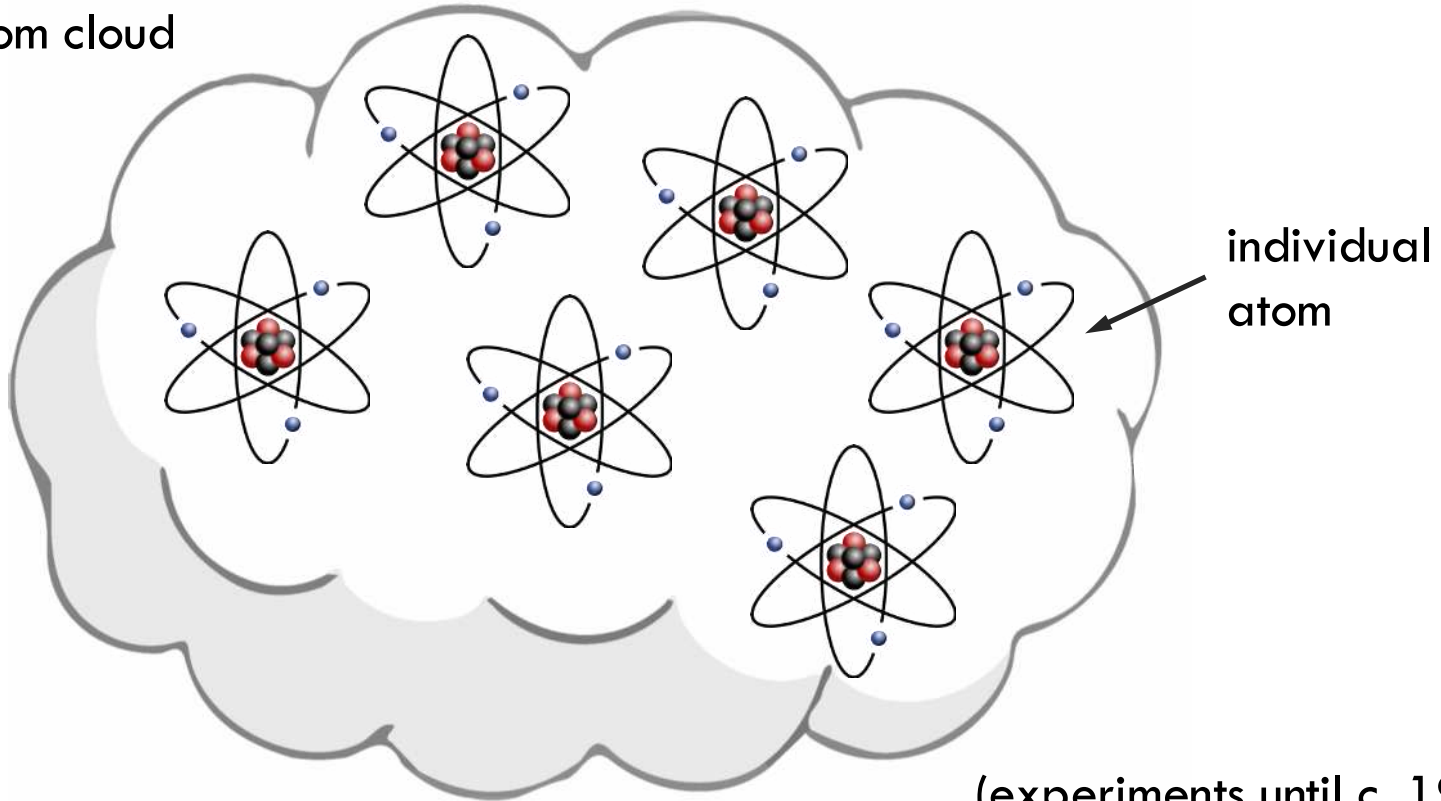
Measurement record is *continuous*: a time series of outcomes
(threshold is essential to declare when jump has occurred)

What is deterministic/predictable and what is not?

1. The onset of an individual jump evolution is not predictable; although it is knowable from the null-event readout signal.
2. The individual jump evolution is deterministic (predictable if known to have started) for as long as it continues; it follows exactly the same path in every occurrence.
3. Whether an individual jump evolution completes (continues all the way to state $|D\rangle$) is not predictable, which is why the passage quoted from our paper references the COMPLETED quantum jump. Although not predictable, noncompletion of a jump is knowable from the readout signal.

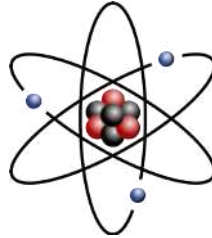
Ensemble of quantum systems

e.g., atom cloud



(experiments until c. 1980)

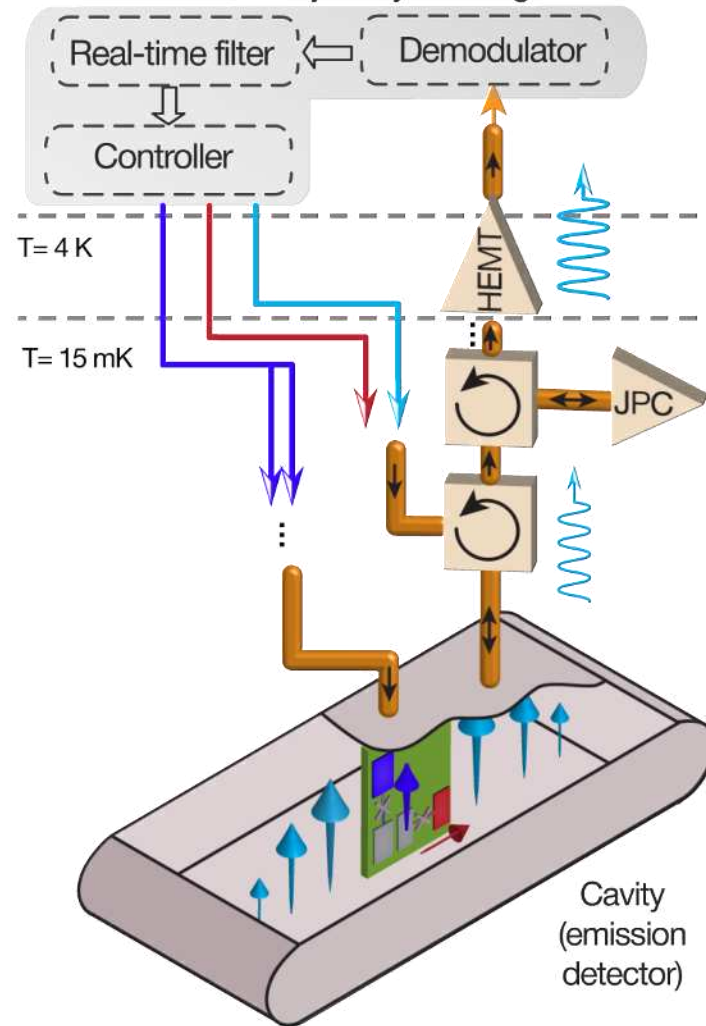
Individual quantum system



Can there be, despite the indeterminism of quantum physics,
a possibility to know
if a quantum jump is about to occur or not?

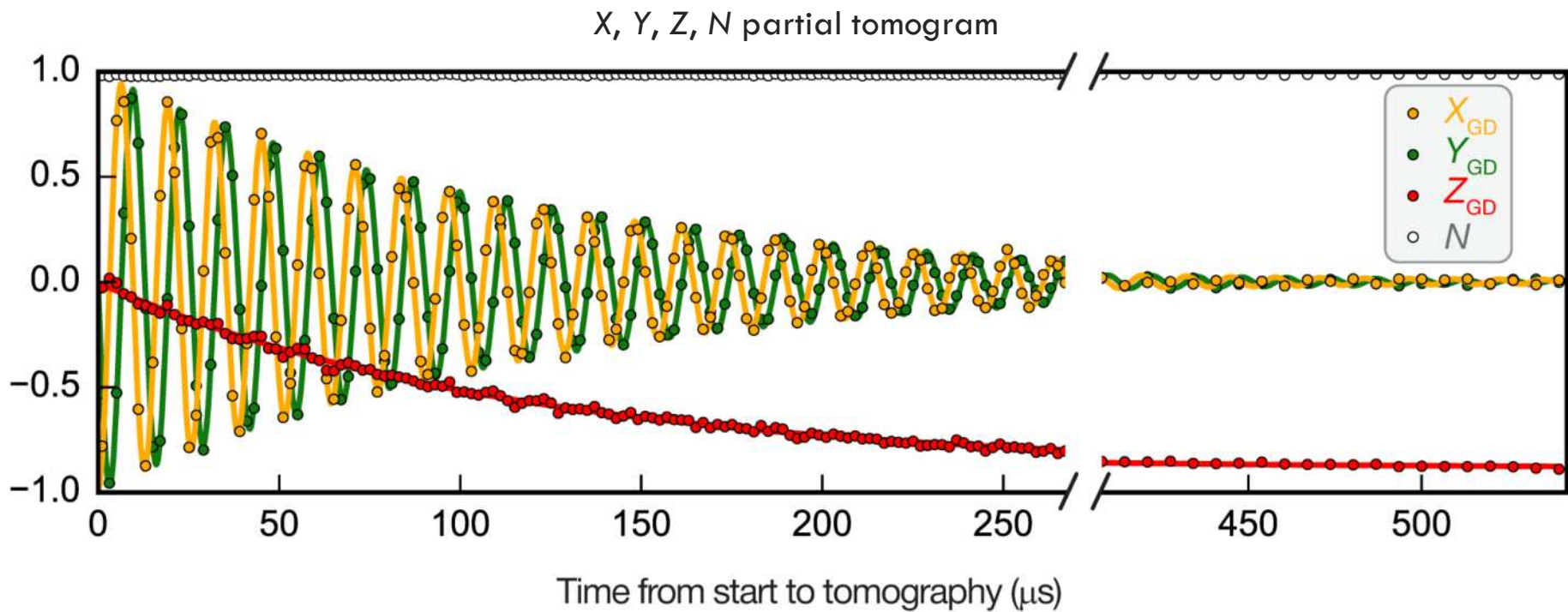
Schematic of the experiment

T= 300 K FPGA trajectory tracking and control



Control experiments

Coherences: free evolution

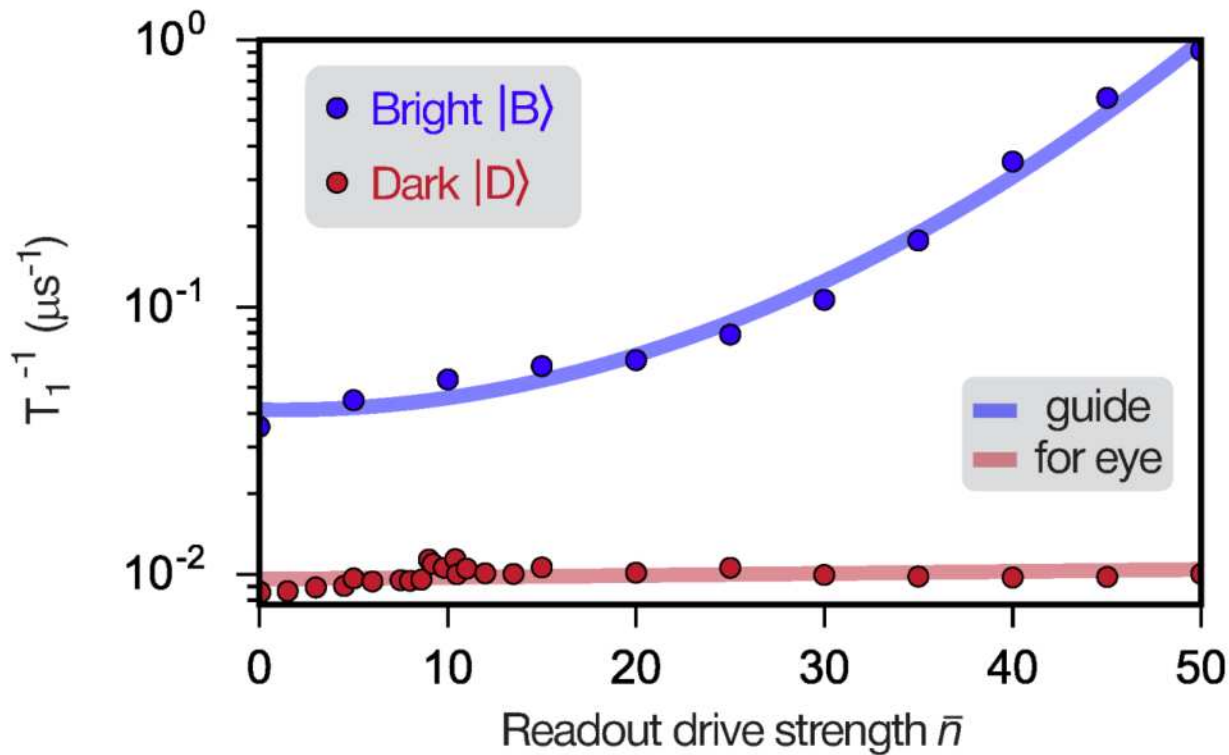
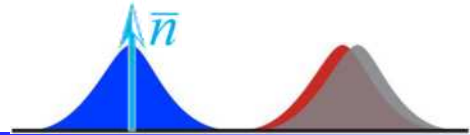


$$T_1^{DG} (\bar{n} = 0) = 116 \pm 5 \mu s$$

$$T_{2R}^{DG} (\bar{n} = 0) = 120 \pm 5 \mu s$$

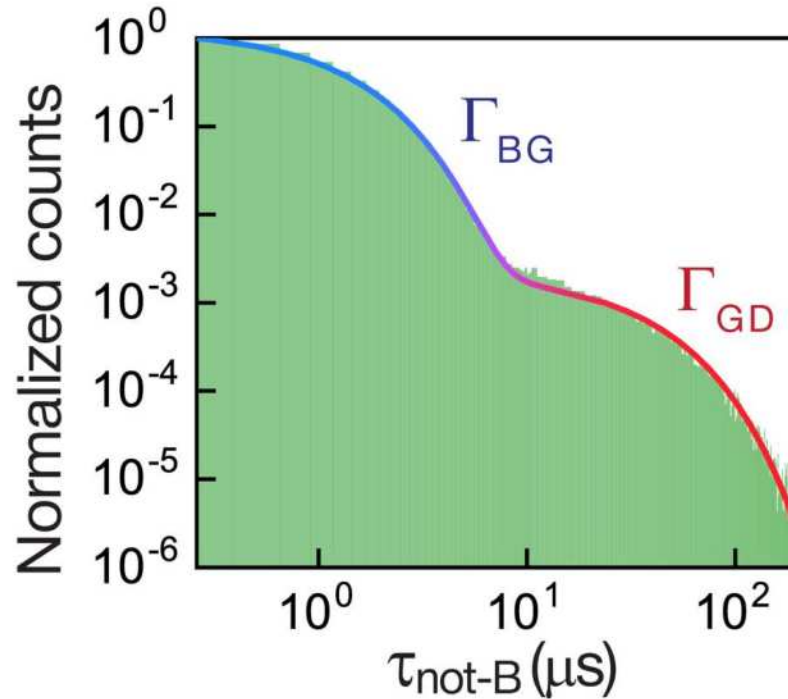
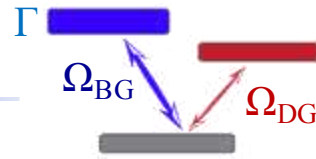
* Detuned generator

T_1 vs. readout strength



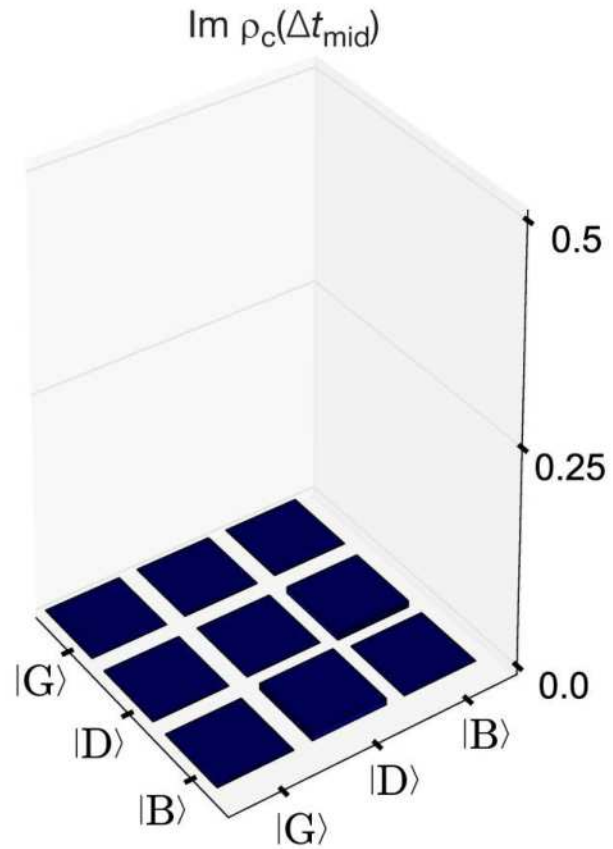
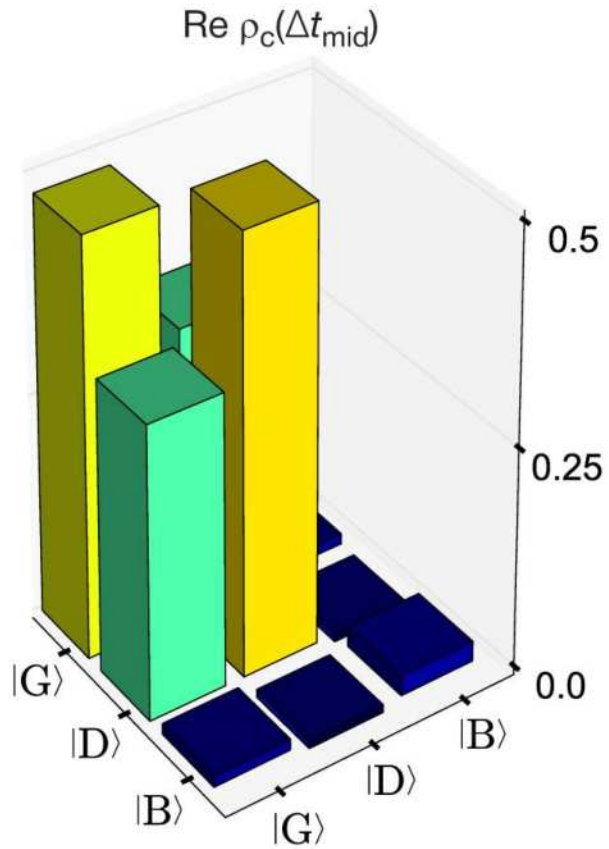
Unconditioned quantum jumps

Unconditioned quantum jumps



* Note the *log-log* scale

Mid-flight tomogram



Based on 4×10^6 catch events

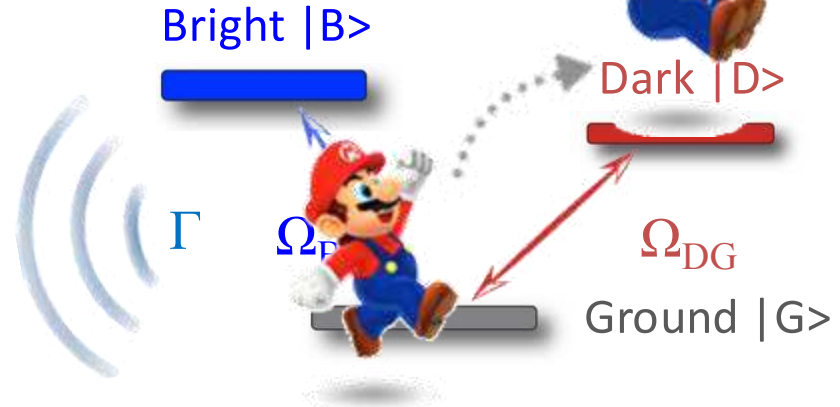
Original observation of quantum jumps

Photodetector

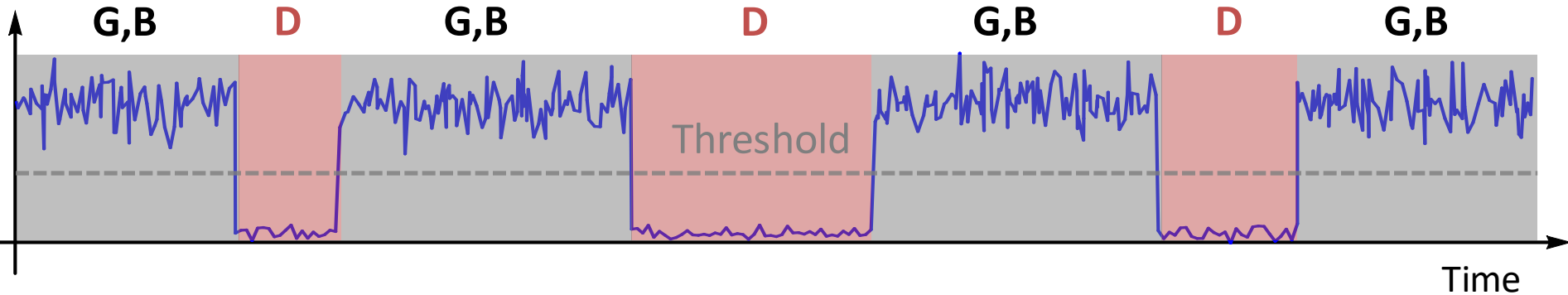
Collection efficiency $\eta \sim 10^{-3}$



$$\Gamma \gg \Omega_{BG} \gg \Omega_{DG}$$



Measurement record (fluorescence)



Berquist et al., PRL (1986)

Sauter et al., PRL (1986)

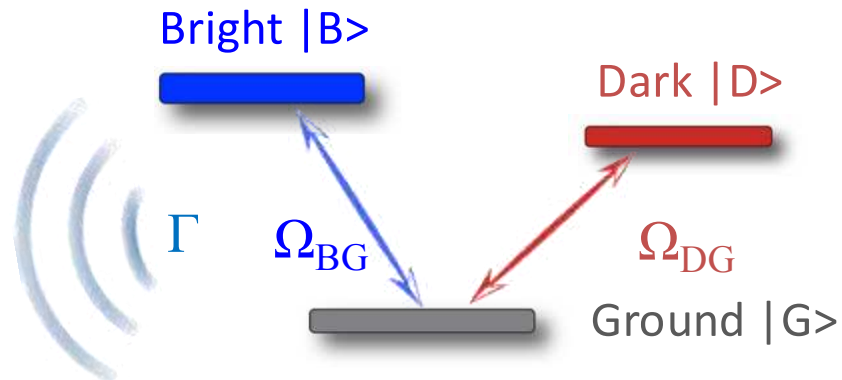
Nagourney et al., PRL (1986)

Cook, *What are quantum jumps?* Phys. Scripta (1988)

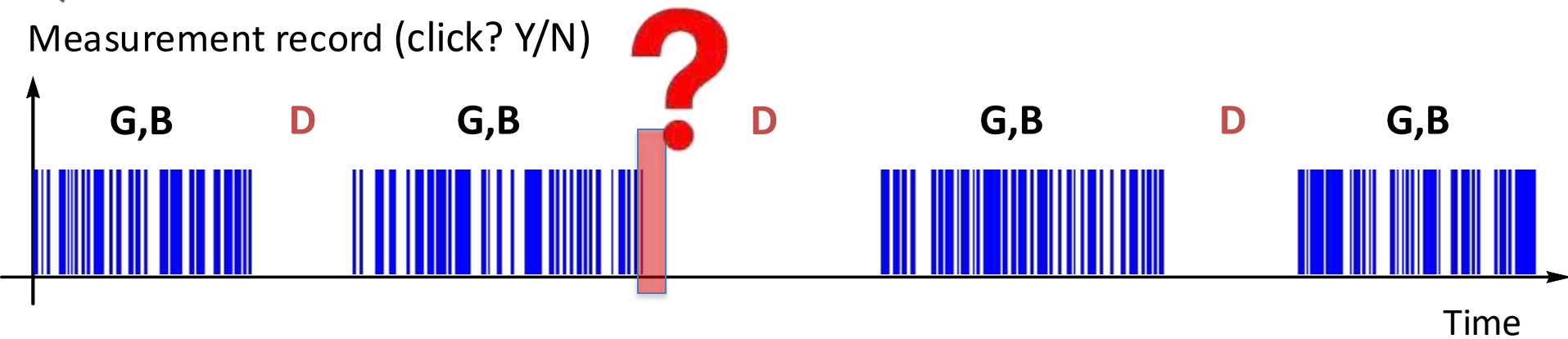
Omniscient observer

Hypothetical efficiency $\eta = 1$

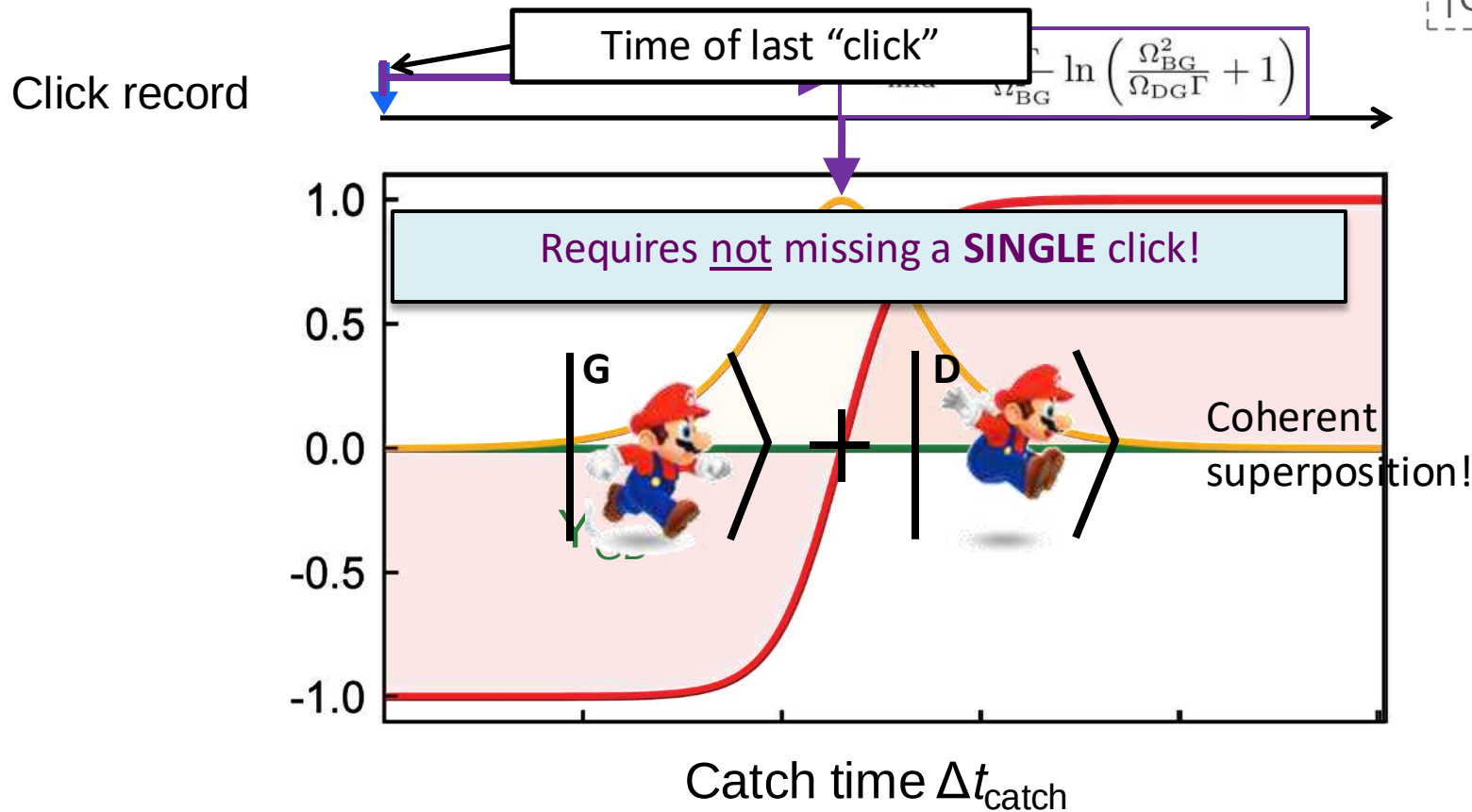
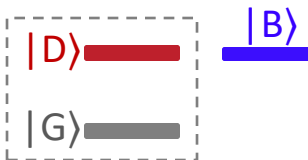
$$BW_{\text{det}} \gg \frac{\Omega_{\text{BG}}^2}{\Gamma} \quad \Gamma \gg \Omega_{\text{BG}} \gg \Omega_{\text{DG}}$$



Measurement record (click? Y/N)



Quantum jump from $|G\rangle$ to $|D\rangle$: prediction

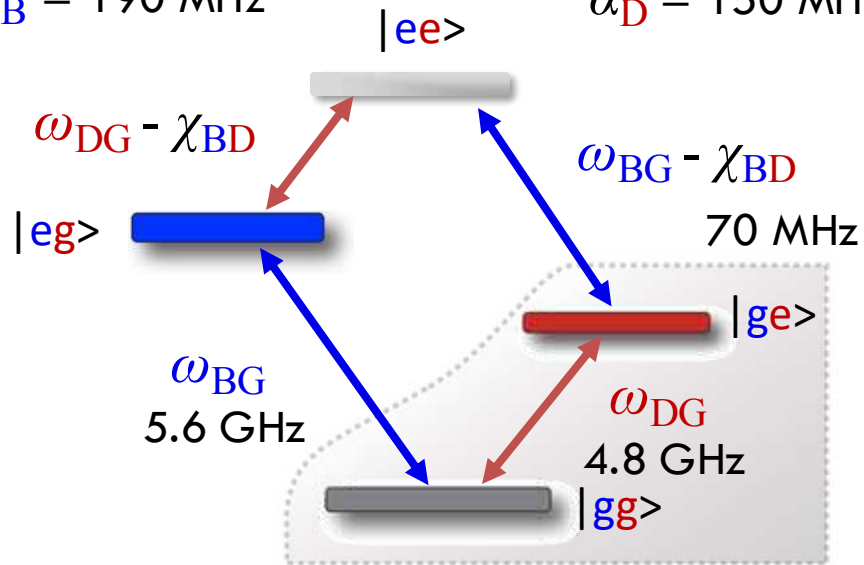


cQED implementation of V-system

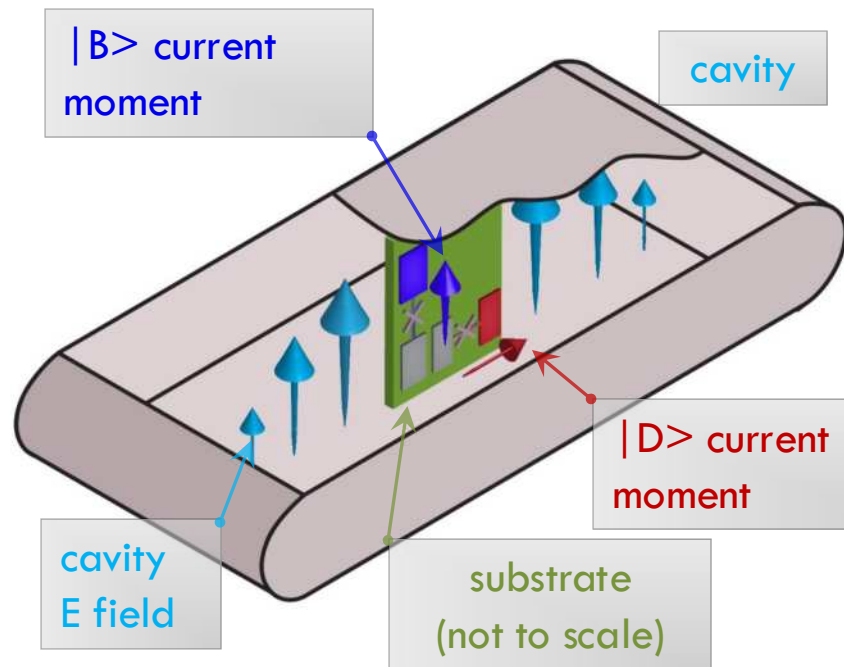
$$\hat{H} = \omega_{\text{DG}} \hat{\sigma}_Z \otimes \hat{I} + \omega_{\text{BG}} \hat{I} \otimes \hat{\sigma}_Z - \chi_{\text{BD}} \hat{\sigma}_Z \otimes \hat{\sigma}_Z$$

$$\alpha_{\text{B}} = 190 \text{ MHz}$$

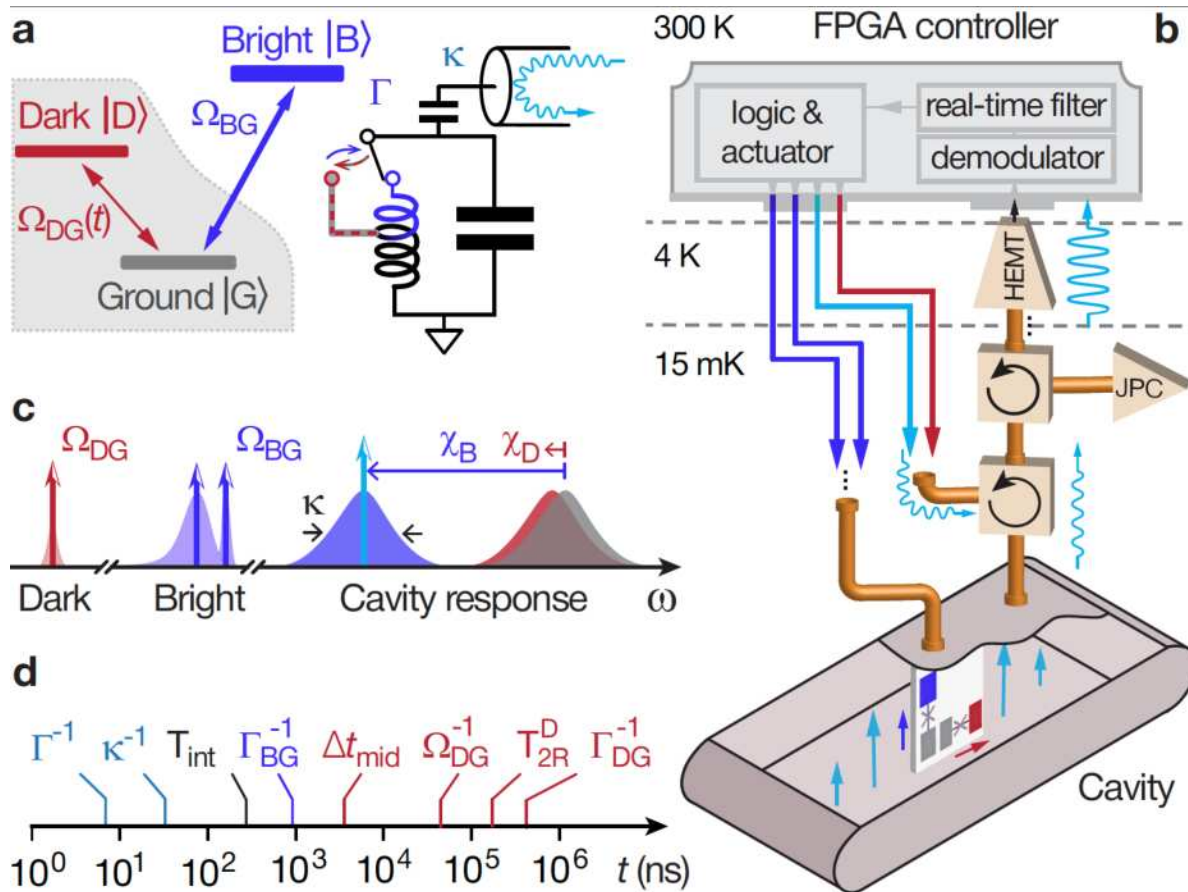
$$\alpha_{\text{D}} = 150 \text{ MHz}$$



Schematic representation of design



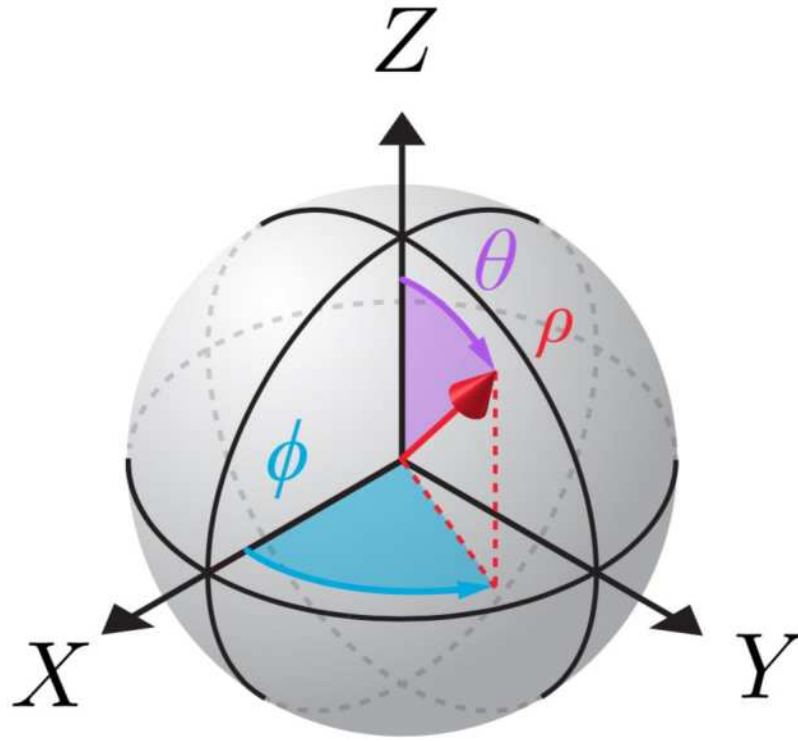
Setup of the experiment



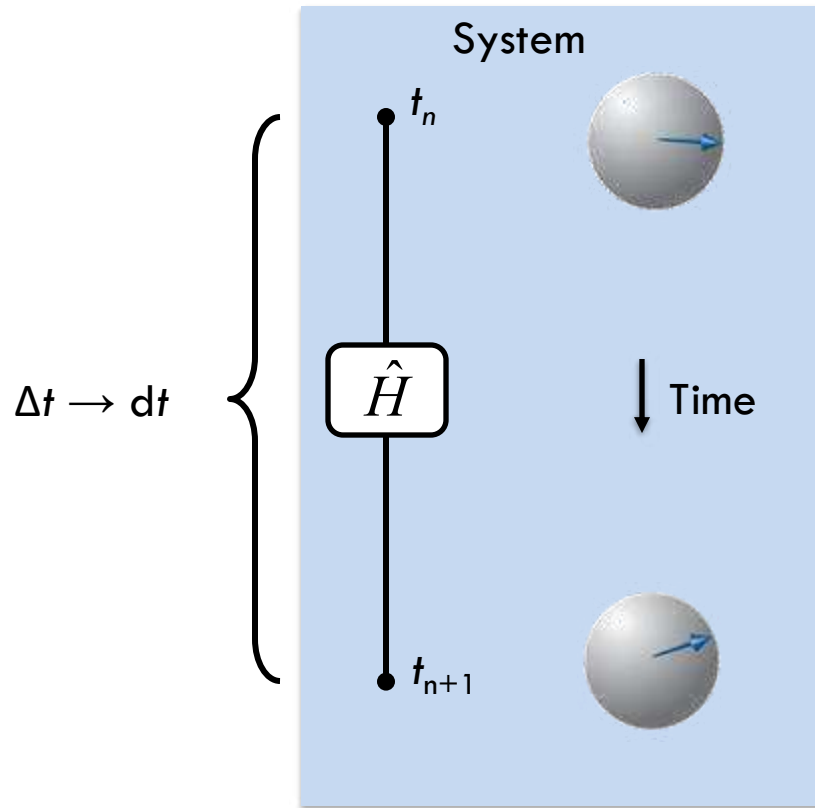
A primer on trajectory theory

Qubit evolution: finite-difference model

(theoretical-minimum example)

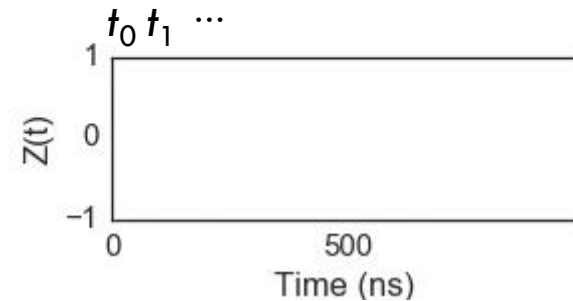
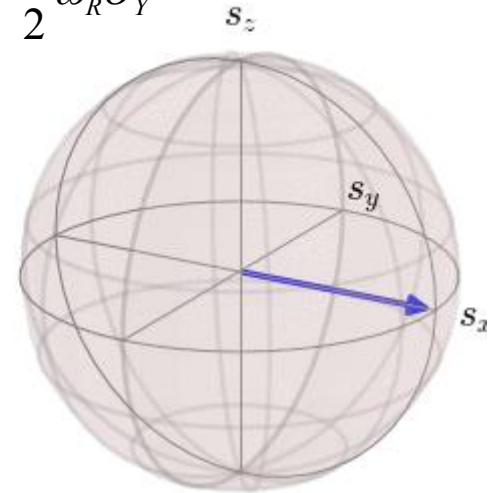


Isolated system

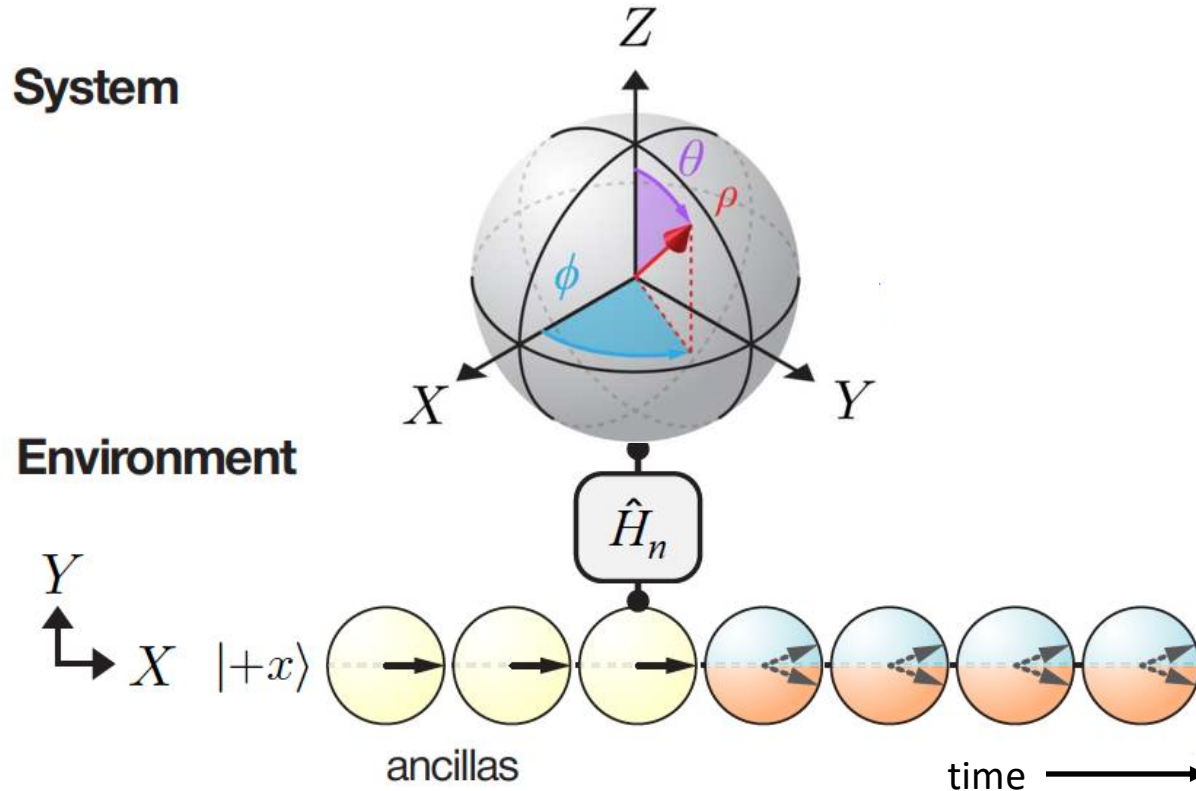


Example

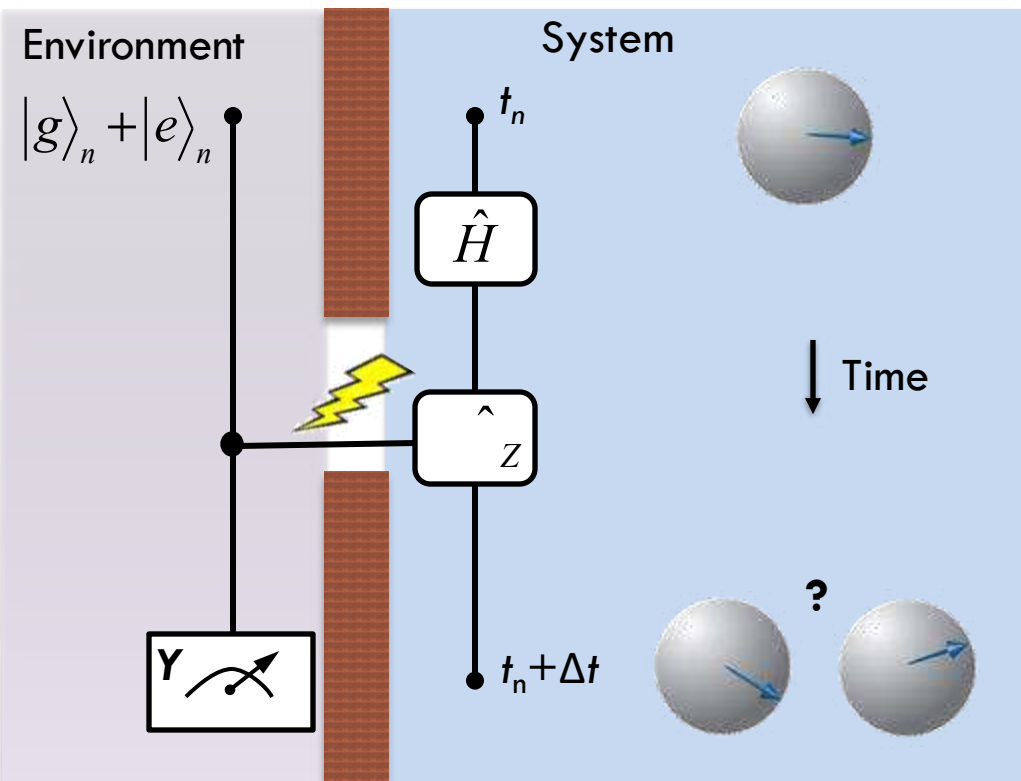
$$\hat{H} = -\frac{1}{2} \omega_R \hat{\sigma}_Y$$



Model of an open quantum system



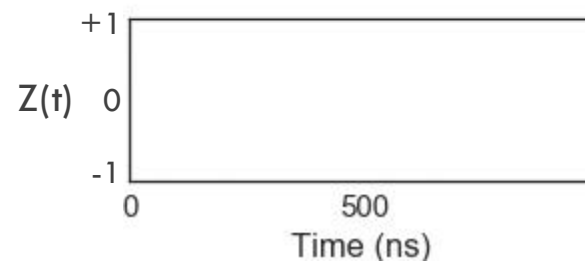
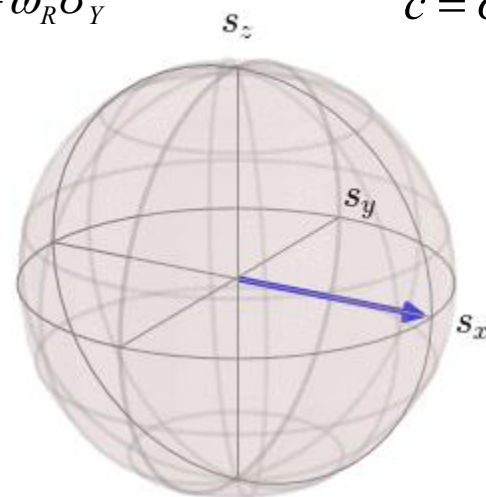
Open system



Example

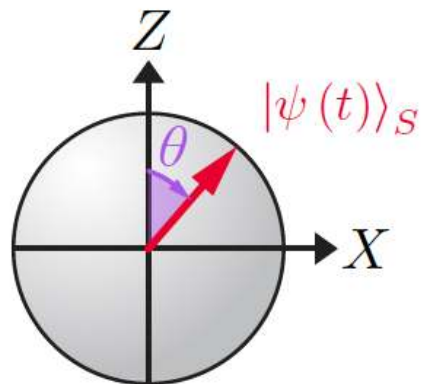
$$\hat{H} = -\frac{1}{2} \omega_R \hat{\sigma}_Y$$

$$\hat{c} = \hat{\sigma}_Z$$

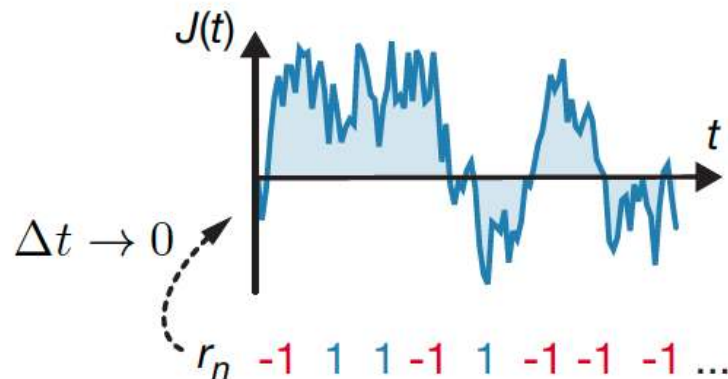


Model of a monitored quantum system

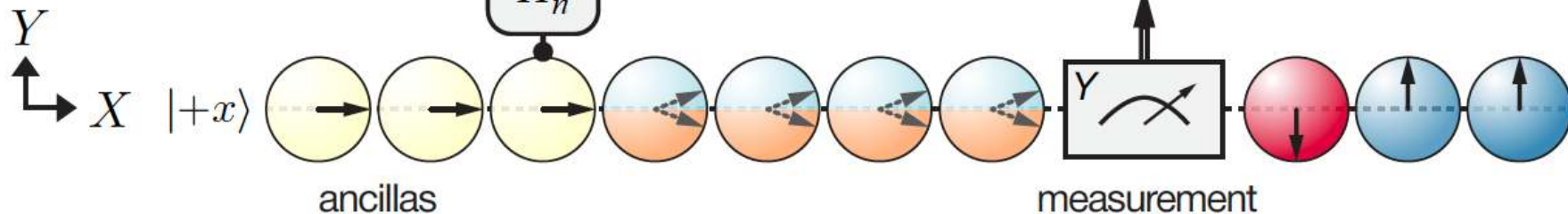
System



Measurement record

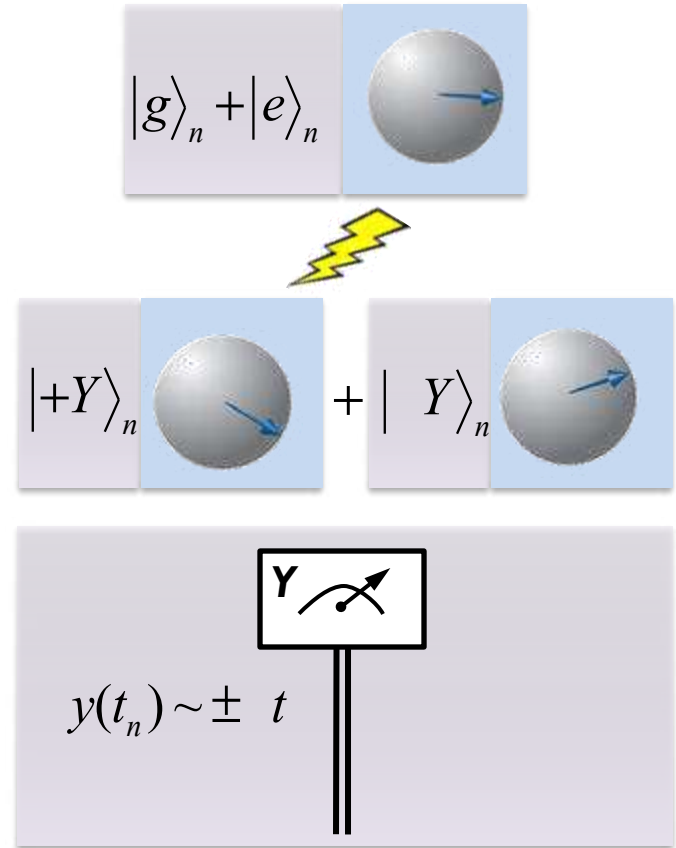
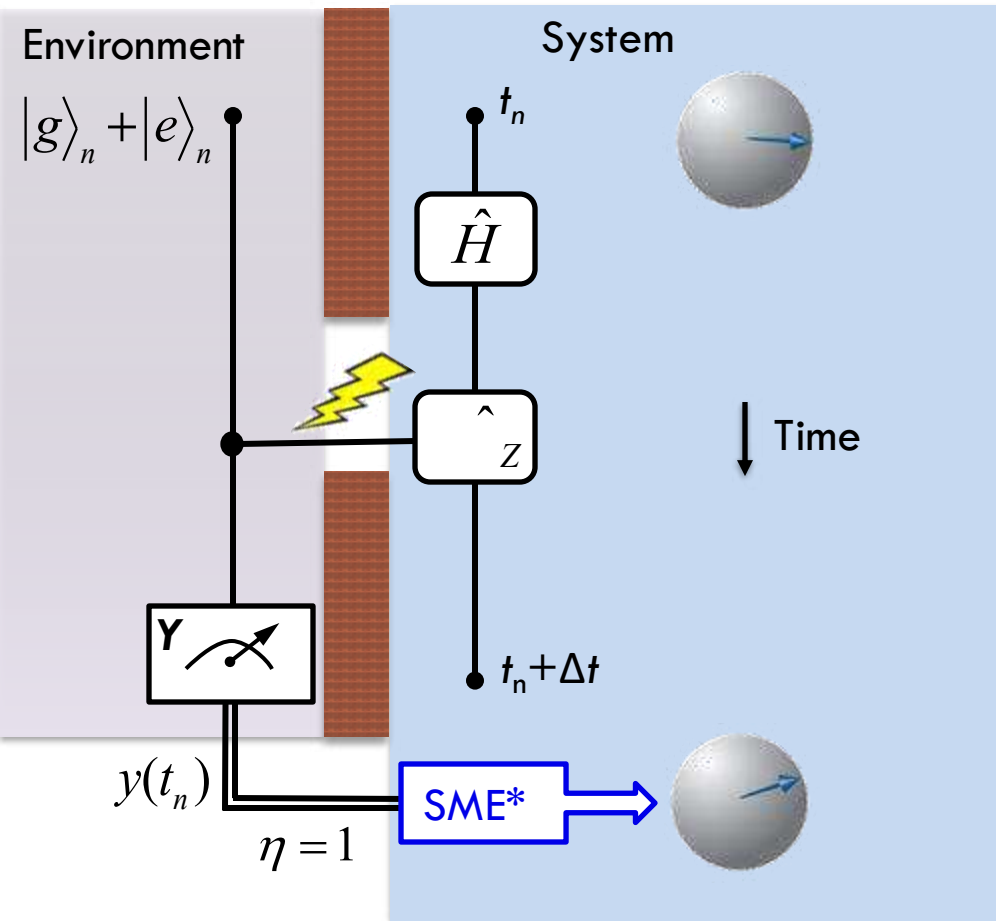


Environment



time $\longrightarrow$

System + omniscient observer



* Carmichael, H. *An Open Systems Approach to Quantum Optics* (Springer-Verlag, 1993)

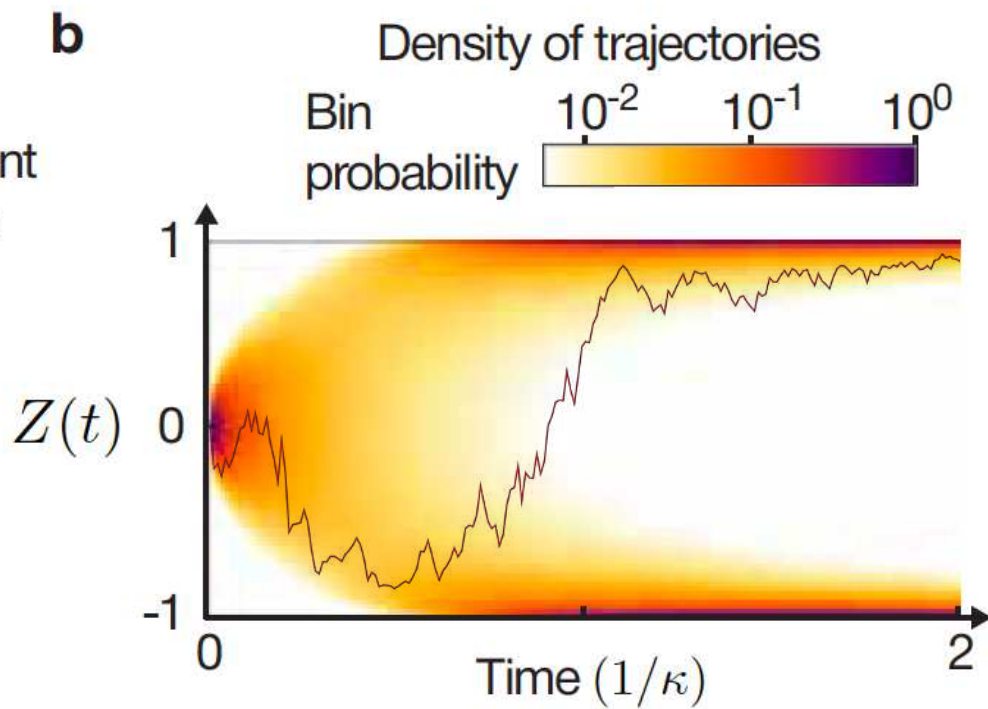
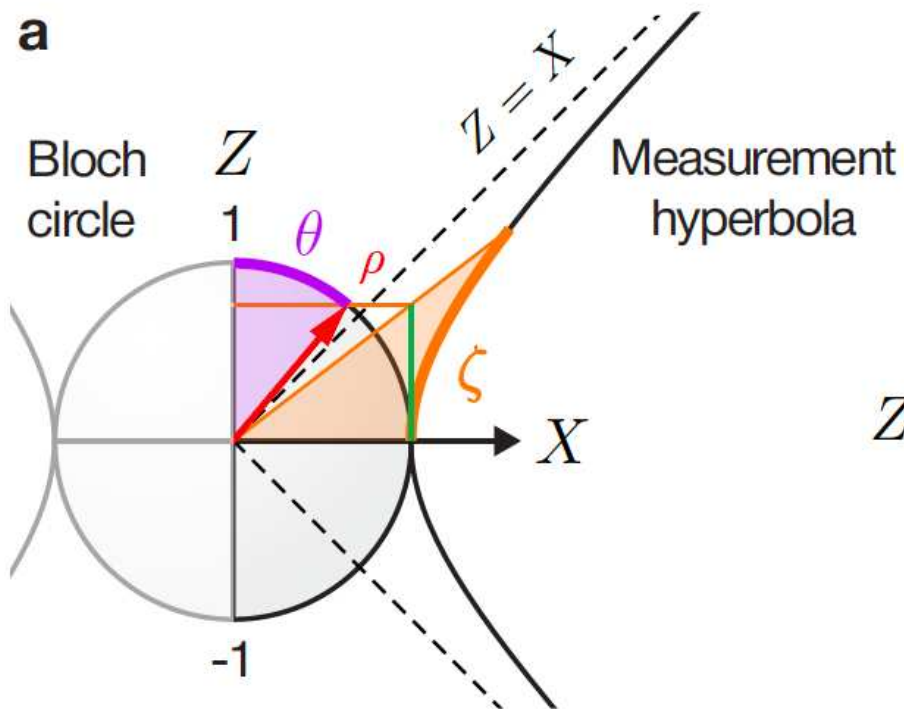
The diagram illustrates a quantum system and its interaction with an environment. The **Environment** (left, purple background) starts with a state $|g\rangle_n + |e\rangle_n$. It interacts with the **System** (right, blue background) via a coupling (yellow lightning bolt). The system evolves under a Hamiltonian $\hat{H}$ from time t_n to $t_n + \Delta t$. A measurement $\hat{Z}$ is performed on the system, and the resulting signal $y(t_n)$ is fed back into the environment through a block labeled $\eta = 1$ and SME^* . A vertical arrow labeled **Time** indicates the progression from the top to the bottom of the diagram.

$\hat{H} = -\frac{1}{2}\omega_R\hat{\sigma}_Y$
 $\hat{c} = \hat{\sigma}_Z$
 homodyne

The figure shows a Bloch sphere with axes labeled s_x , s_y , and s_z . A blue arrow points from the center of the sphere along the s_x axis. Below the sphere is a plot of $Z(t)$ versus Time (ns). The plot shows a constant value of 0 over the time range from 0 to 1000 ns.

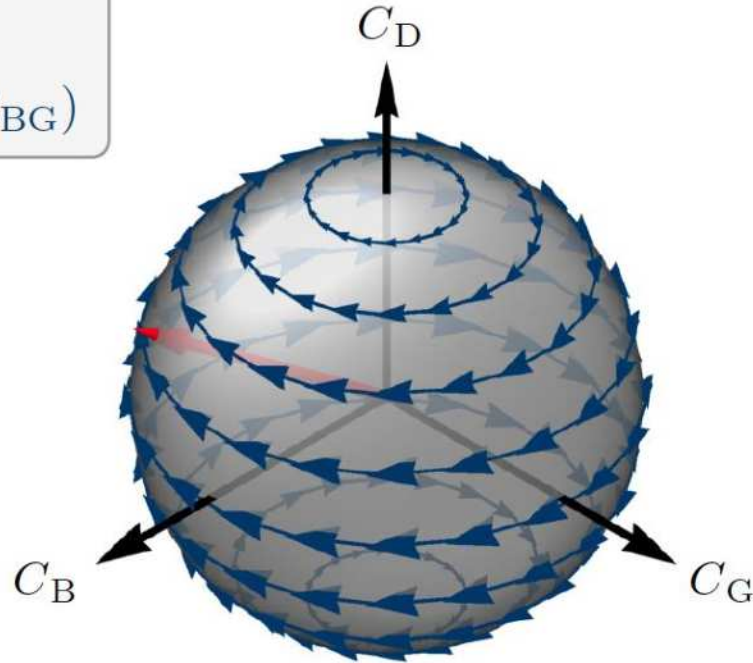
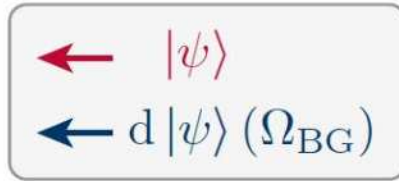
* Carmichael, H. *An Open Systems Approach to Quantum Optics* (Springer-Verlag, 1993)

Continuous quantum measurement

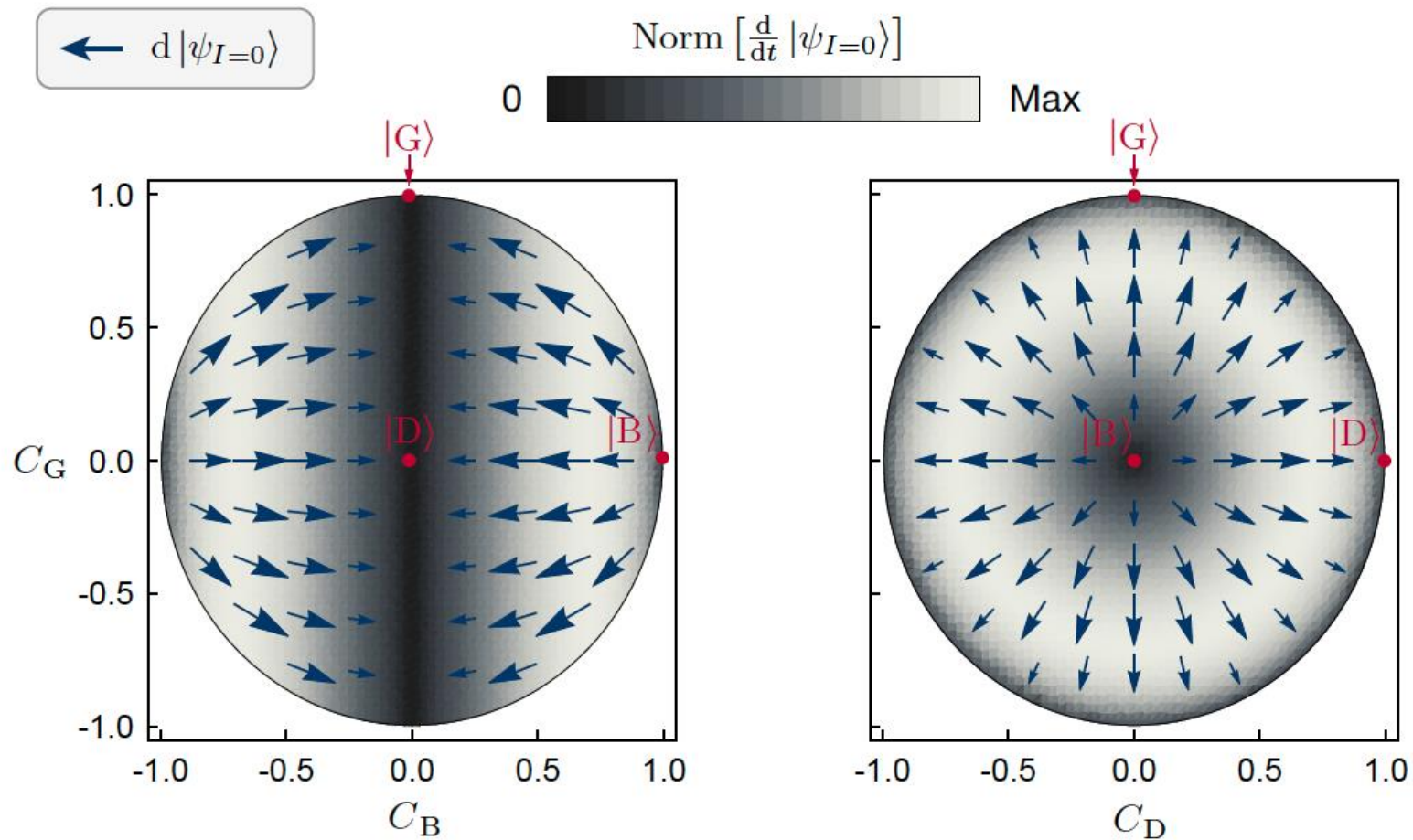


Geometrical representation

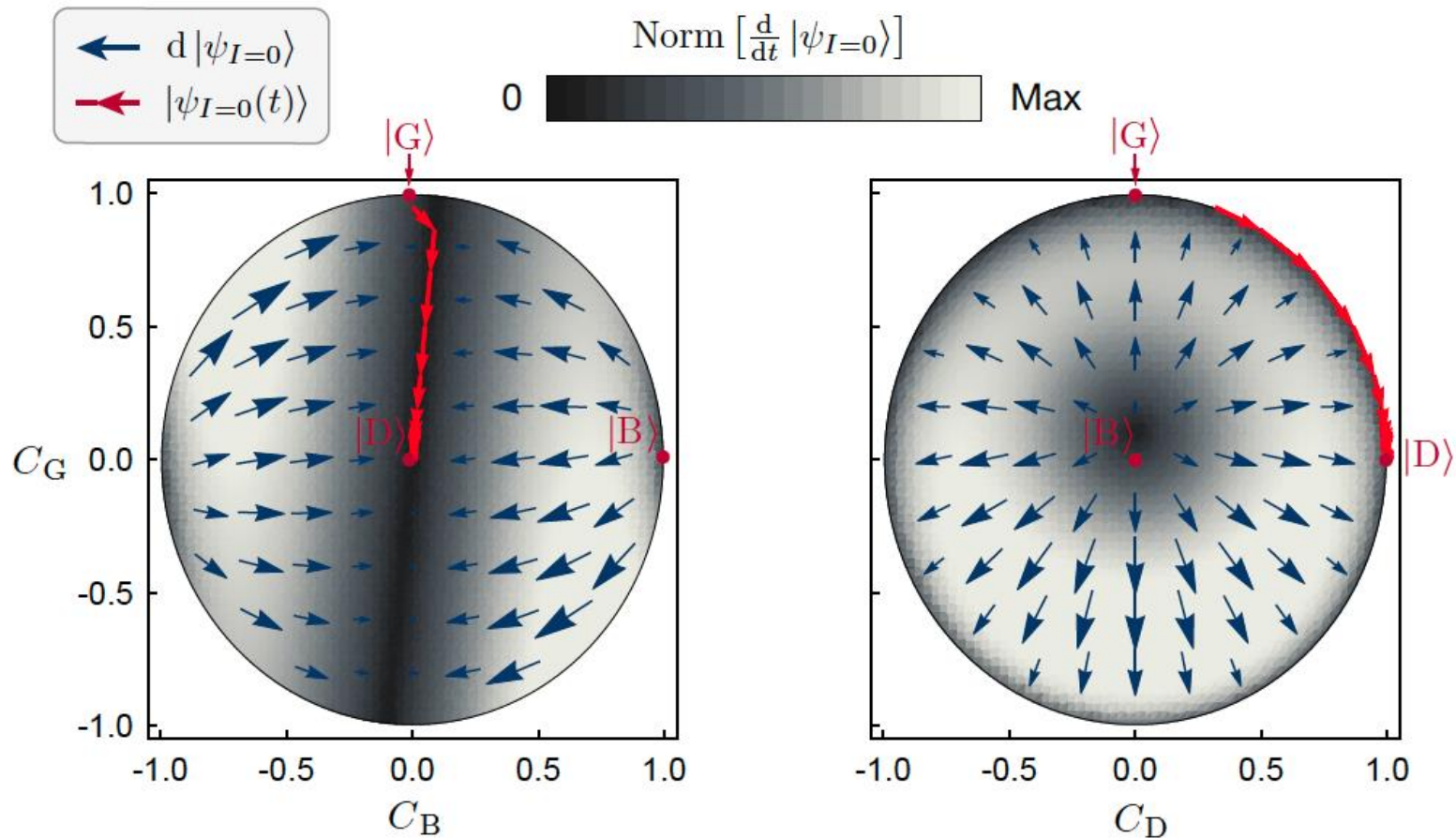
$$|\psi\rangle = C_B |B\rangle + C_G |G\rangle + C_D |D\rangle$$



Geometrical representation: B/not-B

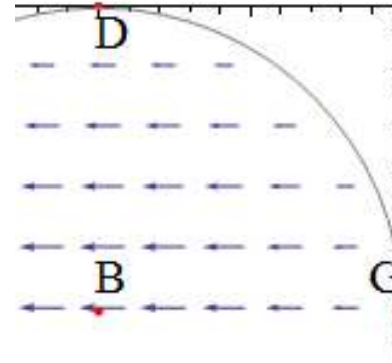
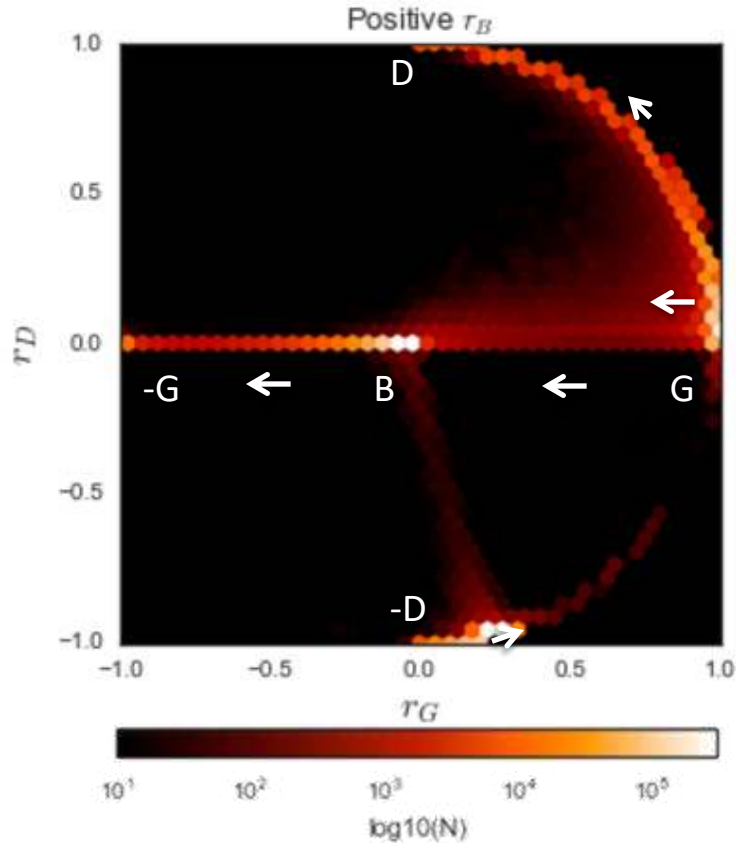


Geometrical representation: B/not-B and Ω_{BG}

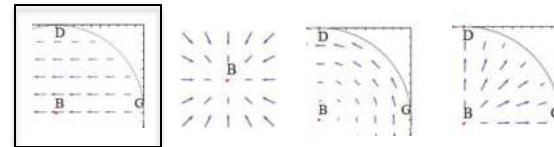


Forces of wind: controller vs. observer

Hilbert space

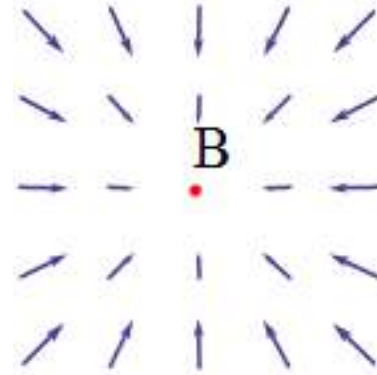
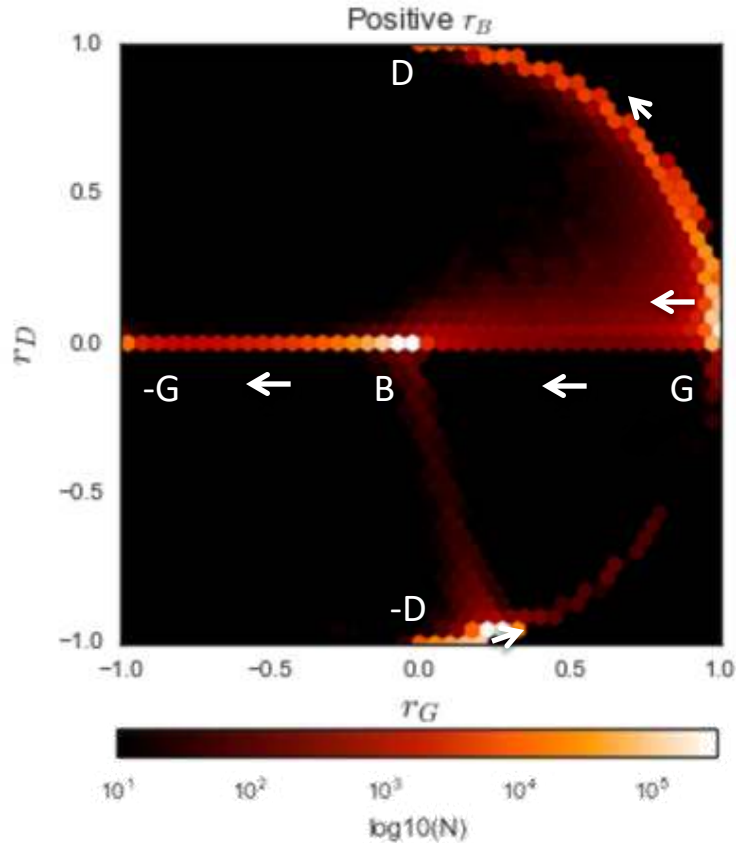


Ω_B

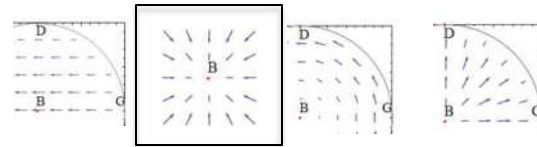


$$\Gamma_m^{-1} = 0.02 \quad 2\pi \Omega_B^{-1} = 1.5 \quad 2\pi \Omega_D^{-1} = 60 \text{ (}\mu\text{s)}$$

Forces of wind: controller vs. observer

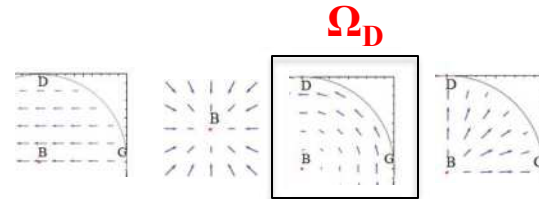
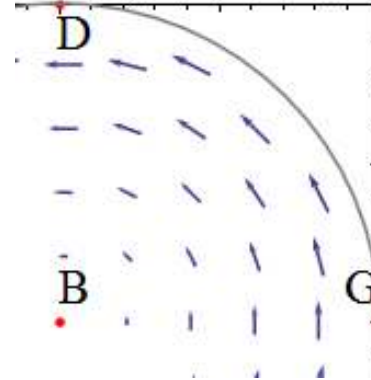
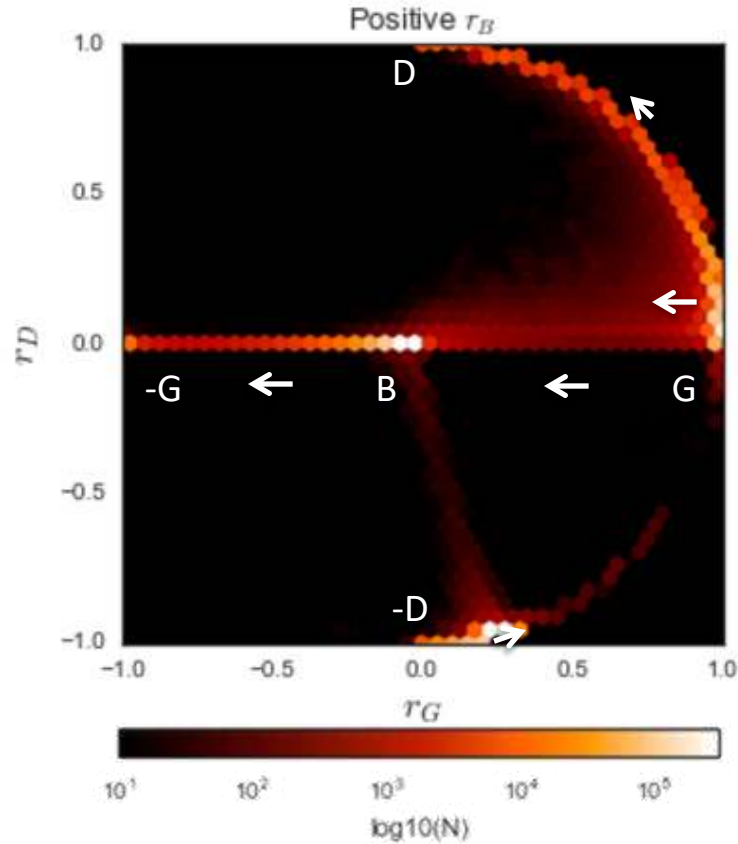


$$\Gamma_m : dy \sim B$$



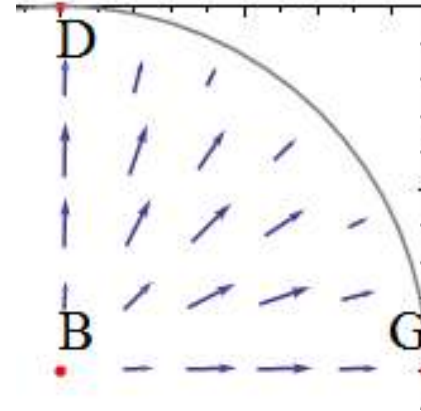
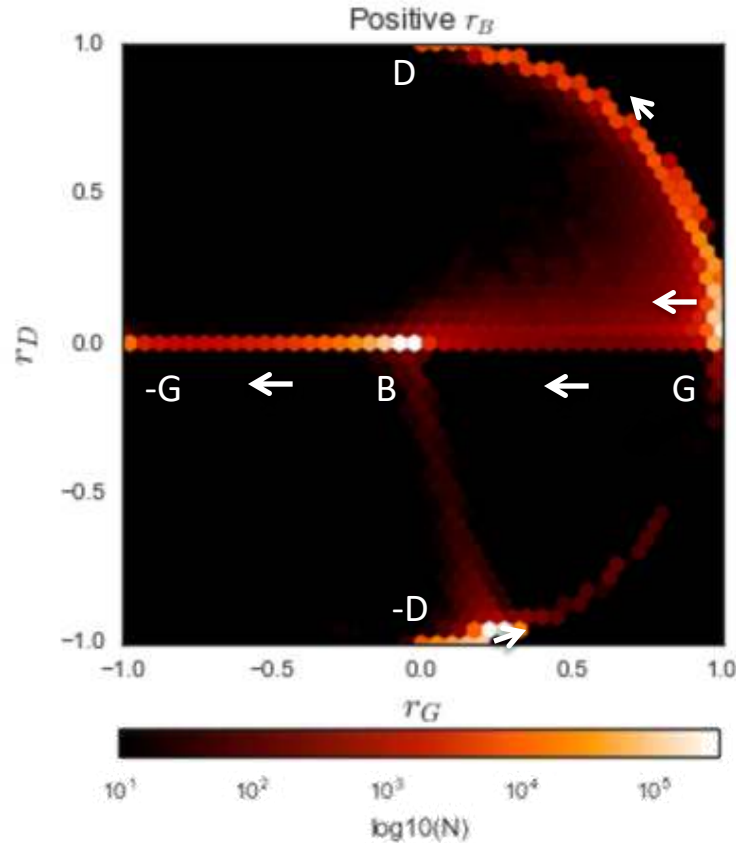
$$\Gamma_m^{-1} = 0.02 \quad 2\pi \Omega_B^{-1} = 1.5 \quad 2\pi \Omega_D^{-1} = 60 \text{ (}\mu\text{s)}$$

Forces of wind: controller vs. observer

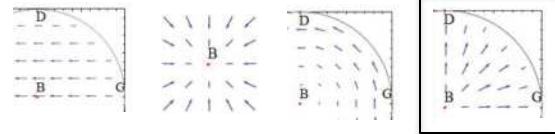


$$\Gamma_m^{-1} = 0.02 \quad 2\pi \Omega_B^{-1} = 1.5 \quad 2\pi \Omega_D^{-1} = 60 \text{ (}\mu\text{s)}$$

Forces of wind: controller vs. observer

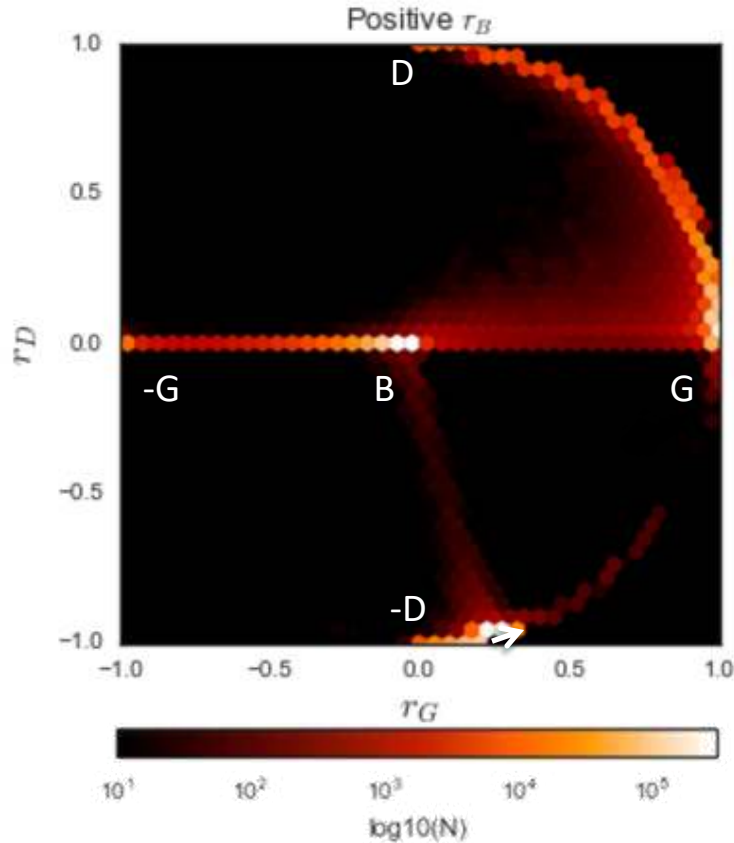


$\Gamma_m : dy \sim \text{not } B$

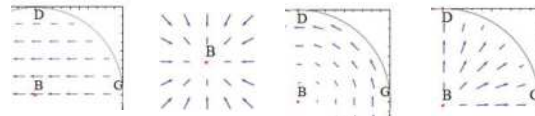


$$\Gamma_m^{-1} = 0.02 \quad 2\pi \Omega_B^{-1} = 1.5 \quad 2\pi \Omega_D^{-1} = 60 \text{ } (\mu\text{s})$$

Forces of wind: controller vs. observer



$\uparrow$ measurement
 $\Rightarrow \uparrow$ PREDICTABLE dynamics!

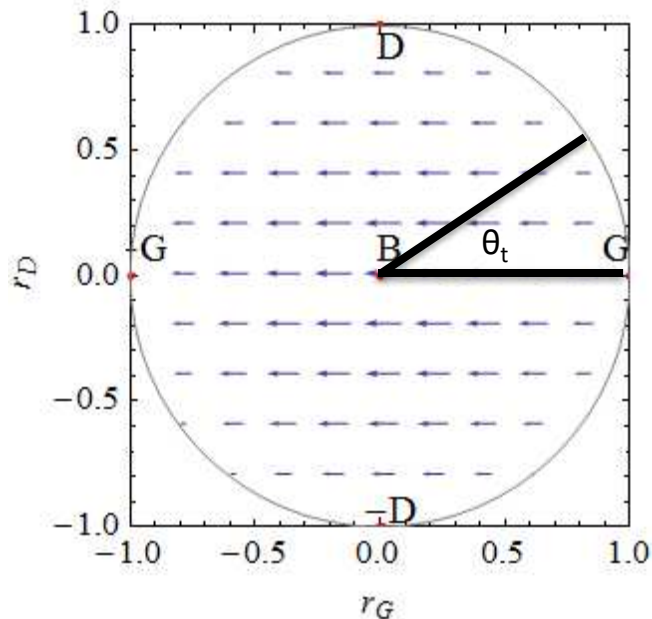


$$\Gamma_m^{-1} = 0.02 \quad 2\pi \Omega_B^{-1} = 1.5 \quad 2\pi \Omega_D^{-1} = 60 \text{ } (\mu\text{s})$$

R-utrit: sphere representation

Representation

$$|\psi(t)\rangle = \begin{pmatrix} r_b(t) \\ r_g(t) \\ r_d(t) \end{pmatrix} = \begin{pmatrix} \sin(\theta_t) \sin(\phi_t) \\ \sin(\theta_t) \cos(\phi_t) \\ \cos(\theta_t) \end{pmatrix}$$



Evolution under conditioned linear SSE

$$|\tilde{\psi}(t+dt)\rangle = e^{-\frac{1}{2}\hat{M}^\dagger \hat{M} dt} e^{-i\hat{H} dt} |\tilde{\psi}(t)\rangle$$

$$\frac{d}{dt}\phi(t) = \frac{\Omega}{2} - \frac{1}{4}\Gamma \sin(2\phi(t))$$

$$\frac{d}{dt}\theta(t) = -\frac{1}{4}\Gamma \sin^2(\phi(t)) \sin(2\theta(t))$$

$$P_D(t) = \cos^2(\theta(t))$$

$$= \left[\frac{1}{2} \left(1 + \tanh \left[\frac{\Omega^2}{2\Gamma} t + t_0 \right] \right) \right]$$

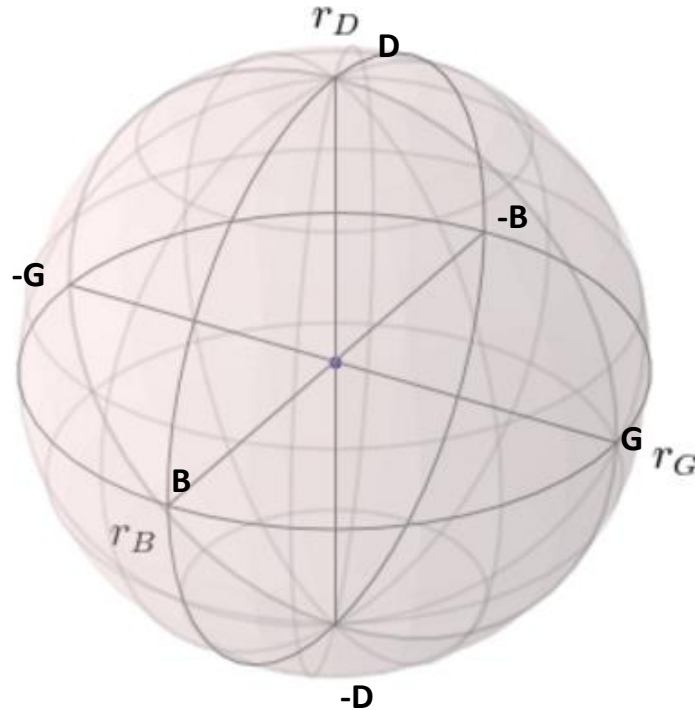
$\mathbb{R}$ -qutrit: sphere representation

$$|\psi(t)\rangle = r_G |G\rangle + r_B |B\rangle + r_D |D\rangle$$

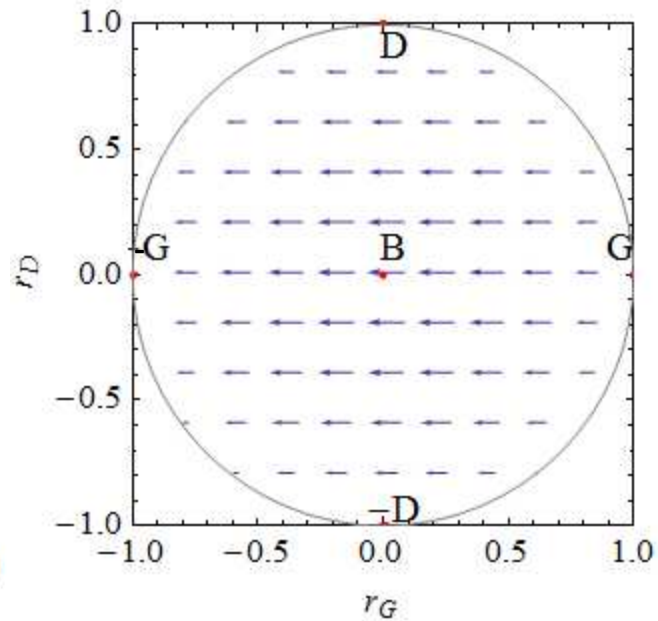
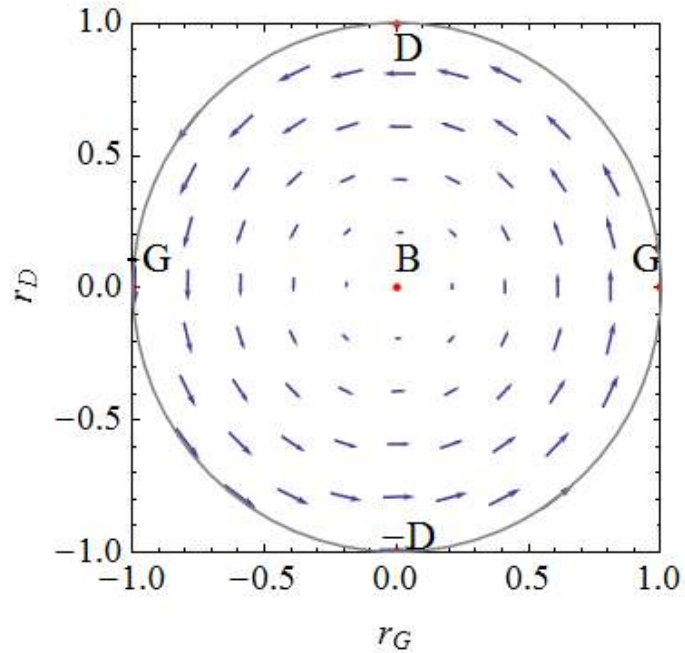
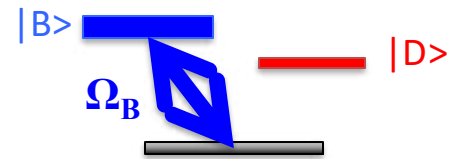
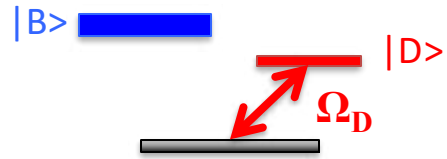
$$r_G^2 + r_B^2 + r_D^2 = 1$$

$$H \sim \mathfrak{so}(3)$$

$$U \sim \mathrm{SO}(3)$$

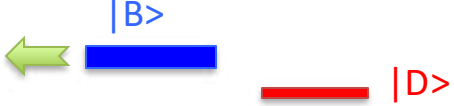


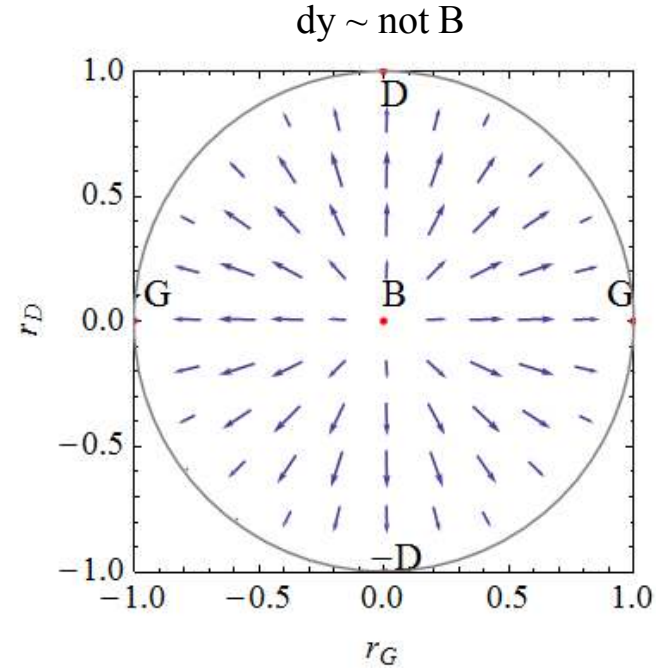
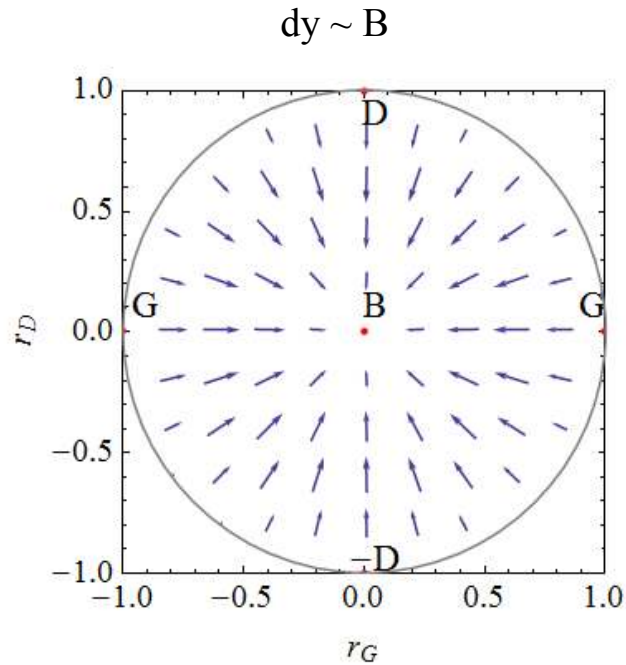
$d\rho$: Deterministic drives



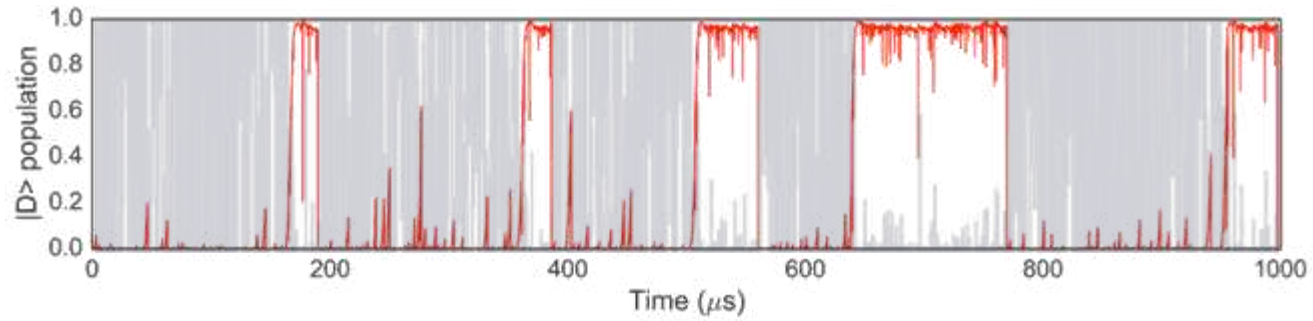
$d\rho$: Measurement – stochastic field

$$J_m = \Gamma_m^{1/2} |B\rangle\langle B|$$



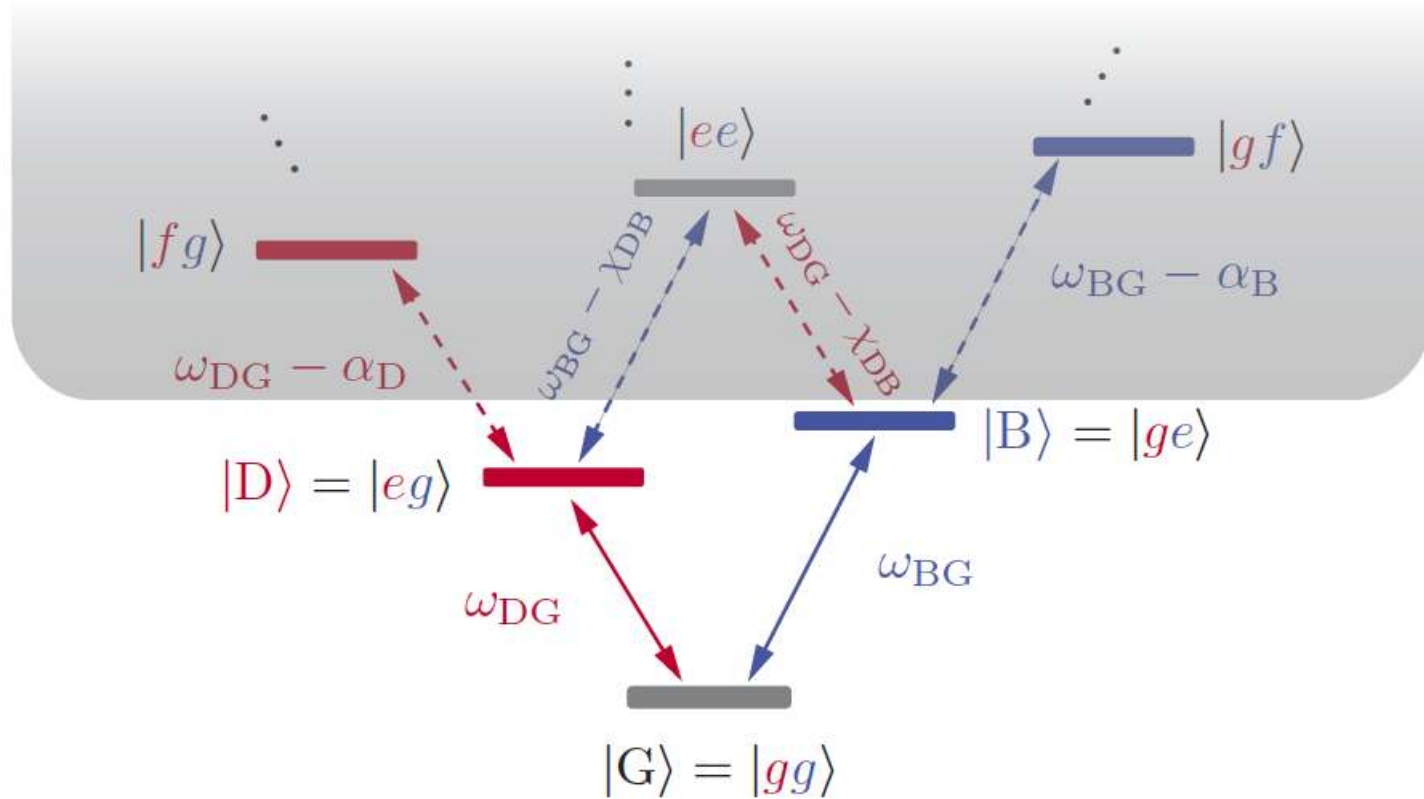


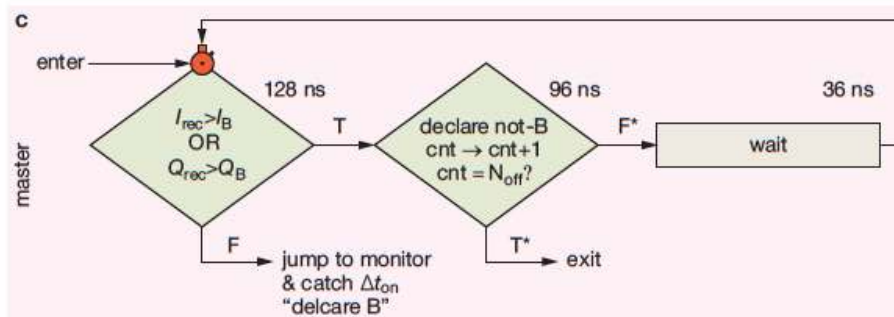
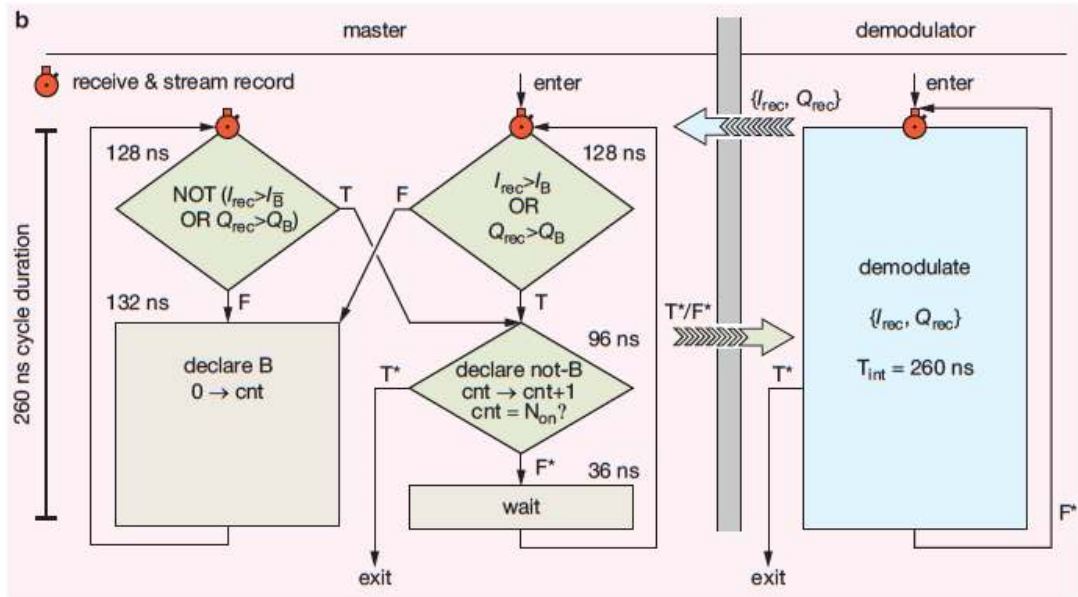
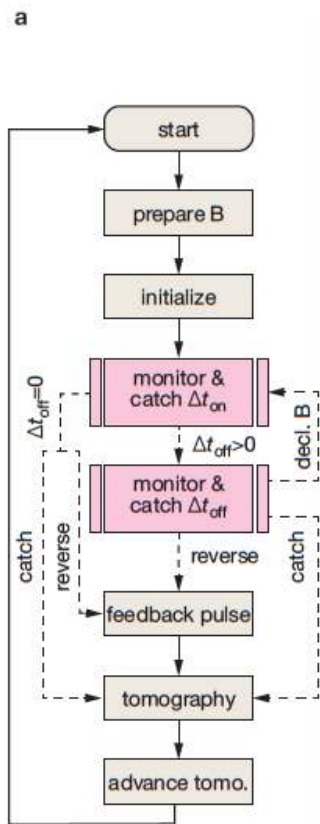
Histogram of dynamics

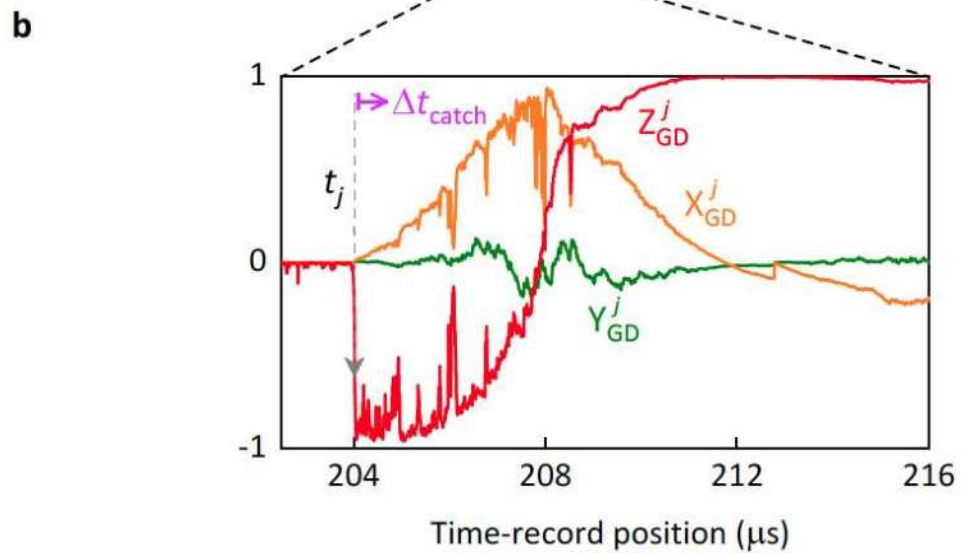
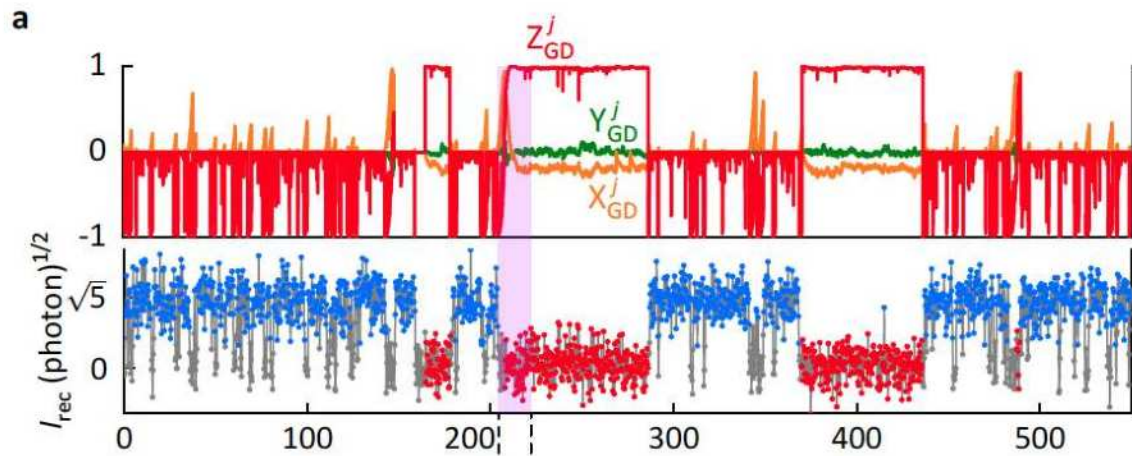


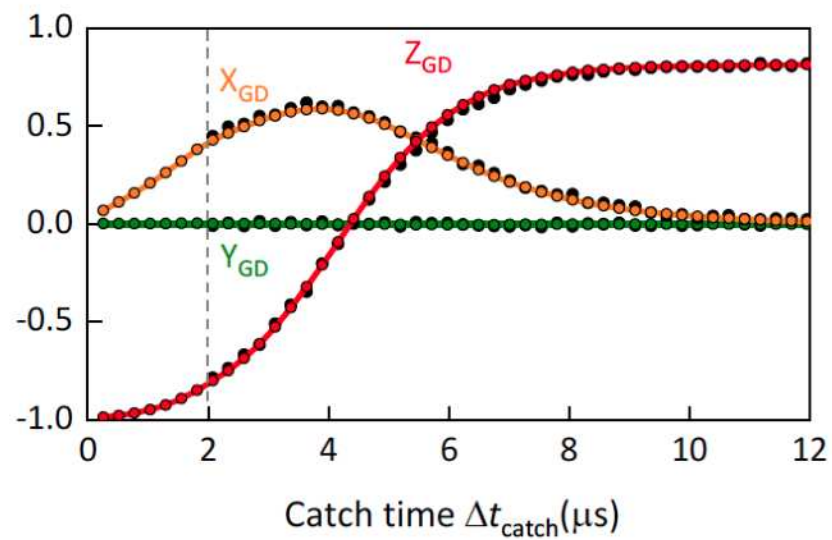
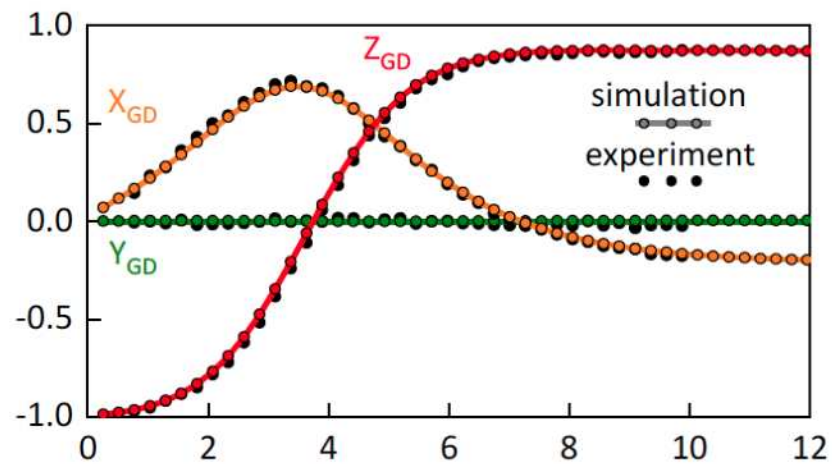
$$\Gamma_m^{-1} = 0.02 \mu\text{s} \quad 2\pi \Omega_B^{-1} = 1 \mu\text{s} \quad 2\pi \Omega_D^{-1} = 60 \mu\text{s}$$

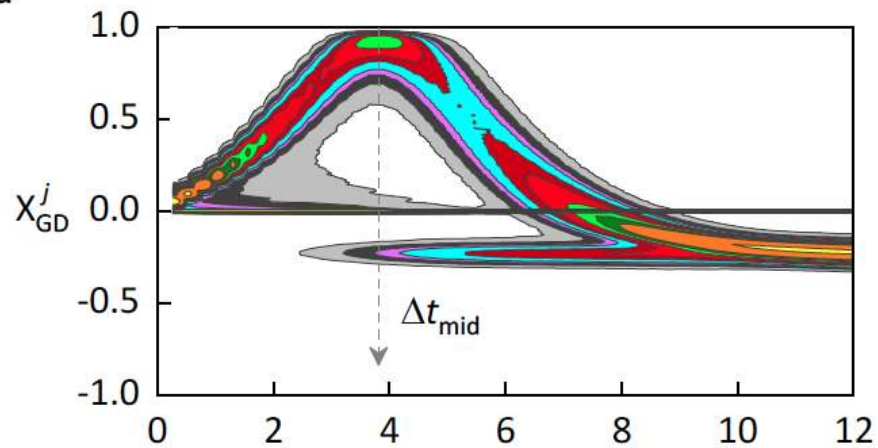
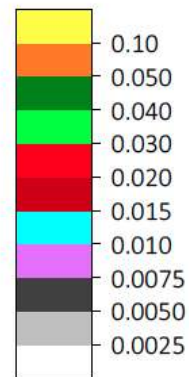
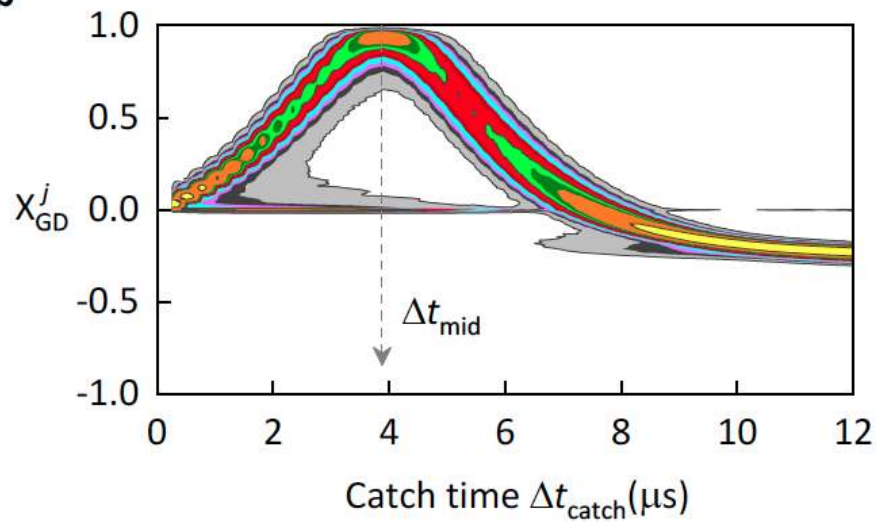
Energy-level diagram

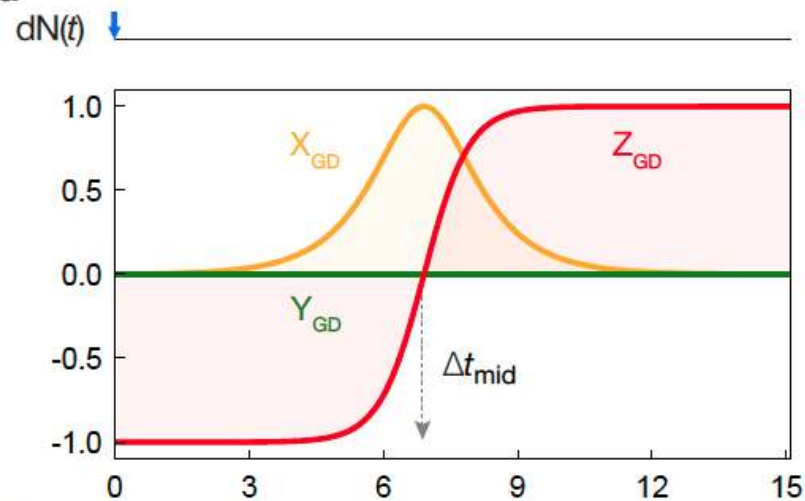
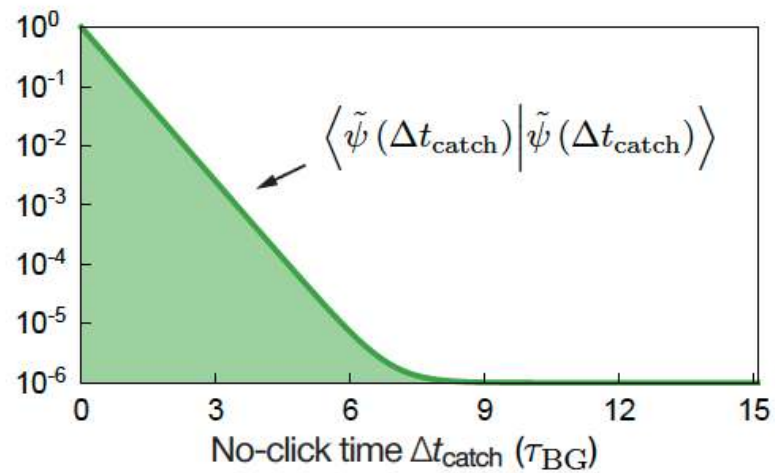




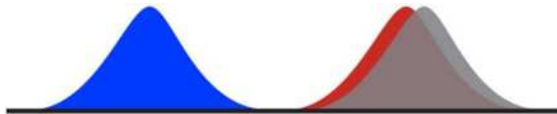
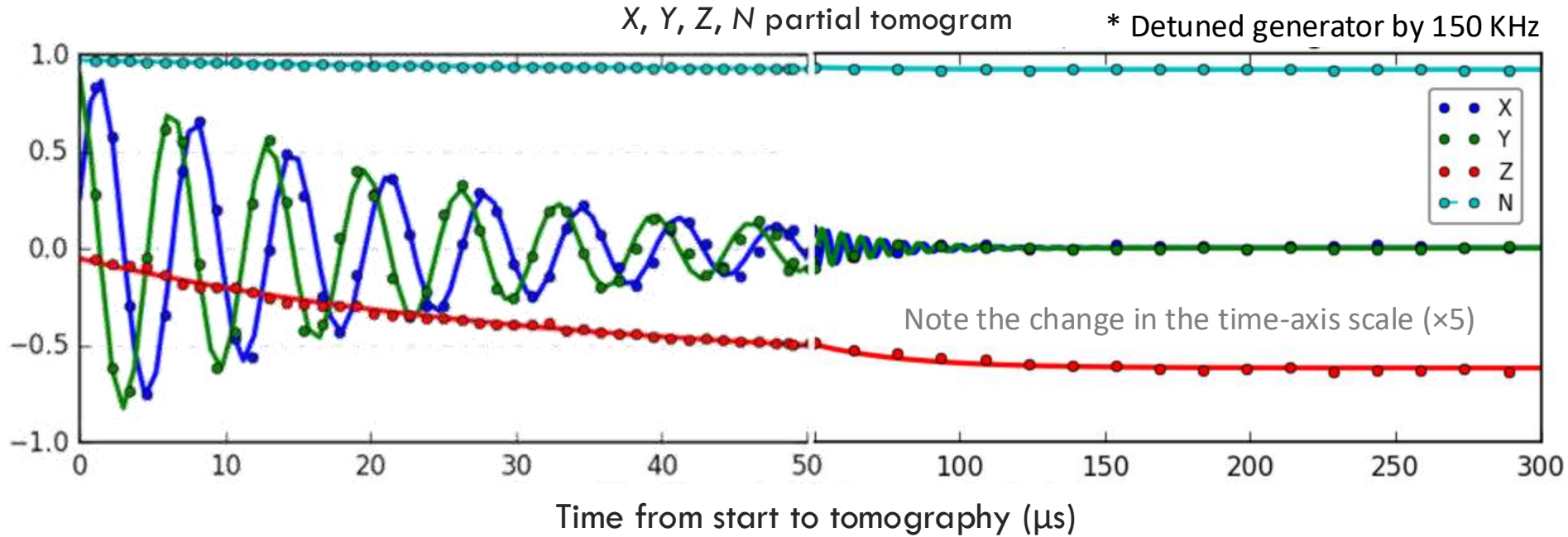




a**b**

a**b**

Coherences: free evolution



$$T_2^{\text{GD}} (\bar{n} = 0) = 23 \pm 2 \mu\text{s}$$

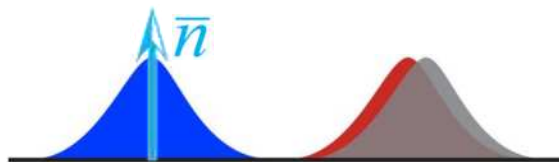
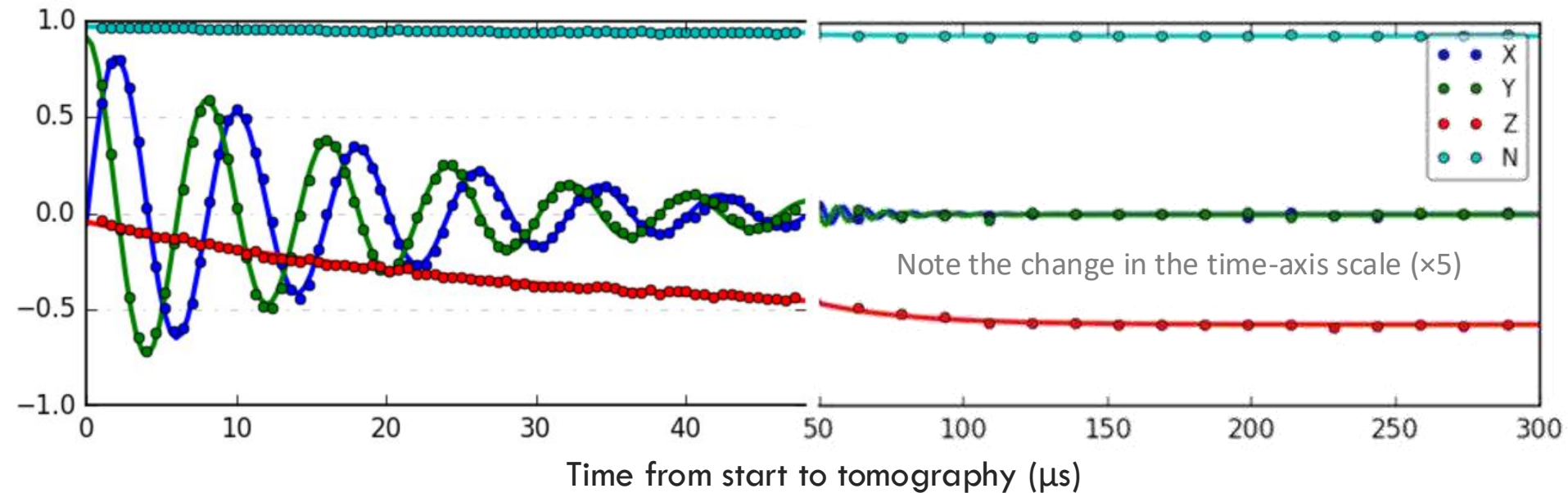
$$\Delta_{\text{det}} (\bar{n} = 0) = 2\pi \times 150 \pm 5 \text{ KHz}$$

$$T_1^{\text{DG}} (\bar{n} = 0) = 32 \pm 1 \mu\text{s}$$

* different thermal cycle from T_1 vs $\bar{n}$ measurement

Coherences: $\bar{n} = 7$ photon drive

X, Y, Z, N partial tomogram

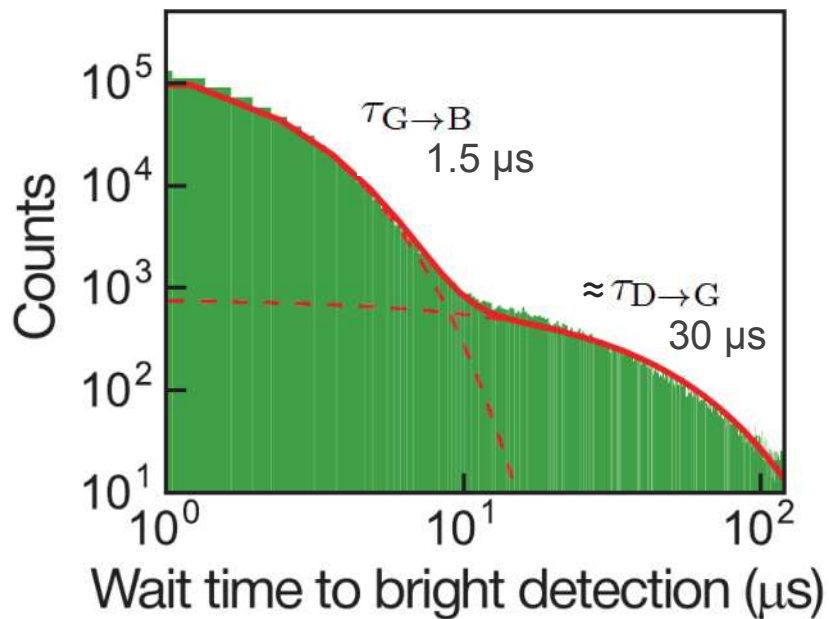
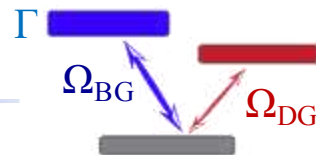


$$T_2^{\text{GD}}(\bar{n} = 7) = 21 \pm 2 \mu\text{s}$$
$$\Delta_{\text{AC}}(\bar{n} = 7) = 2\pi \times 125 \pm 5 \text{ KHz}$$

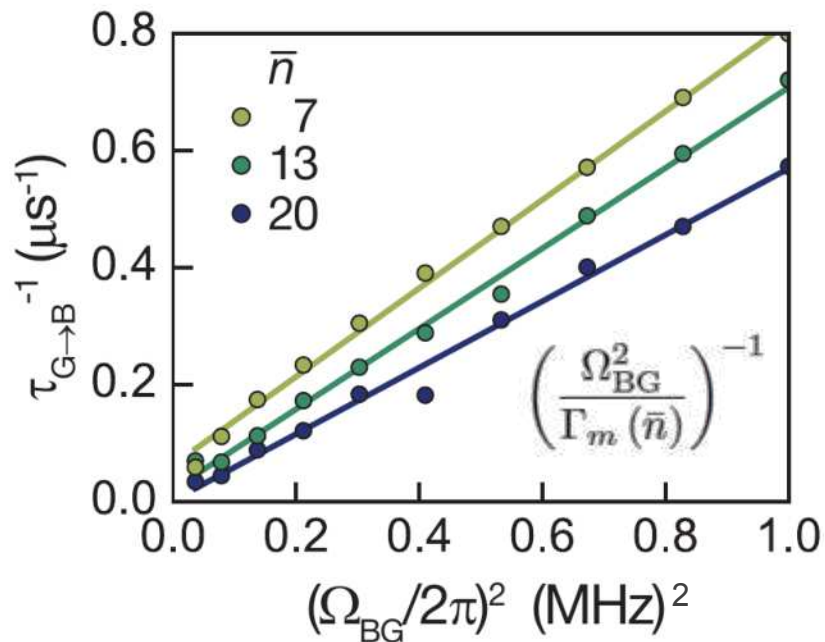
$$T_1^{\text{DG}}(\bar{n} = 7) = 32 \pm 1 \mu\text{s}$$

* different thermal cycle from T_1 vs $\bar{n}$ measurement

Unconditioned quantum jumps



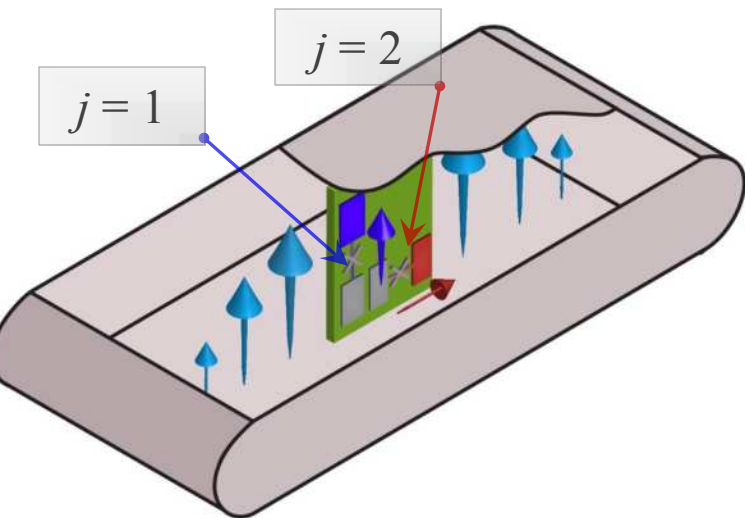
* Note the *log-log* scale



Dispersive shift

χ_{BB} χ_{DD}

χ_{BD} χ_{BC} χ_{DC}



Schematic of design

total number
of junctions
2

$$\chi_{mn} = \sum_{j=1}^J$$

mode label {D,B,C}

junction label {1, 2}

constrained

$$\frac{\hbar^2 \omega_m \omega_n}{4E_j}$$

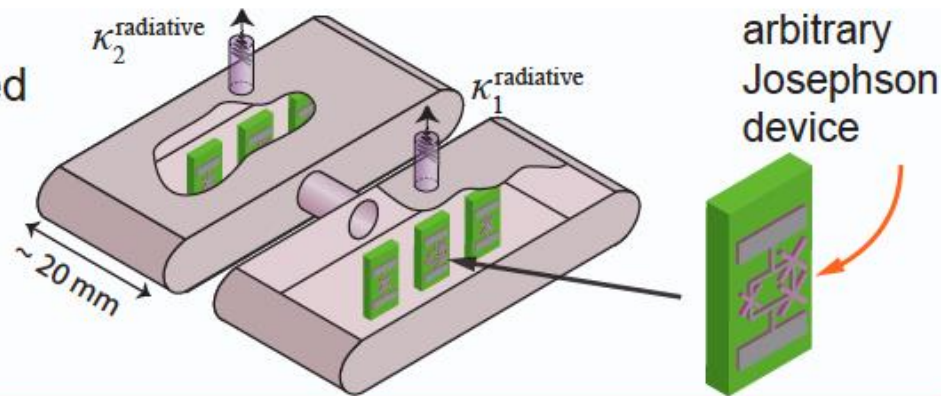
Josephson energy of junction j

subject to
design

$$p_{mj} p_{nj}$$

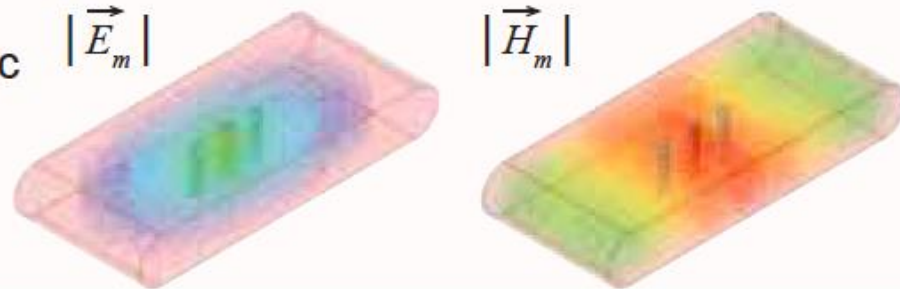
overlap of
energy
participation

a) Josephson system analyzed classically



b) electromagnetic modes

EPR



c) quantum Hamiltonian parameters and losses

$$\hat{H} = \hbar \sum_m (\omega_m + i\kappa_m) \hat{a}_m^\dagger \hat{a}_m - \hbar \sum_{m,n} \chi_{mn} \hat{a}_m^\dagger \hat{a}_m \hat{a}_n^\dagger \hat{a}_n + \dots$$

ω_m	$2\pi \times 9.15 \text{ GHz}$
χ_{mn}	$2\pi \times 6.5 \text{ MHz}$
κ_m	$2\pi \times 2.1 \text{ MHz}$
$\vdots$	$\vdots$

Dispersive readout

Issues

Purcell effect

$$T_1$$

$$T_1^{\text{Purcell}} = \frac{C_q}{\text{Re}[Y(\omega_q)]}$$

Esteve *et al.*, PRB (1986)
Neeley *et al.*, PRB (2008)
Reed, Thesis (2013)

...

Photon-shot-noise
dephasing

$$T$$

$$\Gamma_{\text{th}} = \frac{4\chi^2 n_{\text{th}}^{\text{eff}}}{\kappa_{\text{tot}}} (n_{\text{th}}^{\text{eff}} + 1) \quad \text{large } \kappa_{\text{tot}}/\chi$$

Gambetta *et al.*, PRA (2006)
Schuster, Thesis (2007)
Rigetti *et al.*, PRB (2012)

...

T_1 vs. $\bar{n}$

readout, QND

?

Boissonneault *et al.*, PRA (2009)
Slichter *et al.*, PRL (2012)
Shantanu, MLS (2015)
Sank *et al.*, PRL (2016)

...

Dispersive readout

Issues

Purcell effect

$$T_1$$

Photon-shot-noise
dephasing

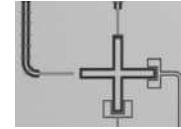
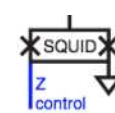
$$T$$

T_1 vs. $\bar{n}$

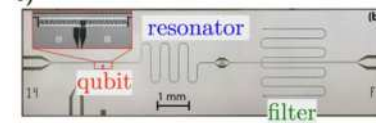
readout, QND

Strategies

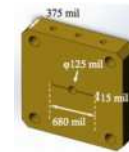
Tunable coupler



Purcell filter



Thermal attenuator

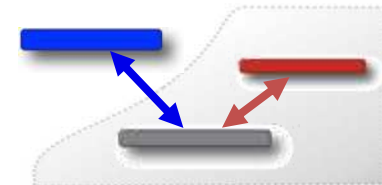


Zhixin

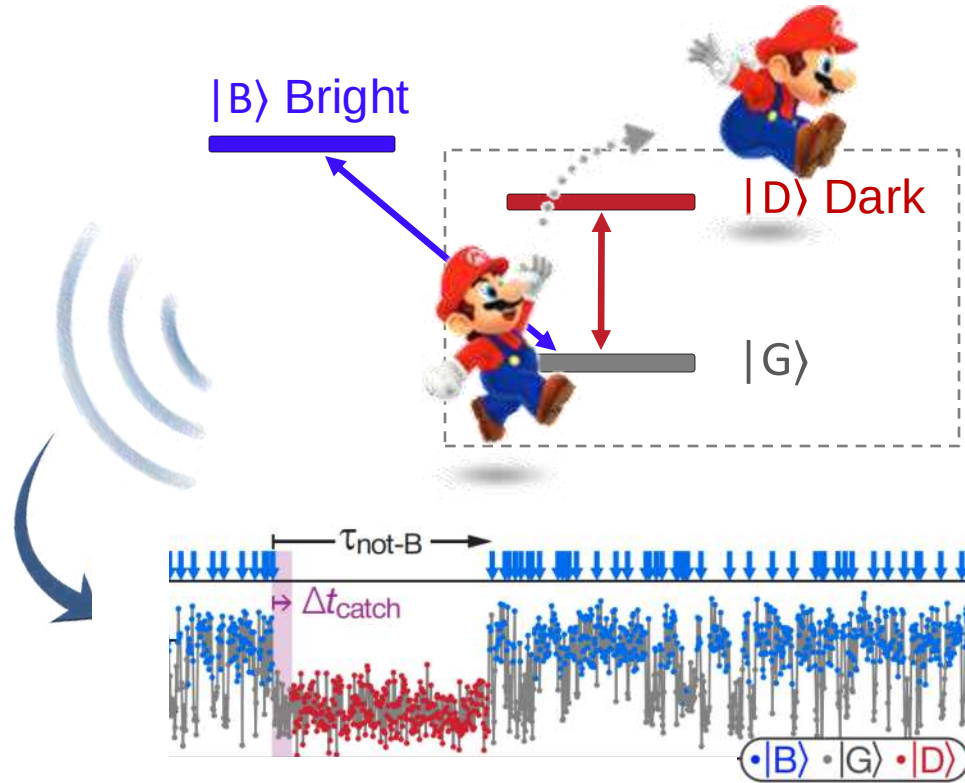
Not dispersive

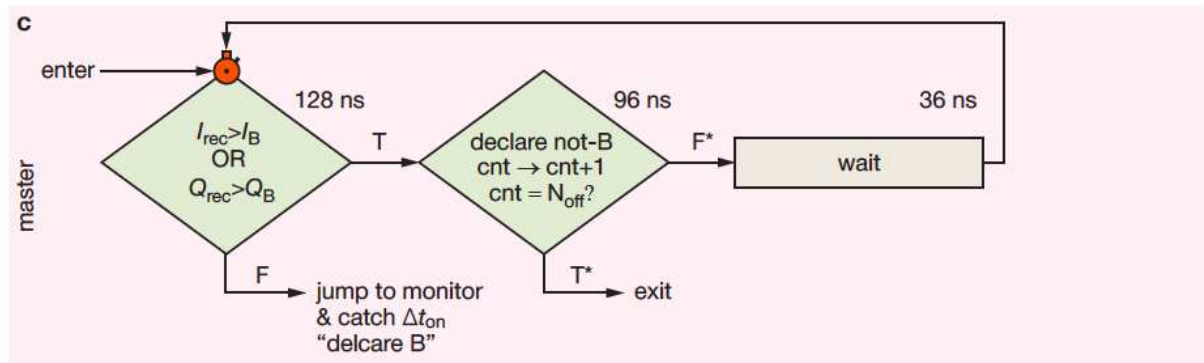
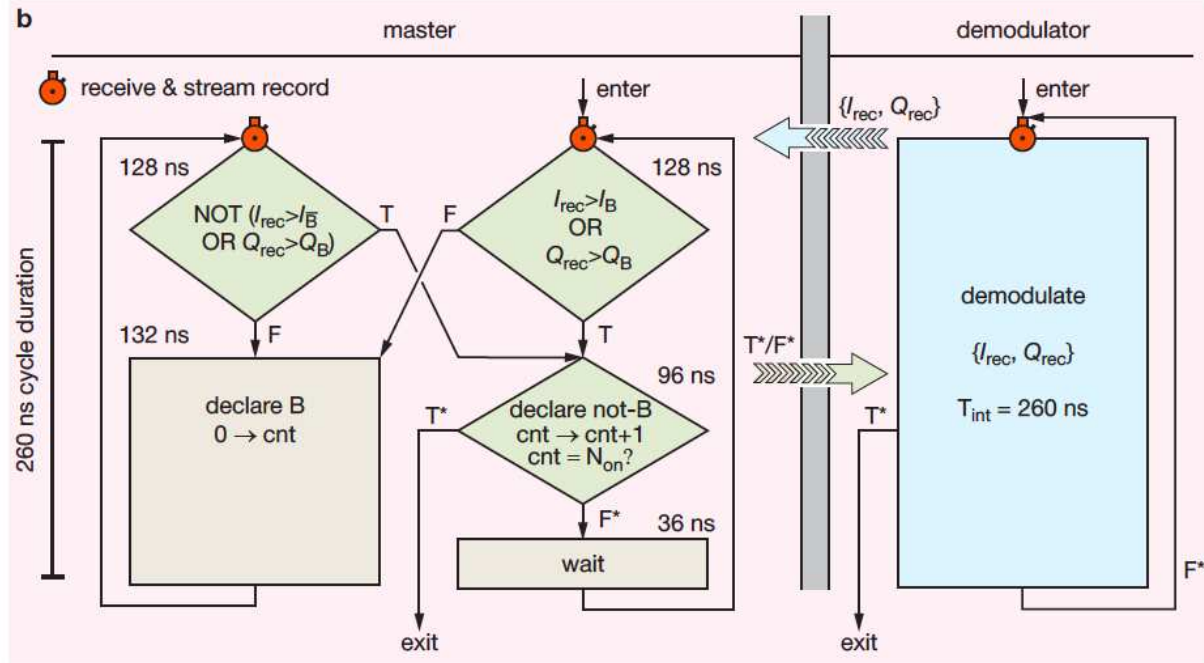
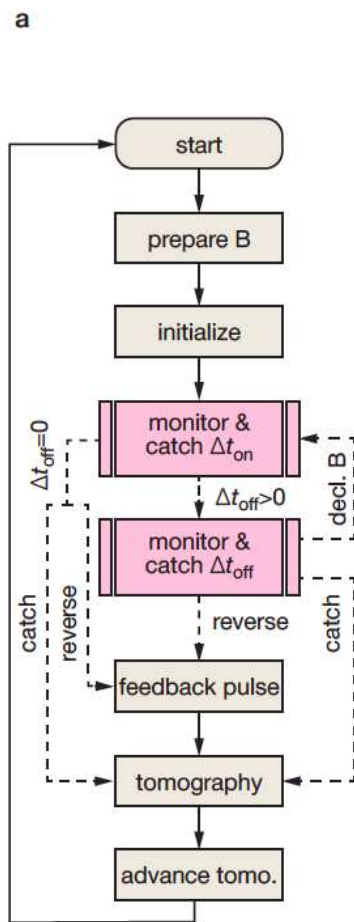
$$g_Z \sigma_Z(\mathbf{b} + \mathbf{b}^\dagger) \quad \text{Steven}$$

Protected qubit



Reversing a quantum jump mid-flight





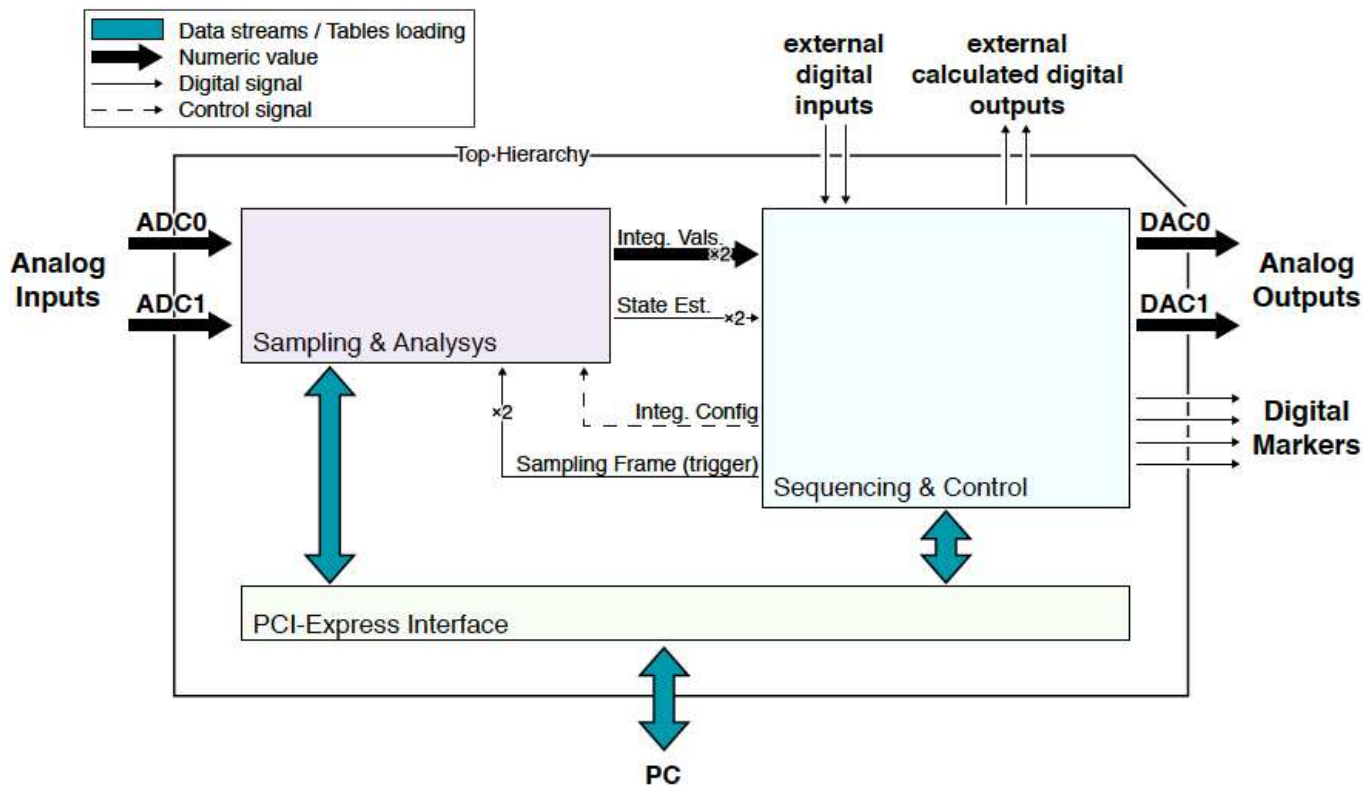


Figure 9: Top Logic Hierarchy. The incoming analog signal is sampled at 1 GHz. The samples are 12 bit deep corresponding to -1V to +1V. Each ADC is controlled independently by the two trigger signals. The integrated IQ values are 16 bit signed integers. A set of integration configurations is loaded before the experiment starts. The sequencer can choose either of them when desired. The sequencing may use these integrated value together with external digital signals to affect the flow. At the end, the main goal is to generate the analog output signals, 16 bit deep values corresponding to -500mV to +500mV.

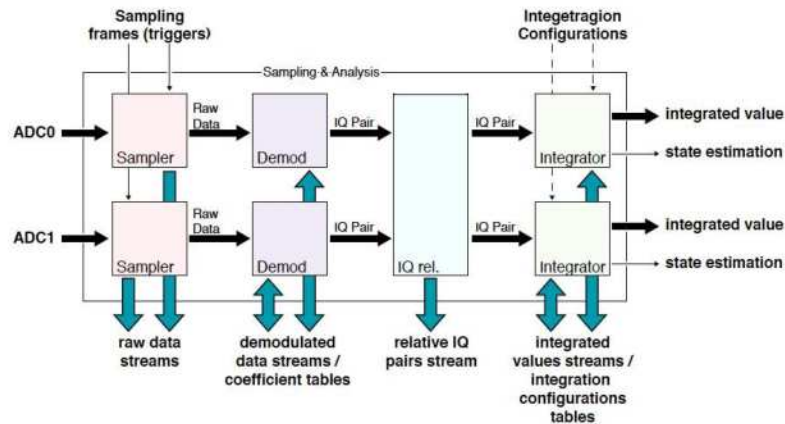


Figure 10: Sampling & Analysis. The raw data is sampled according to the triggers (each channel independently, 12 bit signed integer). It is then demodulated using coefficient table (50 MHz sine and cosine with some amplitude and phase). The results are two IQ-pairs treated as 2.16 signed values. We may then use the first IQ-pair as a reference and rotate the other. The output of the IQ rel. component is also 2.16 signed valued IQ-pairs. It is then integrated according to the desired configuration which includes (for example) the weight for each IQ-pair. All the configurations are stored in tables.

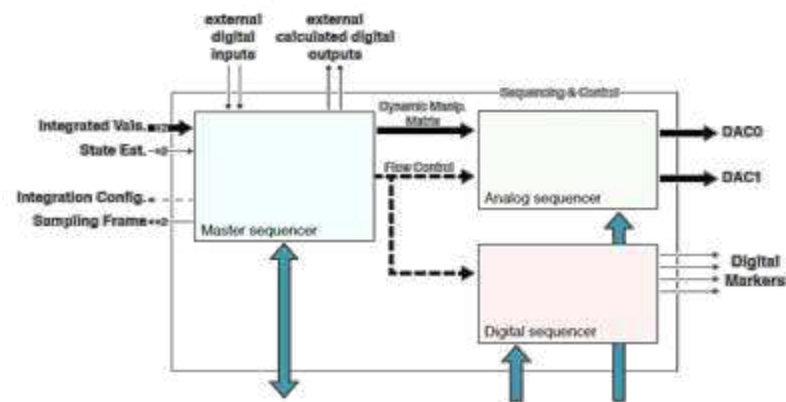
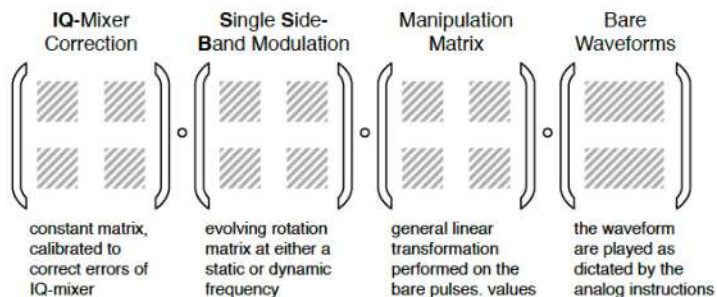


Figure 11: Sequencing & Control. The master sequence produces everything needed to control the flow of the experiment. This includes the internal result which affects branching, manipulation mixers for the analog sequence, updating the integration configuration, triggering data sampling and reporting to the outside world. The job of the analog and digital sequences is much more simple, they go according to their instructions, affected by the flow signals and outputs the results.

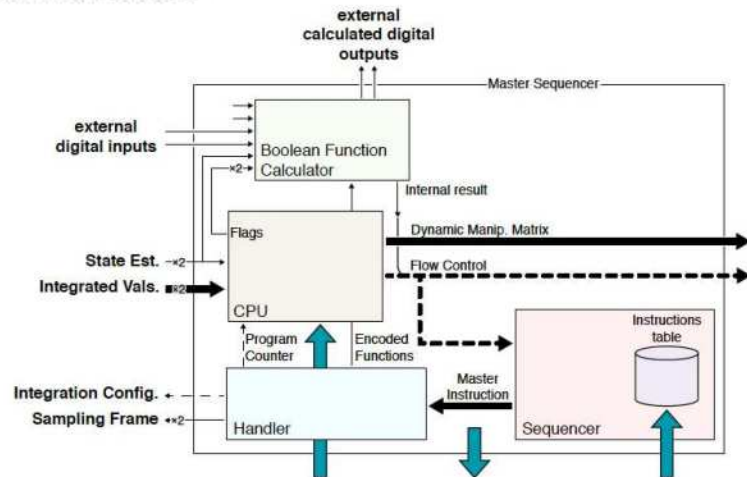
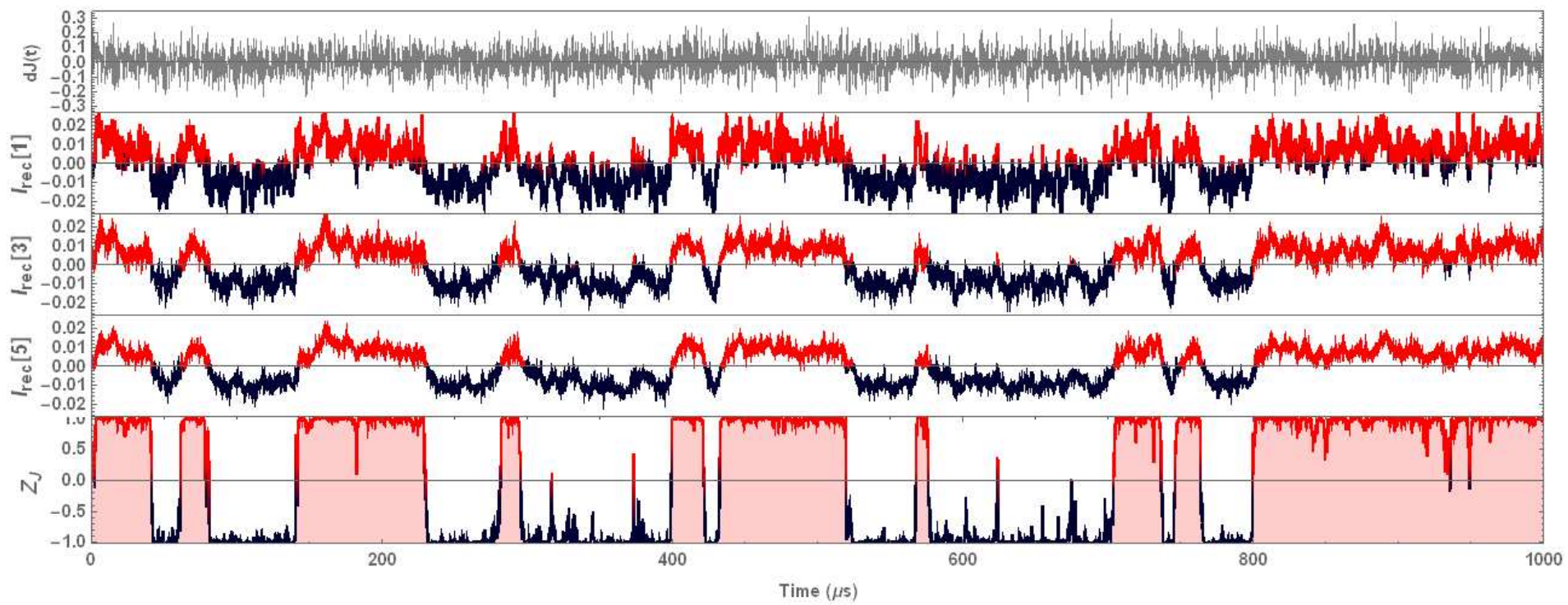
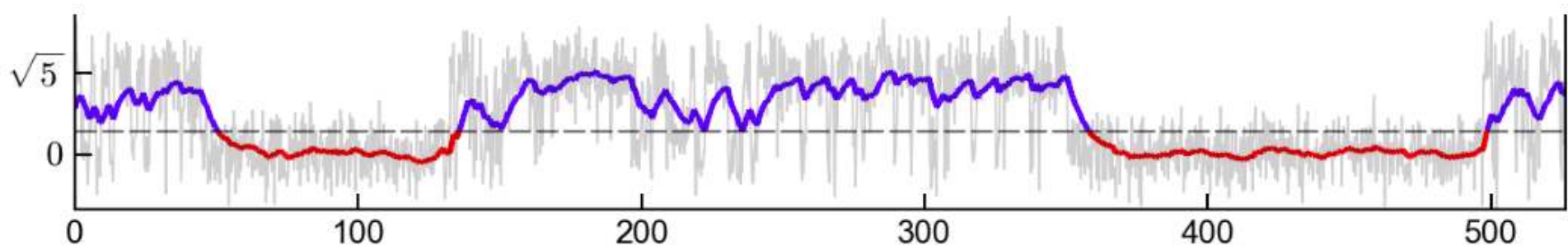
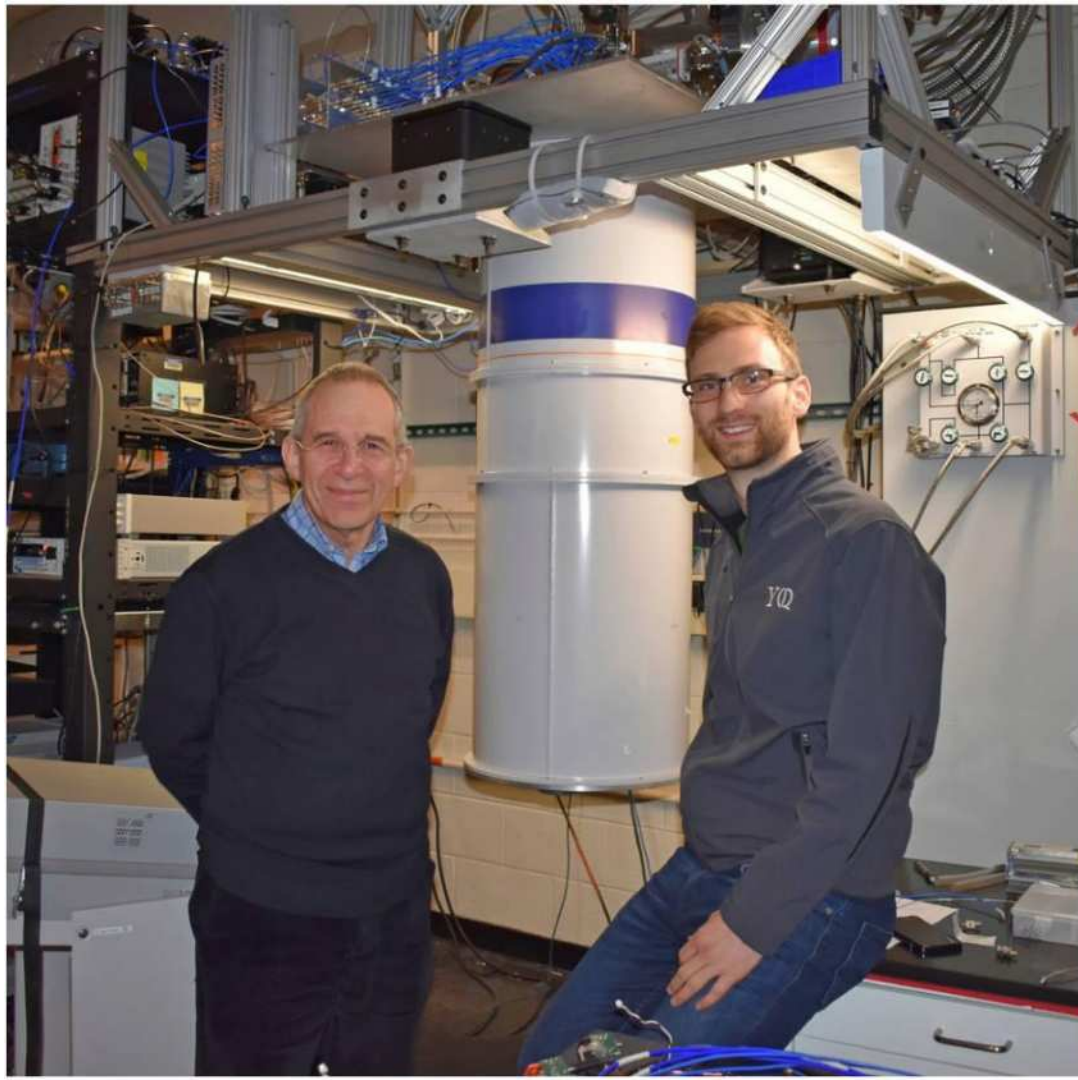


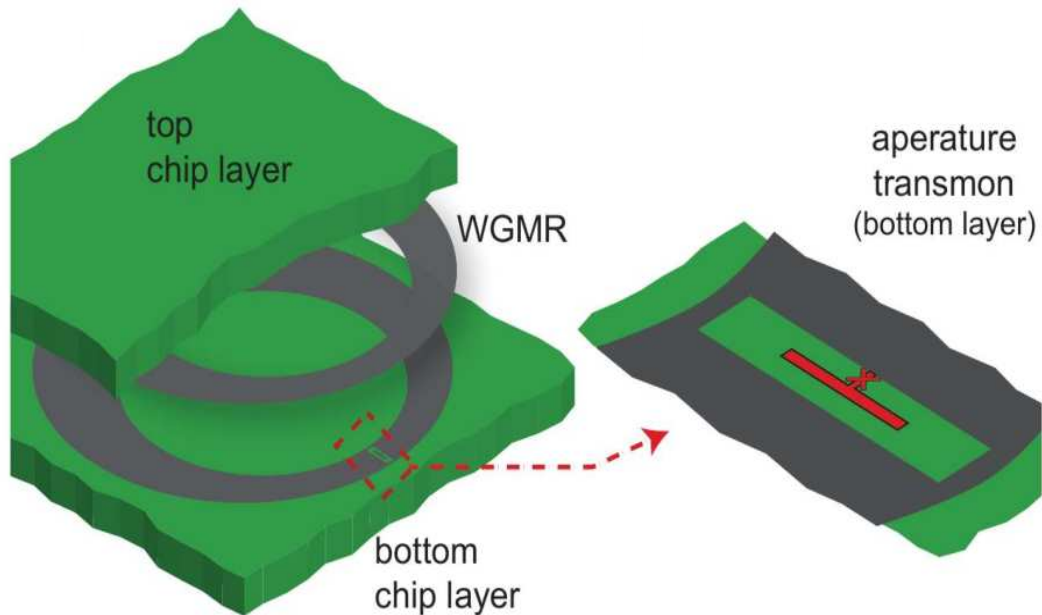
Figure 12: Master Sequencer. The instructions (like any other sequence) are loaded into a table. According to the flow signals and outputs the results. The other components in the system are also affected by the flow signals and outputs the results.







Multilayer (2.5D) cQED

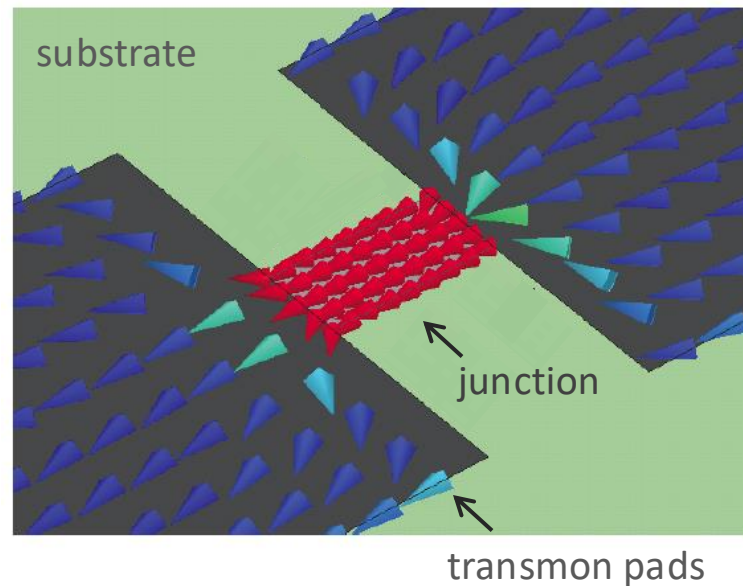


Minev et al., APL (2013)

Minev et al., WO/2016/138395 (2015)

Minev et al., Phys. Rev. App. (2016)

Circuit quantization 2.0 (EPR approach)

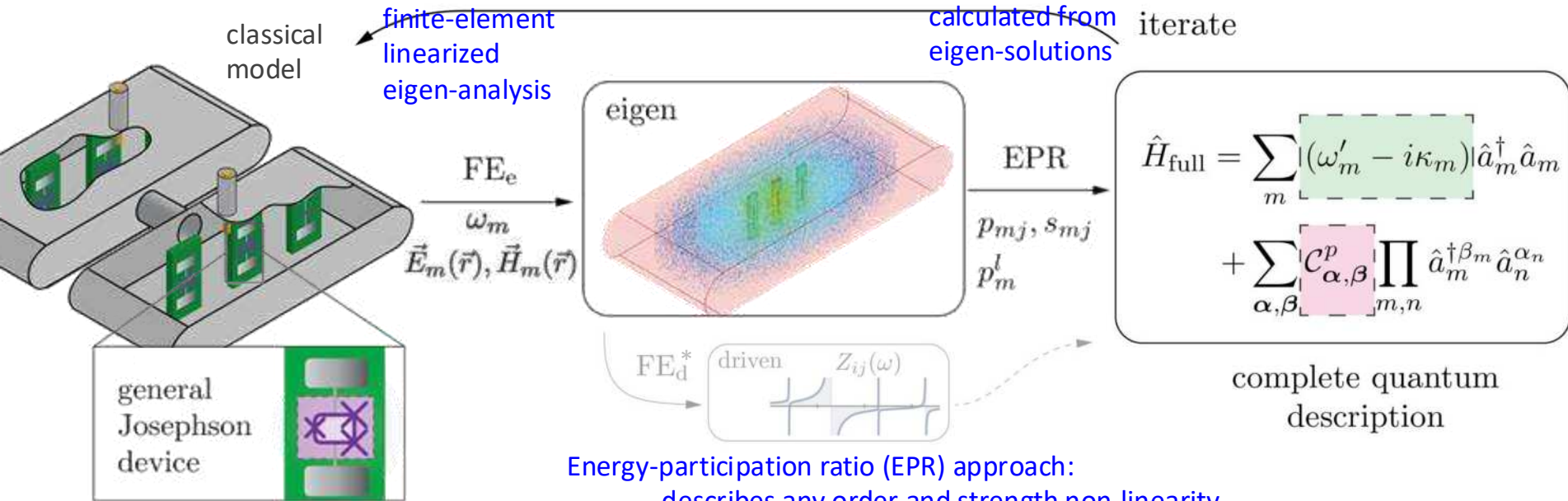


0  Max

Current-density magnitude

Minev *et al.*, in prep. (2018)

Overview of energy approach



Energy-participation ratio (EPR) approach:

- describes any order and strength non-linearity
- describes arbitrary (composite) non-linear inductive devices
- first-principle derivation
- zero approximations (aside from truncation of modes)
- fully automated in python (github.com/zlatko-minev)

Practical limits: Fock and mode basis truncation due to computing power

* Nigg, Paik, *et al.*, PRL (2012),
 Bourassa *et al.* (2012),
 Solgun *et al.* (2014, 2015, 2017), ...



To catch and reverse a quantum jump mid-flight

Nature 570, 200 (2019)
(Thesis: arXiv:1902.10355)



Nintendo Corp.

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